

Beyond Getting Work Done: A Landscape Review of Semantic Scaffolding for Learning with AI

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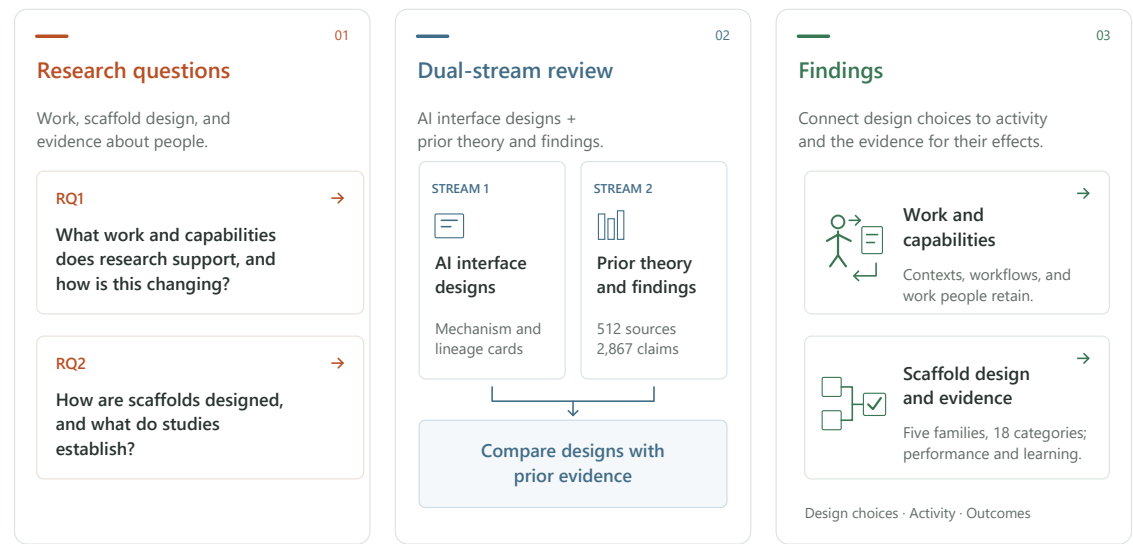


Fig. 1. **Work, capabilities, and scaffold design.** Two research questions guide a dual-stream review of AI interface designs and prior theory and findings. The comparisons examine how features work together, what systems provide and people still do, and what earlier studies tested. The Stream 2 panel shows the synthesis selection: 512 sources and 2,867 claims after source and claim checks.

ABSTRACT

AI tools now help people write, program, analyze data, and design, and many also aim to help people learn through that work. What such a tool gives a person to think with varies: a generated explanation, criteria for judging alternatives, an editable plan, a record of earlier choices. We call these designed structures with task meaning *semantic scaffolding*. We reviewed 4,511 AI-involved papers from 40 venues (2019–2026), together with theories and empirical studies of how support works, and synthesized a design space of 5 mechanism families and 18 categories. Compared with papers on earlier AI, generative-AI papers more often concern producing artifacts and more often give people control over what the system does, after adjusting for venue and year. What a person can edit matters: the task specification, the AI’s interpretation of it, the criteria, or the result. Designs also differ in whether people construct a representation, reason with supplied material, or judge proposed results. Studies that measure learning report gains and costs together: CodeVoyager raised closed-book comprehension without a completion advantage, and required reflection cut completed practice from 3.4 to 2.0 problems in a fixed time. The design space gives researchers named alternatives to build and test.

CCS Concepts: • **Human-centered computing** → **Empirical studies in interaction design**; *Graphical user interfaces*; *Collaborative interaction*; Natural language interfaces.

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1 INTRODUCTION

A writer asks Luminate for story ideas. Instead of one draft, the system proposes dimensions along which ideas can differ and fills each with options [120]. The dimensions help the writer survey the possibilities and compare ideas. They are also generated, and a generated dimension can miss what matters to this writer. The writer now judges two things: the ideas, and the structure offered for judging them.

This is the design problem we study. AI tools increasingly supply the structures people think with, not only the content they produce: criteria, examples, relations between concepts, plans, and records of earlier choices. We use *semantic scaffolding* to describe designed structures with task meaning that a person inspects or acts through. The term joins two lines of work. Scaffolding research asks how support lets a learner do what they could not yet do alone [100, 134]; cognitive-artifact research asks how representations change the task a person performs [92]. The assistance dilemma and ICAP add that what support supplies, and what it leaves for the learner to do, shapes what people learn [27, 64].

Broad labels such as feedback, reflection, and user control hide the choices that matter. Editing a generated answer differs from correcting how the AI interpreted the request before it answered. A learner’s account prepared for a coach differs from a simulated learner used to rehearse coaching. We map these choices across domains such as programming, writing, data analysis, and design, connect them to the work people do, and compare what studies found when the choices were tested.

We reviewed papers from 40 HCI, AI, and learning venues published from 2019 to 2026. Working with language models, we screened 35,480 titles and abstracts, selected 7,372 papers for full-text review, and analyze 4,511 in which AI is part of the system or study and a person inspects or acts through a designed structure. A second body of work, theories and empirical studies of how support works and when it helps, gives the comparisons we use to interpret designs; it includes studies without AI. We ask:

- **RQ1.** What work and human capabilities does semantic-scaffolding research support, and how has this focus changed from 2019 to 2026?
- **RQ2.** How is semantic scaffolding designed, and what does existing evidence show about its support for performance and learning?

We contribute:

- (1) a design space of semantic scaffolding in AI-assisted work, with 5 mechanism families and 18 categories, each illustrated by named systems;
- (2) an account of what people can inspect and change in these designs (the specification, the AI’s interpretation, the criteria, or the result) and of how support passes between learners, peers, teachers, and coaches;
- (3) a comparison of the consequences studies report for performance and learning, tied to the design alternatives they tested.

2 BACKGROUND

2.1 Scaffolding and Cognitive Artifacts

Scaffolding began as an account of tutoring: a tutor takes over the parts of a task a learner cannot yet manage, so the learner can do the rest [134]. Developmental scaffolding responds to the learner, reduces help as the learner improves, and hands over responsibility [124]. Reiser separates structuring a task from problematizing it, drawing attention to what the learner must reason about [105], and inquiry-support guidelines turn these ideas into design [104]. Support can also be spread across people, artifacts, and tools [10, 100]. Pea argues that support meant to stay should be kept apart from scaffolding that fades. Sherin et al. extend the metaphor into a comparative account of scaffolded activity [114], and Puntambekar and Hübcher warn that calling a tool a scaffold does not make its help responsive to the learner [103].

Cognitive artifacts are support meant to stay. Norman uses the term for designed devices that store, display, or transform information for a task [92]. He distinguishes what the person and the artifact accomplish together from how the artifact changes the person's task: a checklist moves work into preparing, consulting, and interpreting the list. The same content supports different operations when it is arranged differently [110, 144]. People also change the world to help themselves reason [62], and several representations can aid understanding while adding the work of coordinating them [2].

We use *semantic scaffolding* for designed structures with task meaning—concepts, relations, examples, criteria—that a person inspects or acts through, whether they fade or stay. Zong et al. use the term for an LLM technique that explains domain groupings of data to blind and low-vision readers [149]; we use it more broadly. Keeping the two kinds of support apart matters for evaluation: performance with a tool and capability without it are different outcomes, and only the second shows growth toward independence [100, 103].

2.2 Representations in AI Interfaces

Recent design spaces describe these representational choices directly. Min et al. separate an abstraction's schema from its representation and examine granularity, presentation, and transformation [87]. Kim et al. organize how generative interfaces elicit preferences along six axes, including when a request appears and which part of an artifact it concerns [60]. Systems show the range. In Graphologue, selecting a concept extends a diagram with related concepts and links the person can follow and question [56]. In AI Chains, intermediate steps expose how a problem, a suggestion, and drafted feedback connect, and people can change each step [137]. In HaLLMark, a highlight marks authorship and links a passage to the prompt that produced it [49]. A cue can show that an action exists without showing what it does [93]: in Codestrates v2, a file-like tree organized code, and its order also governed execution, which one participant discovered only in use [18].

People and technology change together. Mackay describes users who repurposed email filters, built to sort messages before reading, into tools for filing them afterwards; the developers then supported both uses [81]. Interfaces have long adapted controls to a person's abilities [42, 132]. LLMs add support generated during use: diagrams that grow during exploration [56], views people recompose [86], and representations generated as tasks change [23]. A generated view can perform an operation, expose relations, and become the control for the next step. The design questions are who builds it, who can revise it, and what work its use adds or removes.

2.3 Offloading and Control in AI-Assisted Work

Distributed cognition treats people, tools, and artifacts as one cognitive system [47]. The extended-mind argument counts reliably consulted resources as part of memory [30], and material structures hold relations otherwise kept in mind [55]. Offloading changes what people retain: expected access to information shifts memory toward where to find it [117], and automation can leave operators with the hardest cases and little practice [8]. Mixed-initiative interaction combines automation with direct manipulation [50]. Interactive machine learning makes the user–system loop the object of design [3], explainable-AI frameworks separate understanding, trust, debugging, and decision support [90], and teaming and learning-support research adds capability and transfer [22, 44]. Because people trade accuracy against effort [99], making some operations easy also changes strategies. Generative systems that expose prompts, plans, alternatives, and evidence [24, 63, 120, 136] also produce task criteria and explanations on demand, so people may need to check the guidance as well as the answer.

2.4 Landscape Reviews and Design Spaces

Design spaces organize recurring choices and the criteria for weighing them [82], and how they are drawn shapes what designers explore [14, 70]. HCI landscapes relate creativity support tools to practice and evaluation [38], organize traces of creative activity [46], and examine generative tools in artists’ work [63]. Citeology, Royal Constellations, and HCI Research Trails make citation relationships navigable [19, 85, 89]; we also record what a cited source does in a paper: frame it, explain a mechanism, shape a design, or provide a measure.

Earlier syntheses relate contingent support and learner activity to learning [11, 124]. Self-explanation research identifies practice exposure and redundant prompting as conditions that matter [107]. The assistance dilemma and ICAP frame when to give help and what engagement it invites [27, 64], and expertise reversal and desirable difficulties qualify what a practice result means [15, 58]. Krakauer distinguishes artifacts that leave capability behind when removed from those whose advantage lasts only while they are in use [65]. Human–AI guidelines already address intelligibility and correction [4]. Our addition is a comparison of concrete design choices across domains, linked to the activity each asks of people and to what studies measured when they tested it.

3 METHODS

We reviewed two bodies of work (Figure 2). Stream 1 is a systematic corpus of AI-involved systems and studies from fixed venues and years; it supplies the designs we map. Stream 2 collects theories and empirical studies of how support works, including studies without AI; we interpret the designs against it. The two are counted separately. Scoping-review guidance shaped selection, extraction, and reporting [6, 71, 122], and earlier HCI landscape reviews shaped the venue frame and synthesis [38, 46].

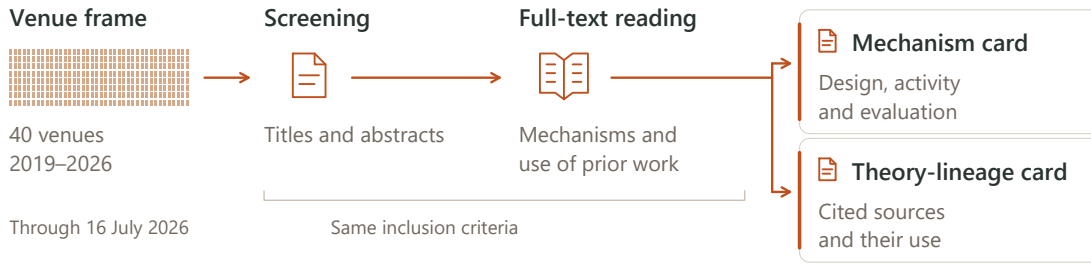
3.1 Literature Review

3.1.1 Venues. Stream 1 covers 40 venues in HCI, design and creativity, visualization, graphics, learning and education, software engineering, and AI and language, published from 2019 to 2026 (Figure 3). Papers are dated by their version of record; the selection figure, the design space, and the trend figures cover papers published by 16 July 2026. Ineligible publication types were excluded by rule, and citation counts played no part.

3.1.2 Screening. Two language models separately screened the 35,480 records with an abstract; the 2,498 without one were not screened. Their verdicts, a review of their disagreements, and (for the first reading list) PDF availability

Literature selection and evidence extraction

Stream 1 · AI-involved designs



Stream 2 · Prior theory and findings

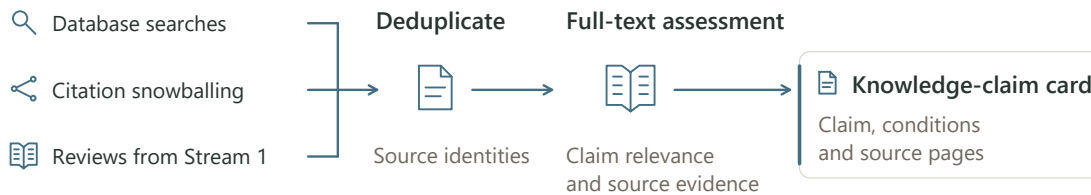


Fig. 2. **Literature selection and evidence extraction.** Stream 1 runs from a venue frame through screening and full-text reading to mechanism and theory-lineage cards. Stream 2 combines three discovery routes, removes duplicates, and records each source's claims on a knowledge-claim card.

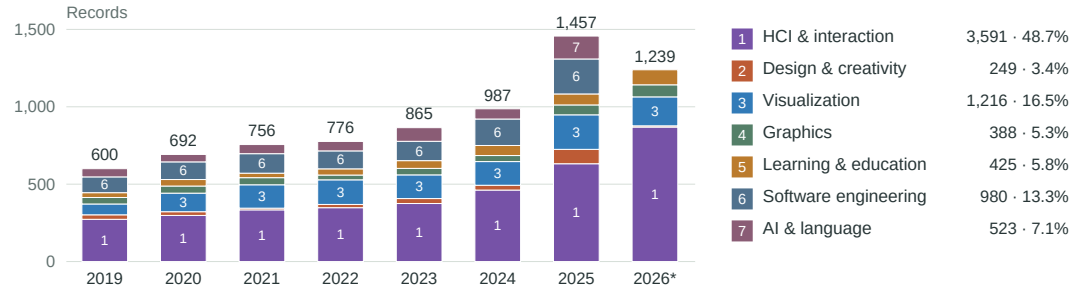
determined the 7,372 records selected for full-text reading. One model read each paper and recorded, with passages and pages, whether it met the criteria.

The AI requirement was added partway through full-text reading, so three separate model passes rechecked the 1,154 earlier cards that left the question open. A paper left the corpus only when all three found no AI, with low uncertainty and a quoted passage (492 cards). The corpus holds 4,511 papers, and 1,076 of them form a learning subset: their support has a documented learning purpose, or the study examines its learning consequences. Another 474 papers without AI form a labeled comparison set outside the systematic corpus counts; the text marks any it uses as an example.

3.1.3 Inclusion Criteria. Stream 1 includes a paper when three conditions hold, at screening and at full text (Figure 4): AI is part of the system or the study; the paper implements or studies a designed structure with task meaning; and a person inspects or acts through that structure. AI is part of a paper when an AI system is built, used, evaluated, or simulated (including Wizard-of-Oz systems presented as AI), or when AI is what the study designs, explains, or labels. Rewording a chatbot's replies (hedges, tone, length) is not a designed structure. Workshop templates, cards, storyboards, and probes used to elicit views count only when the paper builds and evaluates them as its contribution. Reviews, surveys, and meta-analyses go to Stream 2. A design can qualify without a user study; its benefits then count as intended uses, not measured effects. Learning purpose, measured outcomes, adaptation, fading, and the transfer of responsibility are recorded but do not decide inclusion.

Selected records by year and venue group

7,372 records selected for full-text reading; venue groups describe disciplines, not application domains.



Each venue: selected records and share of the 7,372 selections

1 CHI	1,548 · 21.0%	1 ECSCW	10 · 0.1%	2 Design Studies	13 · 0.2%	5 AIED	58 · 0.8%
1 UIST	306 · 4.2%	1 GROUP	6 · 0.1%	2 IDETC	151 · 2.0%	5 IJAIED	37 · 0.5%
1 IUI	359 · 4.9%	1 ASSETS	84 · 1.1%	3 VIS	394 · 5.3%	5 EDM	103 · 1.4%
1 DIS	232 · 3.1%	1 HRI	187 · 2.5%	3 EuroVis	111 · 1.5%	6 CHASE	55 · 0.7%
1 CSCW	258 · 3.5%	1 IVA	82 · 1.1%	3 TVCG	635 · 8.6%	6 ICSE	321 · 4.4%
1 C&C	67 · 0.9%	1 CHIIR	56 · 0.8%	3 VL/HCC	76 · 1.0%	6 FSE	238 · 3.2%
1 TOCHI	45 · 0.6%	1 HCOMP	36 · 0.5%	4 SIGGRAPH	295 · 4.0%	6 ASE	366 · 5.0%
1 TiIS	102 · 1.4%	1 IJHCS	127 · 1.7%	4 SIGGRAPH Asia	93 · 1.3%	7 ACL	191 · 2.6%
1 EICS	33 · 0.4%	2 ICCS	53 · 0.7%	5 L@S	41 · 0.6%	7 EMNLP	255 · 3.5%
1 HAI	53 · 0.7%	2 Design Science	32 · 0.4%	5 LAK	186 · 2.5%	7 RecSys	77 · 1.0%

*2026 is partial: publication cutoff 16 July. Zero selections can reflect unavailable proceedings.

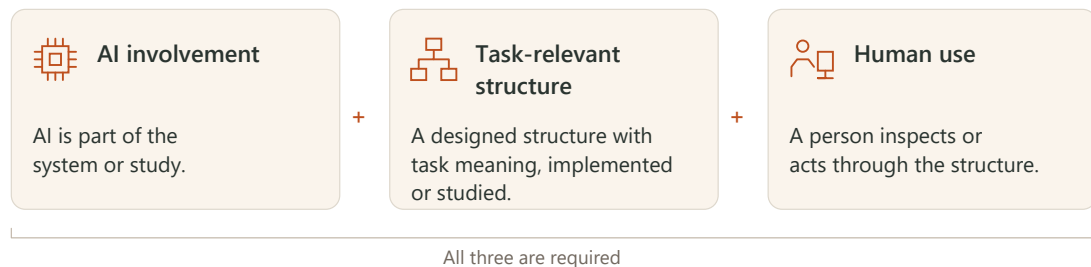
Fig. 3. **Records selected for full-text reading, by publication year and venue group.** The lower panel gives each venue's count and share of the 7,372 selections. The 2026 bar covers papers published by 16 July.

Inclusion criteria

Stream 1

Title/abstract screening

→ Full-text reading



Stream 2 · Prior theory and findings

Relevant, transferable claim

→

Full-text support and conditions

Fig. 4. **Inclusion criteria.** Stream 1 requires all three conditions at screening and full text. Stream 2 assesses whether a claim is relevant and supported by its full text. Learning purpose and measured outcomes are recorded during extraction.

3.1.4 *Theories and Empirical Studies.* Stream 2 gathers theories, design principles, and empirical findings about how support works. Its sources came from database searches, backward and forward citation snowballing [133], and reviews found in Stream 1, and every source received the same full-text assessment of relevance and support. Of 795 candidate works, we read 543 in full and excluded 31 for recorded reasons, such as falling outside the review questions or appearing in a questionable venue. The analysis uses the remaining 512 sources and their 2,867 claims. Books, foundational theory, null findings, and studies without AI are eligible.

3.2 Analysis

3.2.1 *What We Recorded.* A model read each full text and filled a structured evidence card, quoting the passages and pages behind its entries.¹ Scientific-document benchmarks and expert-assisted literature mining offer precedents for such passage-linked records [125, 130]. The Stream 1 unit is a *mechanism*, an operation involving a designed structure with task meaning: AI Chains’ workflow overview and its step editor are two mechanisms [137]. For each mechanism, the card records what the interface represents, how and when it presents that meaning, and what people can do. It also records how the work changed, what the tool took over and what the person kept, and how the design was evaluated: comparison, measures, duration, participants, and results. A second card records how the paper uses each cited source: to frame a problem, explain a mechanism, motivate the design, inform a measure, critique an approach, or in passing. The Stream 2 unit is a *knowledge claim*, a finding or theoretical argument about how support works, for whom, and when; each card holds one to eight claims with their evidence, conditions, and limits. Missing information is recorded as not reported or unclear, and author claims, measured results, and reviewer inferences stay apart (fields in Appendix A).

3.2.2 *Building the Design Space.* Closed fields take predefined values, such as whether learning was measured and for how long; open fields describe mechanisms and human work in each paper’s terms. We merged the 142,877 open-field observations into 1,290 codes and grouped these into 223 categories and 92 families across 18 fields (Appendix A.3). Each grouping links to its observations, so the grouping and the assignments can be checked separately [52]. The mechanism field’s hierarchy is the design space: 5 families and 18 categories (Section 5; codebook in Appendix B). For the four-function view (Figure 12), we assigned each of the 110 mechanism codes to the functions it serves; exploring alternatives serves both representing and evaluating.

We built the vocabulary from the cards completed by 28 August 2026 and coded later cards with it. The design-space counts describe the 2,847 papers of that first set still in the corpus (2,891 for the closed fields); the annual and technology figures draw on the 4,086 papers with usable coding (Appendix A.6). A paper counts once per category, however many mechanisms it has. We classified each paper’s AI as generative or traditional AI/ML by the technology it used, not its year, and adjusted technology comparisons for year and venue, since venue mix alone can create differences (Appendix D).

3.2.3 *Comparing Designs and Studies.* We used the codes to find cases and the source passages to compare them. For each question, we compared the task, the representation supplied, the actions available, the reported use, and the assessed consequence, recording contrary and unclear cases beside supporting ones. We chose examples that fit a mechanism, varying domain and year; citation counts mattered only after fit. We retrieved prior findings and counterexamples through the seven fields coded on Stream 2 claims, such as the form of support and a claim’s conditions.

¹Evidence cards: mechanism, theory-lineage, and knowledge-claim cards.

The lineage cards trace how a cited theory or finding entered a design or a measure (Section 5.5); reference lists and citation counts come from Semantic Scholar (Appendix A.9.2).

For study results, each contrast keeps its population, comparator, outcome, timing, and the assistance available during assessment, and positive, mixed, null, harmful, and unmeasured results stay distinct. A whole-system result is credited to the whole system, and a learning claim counts as tested only when a study measured learning or transfer. We sorted outcomes into process, artifact, and learning, each assessed by the study or reported by participants (Section 5.6). We did not pool effects across studies (Section 8).

3.2.4 Checking the Coding. Scripts checked the cards' required fields and matched their quotations against the source text, and quotations the text layer could not match were checked against page images; counts and plots are computed from the saved data. To test the open-code grouping, a separate model pass coded 1,571 extracted observations in six fields with the existing codes hidden. Agreement was low (adjusted Rand index .079 to .445), and the blind pass grouped more coarsely in five of the six fields (Appendix A.4). The mechanism vocabulary gained 12 codes across the 1,754 papers read after it was built (Appendix E).

3.3 AI Use Disclosure

We used language models to screen, read, and code papers, group codes, develop and critique the cross-case comparisons, write analysis code, and draft and revise text, tables, and figures. Claude Opus 5 and GPT-5.6 Sol screened titles and abstracts. GPT-5.6 Sol, Claude Opus 5, and Claude Opus 5.5 read the full texts and wrote the Stream 1 cards; Claude Opus 5 and GPT-5.6 Sol wrote the Stream 2 cards; Claude Opus 5.5 ran the three-pass AI check. Claude Opus 5 merged labels into codes, and model-assisted review, including Claude Fable 5.1, grouped the codes into categories and families. GPT-6 Sol linked card evidence for the cross-case comparisons, and GPT-6 Astra developed and critiqued them. GPT-6 Astra, working through Codex, drafted the merged manuscript, and Claude Opus 5.5 revised it. Prompts and run records are kept. The authors set the research questions, scope, criteria, and card formats, directed the synthesis, and are responsible for its conclusions; people have not yet checked the coding or the cross-case interpretations (Section 8).

4 FINDINGS: WORKFLOWS AND CAPABILITIES

In GenAI papers, the tool more often generates the artifact itself. This shifts attention to the material people encounter and the decisions they still make. We first place the shift among the work domains papers study, then ask what the person still composes, checks, and decides, and last how support passes through other people. The activity map in Figure 5 distinguishes what people do; Figure 7 then separates work attributed to the tool from work retained by the person.

4.1 Work represented and changes in research attention

Within venues, we detected no annual change in a domain's share from 2019 to 2025 after correction for multiple tests (Table 12). The quantitative pool contains 4,086 AI-involved paper groups; each comparison uses the subset with the required coding and technology classification. GenAI includes earlier generative models and generative hybrids, and traditional AI/ML appears in every year. GenAI becomes the majority from 2024 (Figure 6). Technology comparisons leave out the 639 papers whose technology is unresolved, 2026 is partial because papers were collected in July, and a paper can support several activities.

What does the person do?

Twelve activities that interfaces can support



Fig. 5. What does the person do? Twelve activity groups organize the workflow codes. Each schematic illustrates an action through an interface: finding, inspecting, comparing, judging, directing, creating, articulating, reflecting, monitoring, following a procedure, coordinating, or preparing a system. These activities can occur together; the grid does not imply a sequence. Allocation, supplied support, and measured outcomes are coded separately.

GenAI moves generation to the tool. In 2,119 papers from 2019–2025 with coded delegated work, the tool generates the artifact in 19.3% of GenAI and 1.9% of traditional AI/ML papers (+16.6 points after adjusting for year and venue; 95% CI 12.7 to 20.5), and it more often generates responses and revisions (+9.4). Analysis moves the other way: the tool less often calculates and models (-11.5), evaluates and classifies (-9.7), or compares and finds patterns (-6.7) (Appendix D.2).

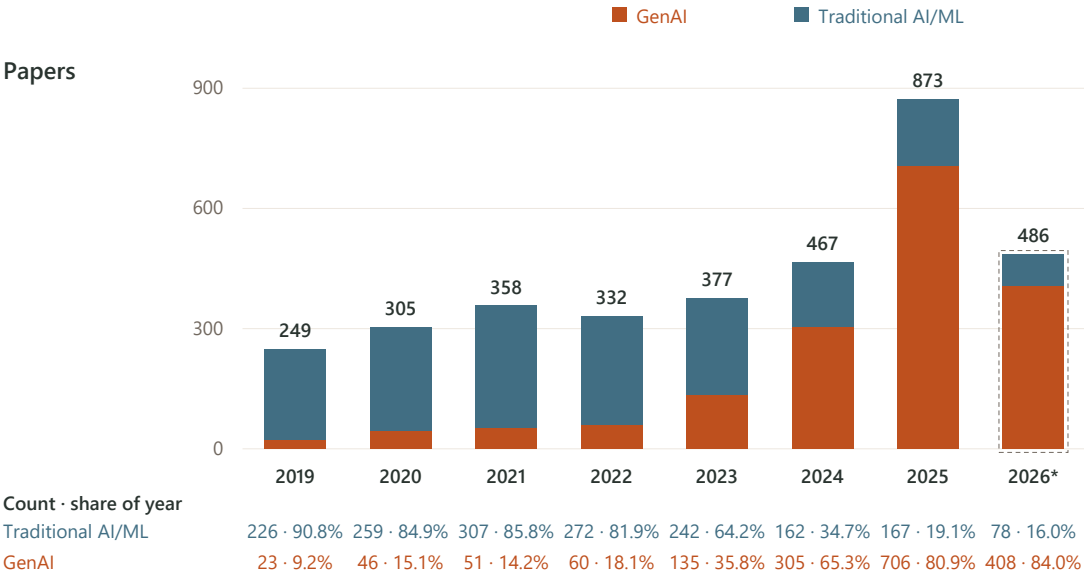
Appendix D gives the annual domain and activity distributions with the paper counts behind each share (Figure 55 and Figure 56). The paired workflow figure uses 2,390 papers with an activity code and assigned technology. Allocation, supplied support, reported outcomes, and study-description codes remain separate: faster work, for example, does not establish automation.

4.2 Delegating work while retaining judgment

Two choices shape what the person still does: how much of the material the system has already composed, and what the person can check it against. CorpusStudio and ABScribe both support writing but differ in the first. CorpusStudio supplies examples and patterns from prior writing, which authors interpret before composing their own prose [31].

AI technologies by year

3,447 classified papers · bar height shows volume; percentages show composition



GenAI includes hybrids and earlier generative models. * 2026 through 16 July; September update pending.

Fig. 6. AI technologies by year. Stacked bar heights encode paper counts, with annual totals above. Rows below give each segment’s count and within-year percentage. Classification follows the technology used; 2026 is partial (collected in July). The figure omits 639 papers without a resolved technology classification.

ABSubscribe generates alternative passages that authors keep, edit, and compare [106]. One gives authors material to write from; the other gives them passages to select and revise.

Figure 7 makes this division of work explicit. A tool can compose material while the person judges its fit, revises it, or checks its evidence. Learning and reflection are distinct from supervising the system: either may be described without the other. These distinctions locate the action to examine in a design; the outcome comparisons later in the paper establish what was measured.

Norman separates what a system accomplishes from the activity of the person using it [92]. In the reviewed interfaces, composing a passage, choosing a criterion, inspecting execution, and deciding whether to keep an alternative are separate operations, and designs divide them differently.

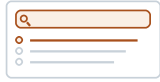
Different references support different checks. A proposed result can be checked against executed values, retrieved examples, the system’s interpretation of an action, or a generated forecast. Leap shows the values that generated code produces as it runs [37]; CorpusStudio retrieves examples from a writing corpus [31]; ONYX shows how it interpreted an action the user taught it [109]; and Rehearsal simulates how a conversation partner might respond [112]. Each grounds a different judgment. A simulated response is material for rehearsal, not a validated prediction of what a particular person will say. A corpus supplies precedents, but its incomplete coverage cannot show that an idea is new.

How is the work divided?

The tool supplies an operation; the person still has work to do.

What the tool does

Perception, search and memory



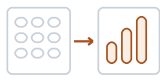
Recognising and extracting information
Searching and retrieving information
Storing and recalling information
Tracking activity and progress

Representation and translation



Representing and transforming information
Comparing and finding patterns

Computation and judgement



Calculating and modelling
Evaluating and classifying

Content generation



Generating new artifacts
Generating responses and revisions

Decision, guidance and action



Guiding the work
Executing actions
Selecting and recommending
Mediating communication

What the person retains

Judgement and decision



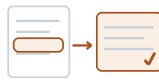
Interpreting and explaining results
Deciding and authorising
Specifying goals and inputs
Applying expertise and context

Doing and revising the work



Creating and carrying out tasks
Editing and correcting outputs
Controlling and refining the process
Training and improving the system

Checking the output



Checking and verifying outputs

Responsibility, social and self-directed work



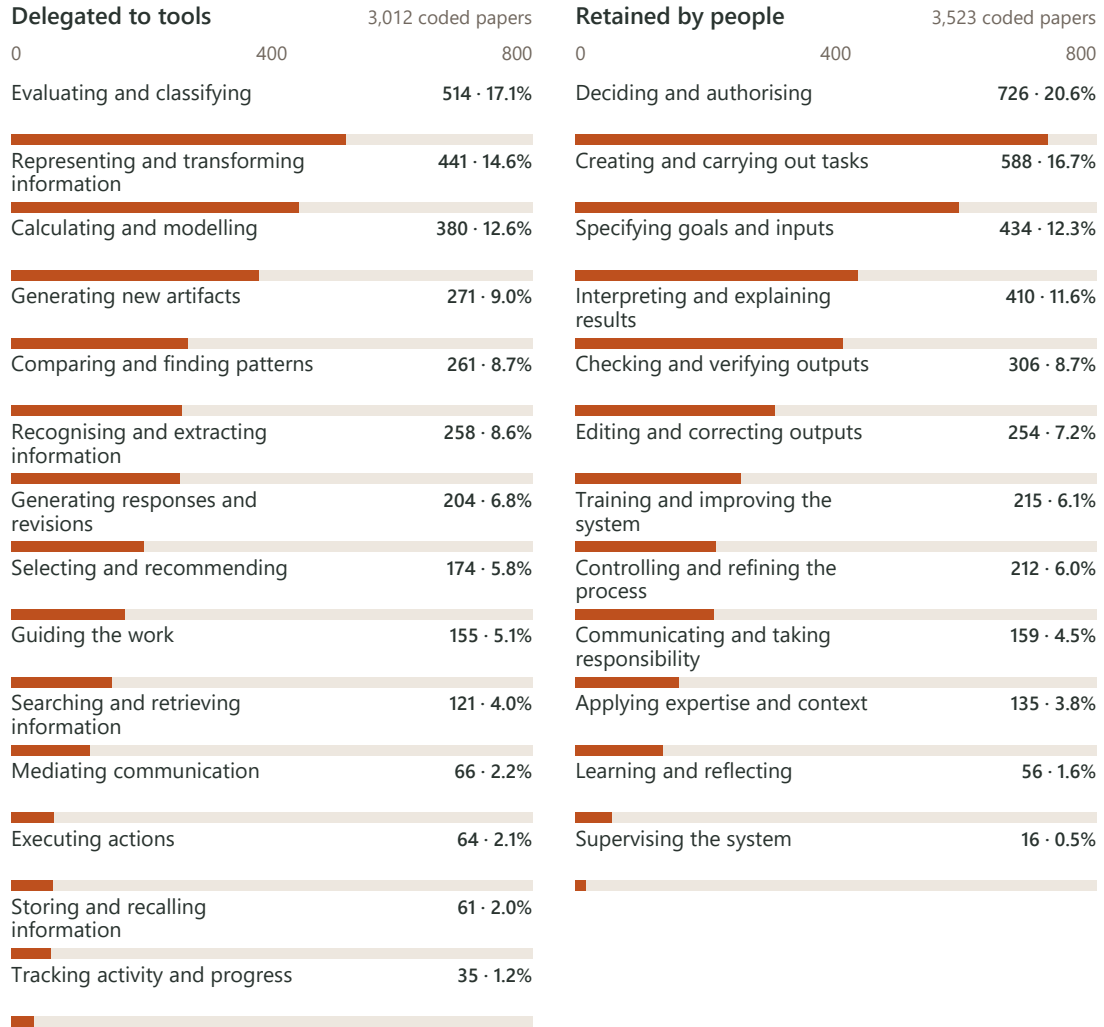
Communicating and taking responsibility
Learning and reflecting
Supervising the system

Fig. 7. How is the work divided? Five families describe operations supplied by tools; four describe work retained by people. The two columns are independent accounts of a design, not paired transfers. Labels distinguish fourteen tool operations and twelve human activities; learning and reflection are separate from supervising a system. Failed, withheld, absent, and unspecified delegation are recorded separately. Browse examples of tool and human work.

A reference helps only when people notice and use it. ONYX's full interface improved script accuracy, yet users missed or wrongly dismissed 44.8% of its follow-up questions [109]. Leap's live values did not significantly change validation success compared with a condition in which people could still run the code themselves [37]. A design

Work delegated to tools and retained by people

Bar length: number of papers · percentages use each field's coded papers



Counts use all papers coded for each field. Allocation conditions and unspecified work are listed with the data.

Fig. 8. Delegated and retained work, ranked by prevalence. Bar lengths encode paper counts on a common scale; labels also give percentages within each panel's coding denominator. A paper can record several roles. Each panel uses papers with usable codes for that role; technology classification is not required.

therefore has to name the error a person should catch, the information that reveals it, and the response that would show the check worked.

ExPeerience shows that generation need not turn learners into passive checkers: it uses AI to create problem contexts, tables, and explanations but leaves writing the SQL solution to the learner [147].

4.3 Coordination and opportunities to learn

Support can pass through another person whose judgment remains necessary. In entrepreneurship coaching, the novice describes a project and selects risks; the interface gives the mentor selected and omitted risks with source traces to prepare a meeting [53]. HTCT instead turns tutor discourse into analytics that a coach selects and discusses with the tutor [17]. One representation carries a person's expressed priorities; the other supplies an analysis of their activity. The immediate interface user, contributor, and intended beneficiary can be different people.

These arrangements require following what crosses roles. The entrepreneurship study observed eleven novices and one mentor in real meetings, with changes in plans and diagnoses. Coaches chose whether to use HTCT. The study followed their tutors' practice for seven weeks across early, delayed, and no-access groups. Both studies give evidence about coaching practice, not about what students learned from a coach's interface. Rehearsal serves a different purpose: its generated counterpart provides a setting to practice conflict responses [112]. A simulated response is practice material, not evidence of how a real recipient understands.

A shared representation does not guarantee a response. In eaSEL, a child's reflective artifact can be shown to a parent with a prompt about the parent's own experience, but the evaluation did not observe the whole parent-child exchange the paper proposes [113]. QMaps, a system without AI, makes the participation issue concrete. Students create questions, link prerequisites, and justify those links. Incorrect links sometimes revealed copied content that students did not understand, while students preferred adding their own questions to revising others' work [139]. Exposing reasoning could reveal a difficulty without ensuring that anyone addressed it.

The social arrangement can also change when software supplies feedback. BrailleBlocks makes a child's construction interpretable to a sighted parent who can offer hints [40]. In WriteAid (Stream 2), some recurring questions became less visible to a teacher when pupils sought software feedback, while the teacher gained time to circulate and new peer conversations occurred [91]. These cases turn on who has the information needed to help. A shared representation should be judged by the response it enables; visibility and collaboration are not learning outcomes in themselves.

Educator-facing support can also target different work. VizProg users diagnosed learners' code and selected feedback during a classroom replay, but that feedback was not delivered to students [143]. M-Powering Teachers supplied mentors with feedback about their own talk; randomized access increased their uptake of students' contributions and reduced their share of talk [32]. QLens lets designers inspect prospective feedback: one recommended path derived from successful traces was judged unsuitable as direct feedback for students [138].

These cases have different evaluation targets: diagnosis and feedback selection, ongoing teaching practice, and the suitability of proposed feedback. M-Powering's result covers the ongoing feedback package as a whole; the study did not record time on the feedback page and did not directly assess student learning. Educators can themselves be the intended learners. Evaluate the response the support is meant to change, and assess students separately when the claim is about student learning. QLens shows that a path students completed successfully can still be unsuitable as instruction.

The next section turns to the interface mechanisms that organize this work and to what evaluations of those designs show.

5 FINDINGS: MECHANISM TAXONOMY

User control is more common in GenAI papers, but the label conceals different design choices. People may revise a task specification, the system's interpretation, evaluation criteria, or the resulting artifact. Plume, for example, lets a

writer preserve selected text while regenerating the surrounding explanation [72]. The object and persistence of an edit determine which decisions remain open to the person.

The 5 mechanism families and their 18 categories organize these choices. We describe the families and how their prevalence differs by technology, then compare concrete designs and what their evaluations establish.

5.1 Mechanism families

A mechanism pairs an operation with its form: a constraint can express a goal, an authored rule, or an adjustable threshold. Figure 9 organizes these forms, and Figure 10 compares their prevalence across technologies; a paper can use several. Table 17 counts the papers in each category and how many of them concern or measure learning, and the category maps in Appendix C.1 lead to particular mechanisms and their source papers.

The families describe different ways to support a decision. The *organization and comparison* family supplies alternatives, relationships, and histories. Graphologue turns text into entities and links [56]; Crystalline gathers options, criteria, and evidence from the pages a developer reads into a comparison table whose cells lead back to their sources [73]. The *guidance and feedback* family supplies recommendations, questions, and task structure. OATutor selects practice and reveals hints [98], while WithYou shortens the pauses it inserts into speech as a learner’s shadowing improves [145]; revealing more help and reducing help are different interventions.

The *explanation and visibility* family exposes sources, rationale, or state. HaLLMark traces copied LLM text to its prompts but relies on writers to mark the passages they consider AI-influenced [49]; automatic provenance captures direct reuse more readily than inspiration. The *user control and expression* family lets people act on the material: Plume can lock text while it regenerates the explanation around it [72]. The *collaboration and perspective* family makes interpretations available to other people: Wikibench supports discussion that revises label definitions and inclusion rules [67].

To use the taxonomy, start from a decision the person must make. Find the mechanism that supplies information or an action for it, then ask what response it requires and which outcome would show success. A provenance link, for example, gives a basis for checking a claim, but the person still judges whether the source supports the wording. The link shows traceability; accurate checking, or learning to judge sources, needs its own observation.

5.2 Mechanisms across workflows and years

GenAI control steers generation rather than correcting output. GenAI papers more often describe user control and expression (+13.7 points after adjusting for year and venue; 95% CI 8.6 to 18.7; Appendix D.2). In the 2,532 papers with facet codes, two control facets show a reliable gap, and both put generated candidates before the person: GenAI systems more often generate alternatives to explore (+4.8; 95% CI 3.0 to 6.6) and complete or continue the person’s own content (+3.0; 1.4 to 4.6; Figure 9). The work people keep leans the same way, toward steering the loop (+3.4; 1.2 to 5.6) and away from improving the system (-5.9; -7.7 to -4.0; Figure 7). Acting on the output and stating criteria are not more common: editing machine output (+1.8; -1.4 to 5.1), direct manipulation of a semantic object (+0.9), and specifying criteria and rubrics (-2.8; -4.9 to -0.8). A choice among generated alternatives still needs criteria and contrasts, which few of these interfaces ask people to state (4.4% of GenAI papers). Section 4.2 and Table 5.3 compare designs that supply them.

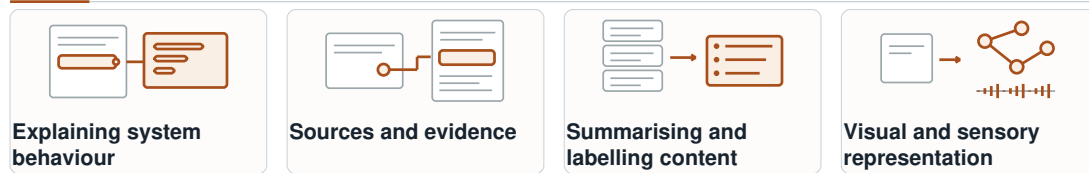
GenAI systems rank items less often. Organization and comparison is less common in GenAI papers (Figure 10). Its largest facet gap is ranked or prioritized surfacing, which orders items so the important ones come first (3.2% of

What does the interface make possible?

Five design families · eighteen categories · 110 retained distinctions

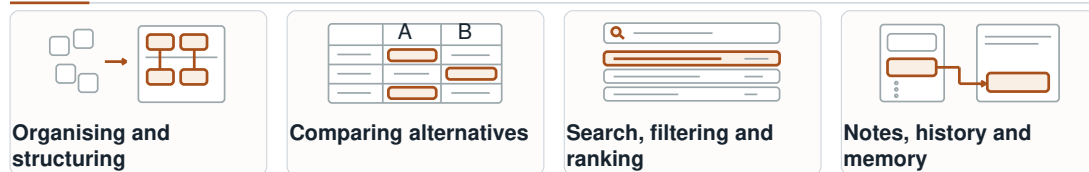
Explanation and visibility

What can a person inspect and understand?



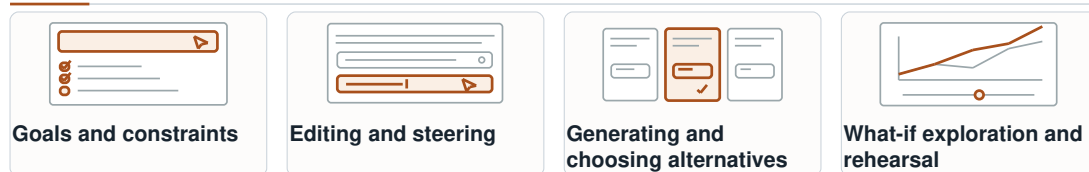
Organisation and comparison

How can a person organise and compare material?



User control and expression

What can a person express, change or try?



Guidance and feedback

How is the next action or reflection supported?



Collaboration and perspective

How can people share work and perspectives?

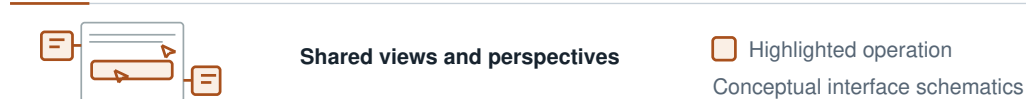


Fig. 9. Five design families organise eighteen categories of semantic scaffolding. Each small interface schematic illustrates a category, with its operation highlighted in orange. A mechanism can combine operations across these branches; layout does not encode prevalence or effectiveness. Complete category maps and source-paper illustrations appear in Appendix C.1. Explore the design atlas; search all 110 distinctions.

GenAI and 11.2% of traditional AI/ML papers; -6.5; 95% CI -8.3 to -4.7), and no other facet in the family differs by more than 2.9 points. Appendix D gives the annual family profiles (Figure 57).

GenAI papers more often offer user control

GenAI versus traditional AI/ML, accounting for publication year and venue

2,444 papers with family coding and classified technology · 2019–2025

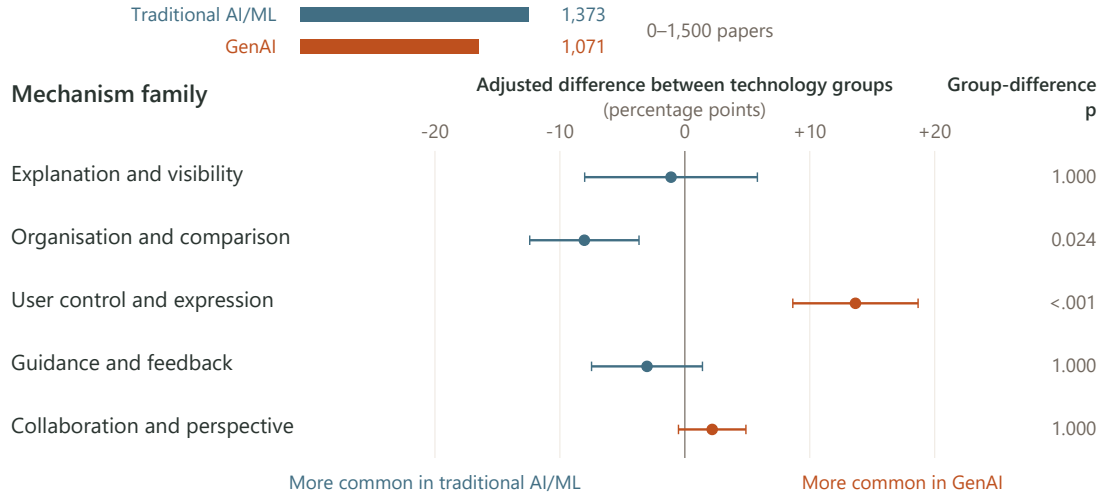


Fig. 10. Between-technology differences in mechanism families among 2,444 papers with usable family coding and classified technology, 2019–2025. Dots show GenAI minus traditional AI/ML prevalence, adjusted for year and venue; lines show pointwise 95% confidence intervals. The p -values test these group differences, with Holm correction across 51 contrasts, rather than change over time. User control retains support in the 2023–2025 and partial-2026 sensitivity windows. Of 3,385 family-coded papers, 540 lack classified technology and 401 classified papers fall outside 2019–2025.

Collaboration and perspective became more common over time. Holding the technology mix fixed at the original analysis’s reference composition (43.75% GenAI), its average annual change was +1.10 percentage points (95% CI [0.52, 1.68], Holm $p = 0.005$). The 2023–2025 sensitivity also showed an increase (+2.67 points per year, CI [1.20, 4.14], Holm $p = 0.007$). Guidance and feedback increased in the primary window (+1.58 points per year, CI [0.70, 2.47], Holm $p = 0.008$), but its recent interval included zero ([-2.73, 6.47]). None of the five tests of change in the GenAI–traditional AI/ML difference survived Holm correction in either window. These temporal estimates (Figure 11) answer a different question from the overall technology-group differences in Figure 10; Appendix D.3 reports both temporal estimands for every family.

From task intent to action.

Mechanisms can also be grouped by the activity they support: *specify intent*, *represent possibilities*, *evaluate alternatives*, and *act and revise* (Figure 12). Appendix C.1 maps all 110 facet codes to these activities; a code can serve several. Annotations, histories, shared views, and handoffs connect the activities across time and people. The functions describe dependencies, not a fixed order: comparing needs alternatives and criteria, and a comparison may prompt a change to either.

Different mechanisms can serve the same function. Intent elicitation, constraints, and shared criteria frame a task. Structured alternatives and source links make possibilities inspectable; comparisons, explanations, and reflection

Collaboration and perspective increase across years

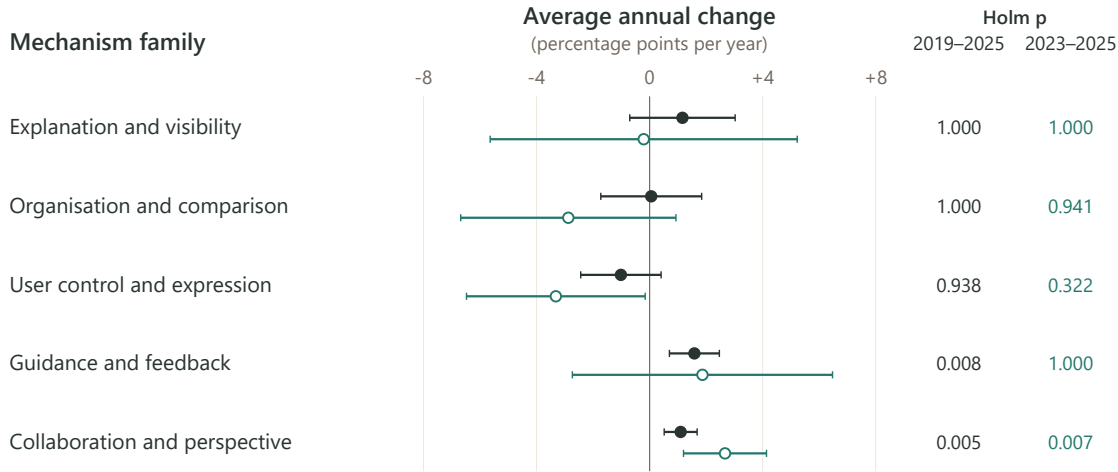
Annual change in coded prevalence, with a fixed mix of AI technologies

Adjusted for venue · 39 venues · Partial 2026 excluded

● 2019–2025 · 2,444 papers

○ 2023–2025 · 1,390 papers

Mechanism family



Lines: 95% confidence intervals · Holm correction across ten temporal tests per window

Fig. 11. Average annual change in coded mechanism-family prevalence: 2019–2025 (2,444 papers) and the 2023–2025 sensitivity (1,390). Dots show percentage points per year; lines show pointwise 95% venue-clustered confidence intervals. The technology mix is held fixed at 43.75% GenAI in both windows. Positive values indicate increasing prevalence. These are temporal slopes, not the overall technology-group differences in Figure 10. Appendix D.3 gives the methods, changing-gap estimates and all ten tests used for Holm correction within each window.

support judgment; selection, editing, and changed constraints enable action. One mechanism can serve two functions: a comparison exposes options and supports choice, and a constraint can frame a task or redirect it. Guidance can enter at any point, and histories keep context across transitions.

Luminate’s dimensions organize alternatives for exploration and evaluation, with return paths to earlier prompt-specific spaces [120]. AI Chains instead feeds one step’s output into the next step’s input, so people can inspect, edit, and re-run intermediate steps [137]. The order in which a system computes need not dictate the order in which people inspect or repair its work.

An ordered workflow can still require return paths. CollabCoder opens shared comparison after independent coding, then supports discussion, decisions, and grouping while letting people restore their original codes [43]. Crystalline instead builds a route back into the material: a reminder marks related evidence on pages the developer has visited but not yet attended to, and three of twelve participants reported using it to guide their research [73]. CollabCoder sets an order for the work; Crystalline returns people to an earlier step when their comparison may have a gap. Transitions therefore deserve the same design care as individual mechanisms: keep the criterion that shaped an alternative, the evidence behind a draft, and a route back to an earlier decision, and test whether people understand a transition, change it, and recover its context.

From task intent to action

Four overlapping functions · conceptual interface sketches

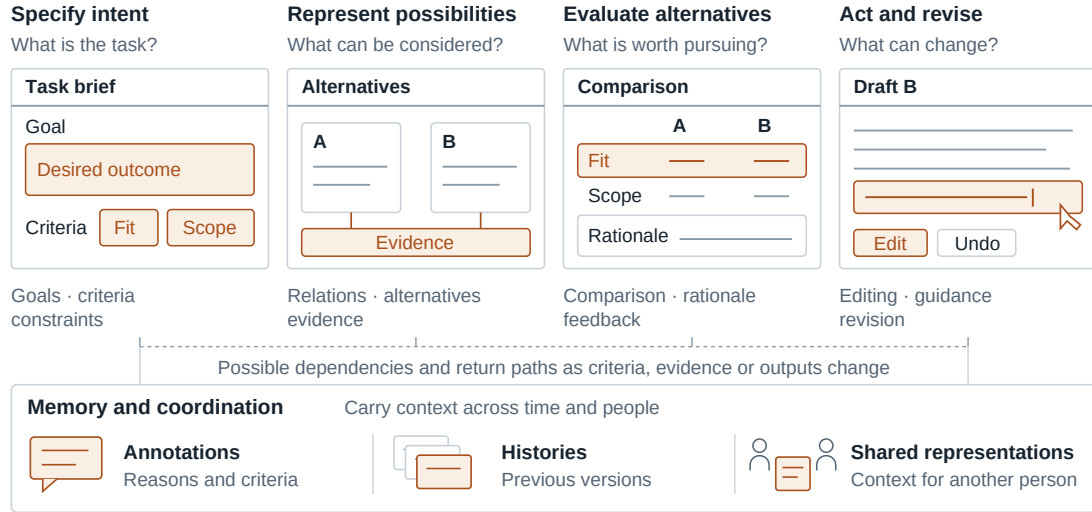


Fig. 12. **From task intent to action.** Four overlapping functions offer an activity-oriented view of the mechanism taxonomy. Memory and coordination connect them across time and people. The sketches show criteria, alternatives, comparison, and revision; the shared strip keeps decisions and their context available across these activities. The accompanying source examples illustrate these relationships [43, 73, 120, 137].

Table 1. **Revision and downstream effects.** What three interfaces preserve, update, or treat as confirmed.

Design choice	Comparison and finding	Conditions and contrary evidence	Design implication
What survives regeneration?	In task decomposition, editing an assumption, plan or code component can regenerate later work while omitting manual edits [59].	The earlier version remains recoverable in a branch; preservation within the revised result is a different property.	Make the fate of earlier edits inspectable when an upstream decision changes.
What does locking text preserve?	Plume retains locked or manually edited dashboard text while using it to inform further generation [72].	An unchanged passage can still influence other text; locking does not isolate its downstream effects.	Distinguish preserving wording from restricting its influence on new content.
What counts as confirmation?	LineageD permits corrections before predicting the next level; advancing implicitly confirms remaining pairings [48].	Explicit and implicit confirmations share the green marking, so the display does not distinguish how a pairing was accepted.	Preserve whether an item was checked explicitly when that distinction matters to later work.

5.3 Revision, preservation, and control

An editable step does not guarantee that people can correct it and keep the later work they still want. Across the cases, the design question is what a human action authorizes the system to reuse or treat as accepted. Table 1 distinguishes saving an old version, preserving valid edits during regeneration, and confirming a result.

Keeping an old version does not carry its edits forward. Kazemitabaar et al.'s data-analysis systems saved earlier versions in branches, but regeneration ignored later manual edits [59]. Participants described avoiding revisions and expecting to redo or reread work; one wanted invalidated parts highlighted instead of regenerated, and another valued the interconnected updates. Recovering an old version differs from preserving the still-valid parts of that work in a new one.

LineageD's pilot showed the cost of predicting a whole lineage at once: correcting one pairing invalidated the rest of the path. The redesign predicts one level at a time, but moving to the next level implicitly confirms pairings the person has not individually confirmed [48]. Plume keeps locked or manually edited text and uses it as context for later generation [72], so a lock preserves wording without limiting its influence on other text or vouching for its correctness. Relating a local step to the whole process is hard in itself: in AI Chains, four of twenty participants asked how a local change would affect the final result [137].

Interfaces promise to keep different things. Automatic propagation can itself help. CoPrompt participants valued automatic coordination, and eight used history and difference views to resolve conflicts or restore versions [36]. Pail's code differences, summaries, and design panel supported inspection, but participants reported difficulty following cascades across them, less so when they paused to read the changes [141]. Selecting part of an artifact does not by itself fix what stays unchanged: FusAIn leaves unselected image regions untouched, whereas Textoshop can change text beyond the selected passage [84, 101]. Blace keeps earlier states for rollback, and DesignManager records the process behind an asset [76, 140]; both support recovery and inspection of past work, while consistency with a revised goal still requires checking the downstream artifacts. A design should name the property it protects and the change that could violate it, and an evaluation should test whether people anticipate that change, detect a violation, and recover the work they want to keep.

Edits can change the specification, the model, or the output. A correction can change what the person asked for, how the system understood it, or what it produced. FusAIn lets designers compose and revise a visual specification [101], and Falx synthesizes candidate charts from examples, which participants described checking against input values they knew [127]. CoAutoML lets novices inspect and edit the task, inputs, target, and evaluation assumptions it proposes before they confirm a machine-learning task [119]. Plume's locks retain part of what the system produced [72]. Outside generative chat interfaces, corrections can also update the model: SmartEye uses a corrected crop as a positive training example for later recommendations [80], and DeepSI fine-tunes BERT from spatial feedback and recomputes its document map, though its authors note limited visibility into the model's internal state [13]. The design question is what a correction changes and which later operation uses it, not whether every control comes before generation.

Generating alternatives and their criteria.

When a system generates the criteria as well as the alternatives, people must judge both. In Luminate, selecting an axis rearranges the responses, whereas adding a dimension revises the responses themselves [120], so the writer in our opening example judges the dimensions as well as the ideas. VIVID generates expected answers and rubrics along with candidate dialogues, and instructors inspect and edit them [28]. Design-rationale research already links questions, options, and criteria [82], and conventional tools sort alternatives into categories that designers choose. Generated criteria raise a narrower question: can people revise a criterion while holding the alternatives fixed, or does the revision also replace what they are judging?

Generated criteria can improve comparison. Selenite gives readers generated overviews of criteria and labels the relevant passages; in a twelve-person paired comparison with unaided reading, participants produced more valid criteria with the overview (12.93 versus 6.79), with higher precision and recall [74]. A criterion can also be easy to act on and

Table 2. Selected AI-involved comparisons of explanation and practice designs.

Comparison	Assessed result	Conditions
Predicted-misconception explanations [21]	Better immediate exercise-prescription decisions after AI removal than unilateral explanations or no AI	Advantage over random foils unresolved; no delayed test
Required reflection after AI feedback [7]	2.0 versus 3.4 problems completed in eight minutes; transfer and code-evaluation advantages unestablished	Fixed time budget; fewer attempts do not establish learning harm
QuizBot versus flashcards [108]	Higher recognition (87.2% versus 65.8%), with an advantage at two months	Equal repetitions; 46 versus 14.3 minutes of practice; dialogue bundled with other support
Adaptive versus generic video recommendations [68]	Small algebra-ability advantage during normal schooling	Low optional uptake; policy effect; closure-period advantage unresolved

still miss the quality it stands for. RECAST lets people revise text against a toxicity model; revisions lowered the model's toxicity scores without reliably improving human ratings of toxicity [135]. When both the model's response and human judgment matter, the evaluation has to measure both.

5.4 Comparing explanation and practice designs

Explanation content, required reflection, dialogue, and optional recommendations change different parts of practice. Table 2 links four AI-involved comparisons to their assessed outcomes.

The practice cost of reflection does not explain every null learning result. In Xiaomai's advanced-mathematics study, requiring error reflection produced neither a detected learning advantage nor a detected difference in attempts [35]. Together with the fixed-time comparison above, this makes the content and uptake of reflection worth examining alongside the practice it displaces.

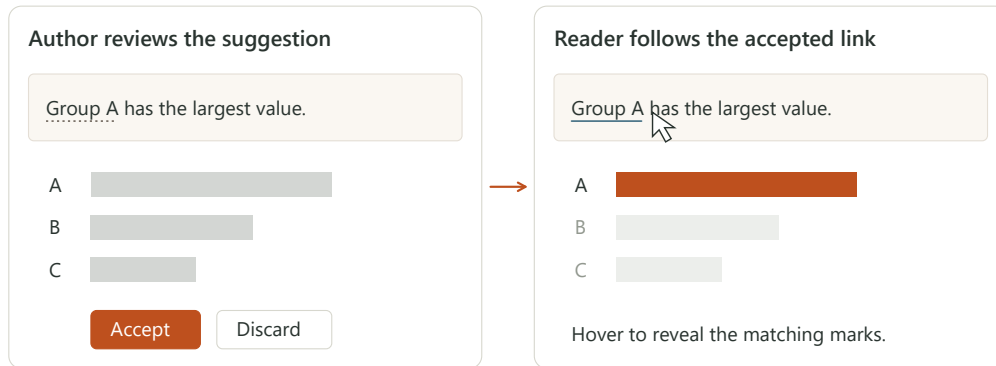
Explanation design also concerns sequence. Incremental XAI supplies factors progressively during model-prediction practice, then tests prediction on unseen cases without the explanation display [16]. This separates the presentation of an explanation from what a person can do once it is removed.

Prior learning research separates creating an explanation from using it. In a randomized study of 120 learners, everyone wrote subgoal labels; those who then practiced with their own labels scored 21.6 of 25 on an immediate test, against 17.1 with expert labels and 18.8 with none. Instructional materials were removed for the test, while the programming environment remained available [83]. This comparison sharpens ExPeerience's question (Section 4.2): which part of the task should the person learn to perform? Learning research has long stressed what support asks learners to do [27, 105]; the interface choice is what to generate and what to leave for the learner.

Writing feedback can describe a revision or judge its quality. ArgRewrite classifies the purposes of a writer's earlier revisions without judging their quality, and asks writers to agree or disagree with those labels before and after revising again [1]. ArgueTutor pairs claim and premise labels with recommendations based on their balance [126], and LegalWriter adds advice on argument components while leaving the semantic fit between claim and conclusion to the writer [131]; both improved the quality of text written with them over their comparison interfaces. A description of a revision and advice about quality make different assertions, and one interface can offer both. A learning claim needs a test of the judgment or revision skill the feedback is meant to build.

Kori makes text–chart links inspectable

AI proposes a connection. The author decides whether to keep it.



The author checks the correspondence; the reader uses it to navigate the data.

Fig. 13. **From an AI suggestion to an interactive reference in Kori.** The author reviews a proposed text–chart connection. After acceptance, hovering over the linked phrase highlights the matching marks by fading the others [69]. Schematic redrawn from the paper’s interface description, with illustrative data.

5.5 How prior knowledge informs design

Prior knowledge enters a design through the material it supplies, the activity it asks of a person, and the measures used to assess it. The same learning principle can motivate different choices at each point. We traced how 2,058 papers, including the non-AI comparison papers, used the prior work they cite (51,815 source records; a work cited by two papers counts twice), and we compare implementations through the passages that document each use.

Tracing sources into designs. The citation map (Figure 14) ranks the sources used by the most papers and follows the highest-ranked source with a known year into the later designs that use it; each source–paper pair counts once. The companion website lets readers search sources, follow connections, and open the passage behind each use.

Making correspondence inspectable. Kori turns the concern about split attention into a visible link between prose and chart marks [69]. The system proposes links, and authors preview, accept, discard, or draw them by hand; once a link is accepted, hovering over its phrase highlights the chart marks it refers to (Figure 13). The author checks each correspondence, and the reader uses it to interpret the data.

Operationalizing a theory is not testing learning. QuizBot cites ICAP [27] as its learning rationale, whereas the Khanmigo methodology turns ICAP into an LLM-judged indicator of cognitive engagement [108, 123]. Deeper use of the theory does not bring stronger evidence of learning: QuizBot tests recognition two months later, while Khanmigo tracks correctness on the next item a student attempts alone. A theory can say what to observe without showing that changing that observation produces learning.

Prior knowledge can shape the evaluation as well as the design. Sibyl adopts Doshi-Velez and Kim’s application-grounded evaluation [33, 150], testing explanations with screeners in child-welfare work. Screeners often rated case-specific explanations helpful and found other forms less useful during those cases, and their feedback exposed problems of terminology and presentation.

Table 3. Measurements of process, artifact, and learning. Timing and available assistance are specified separately for each measure.

Outcome	Assessed in the study	Reported by participants
Process	Task time, action sequences, revisions, errors	Perceived effort, workload, usability, agency
Artifact	Correctness, completeness, external quality ratings	Perceived quality, usefulness, originality
Learning	Scored understanding, skill, retention, transfer	Perceived learning, confidence in ability, self-efficacy

A theory's evidence travels with its comparison. Codigen cites cognitive mapping to motivate active learning and supplies generated programming material on an editable whiteboard [34]. The cited medical study compares maps with notes organized around the same five problem-solving elements, with the assigned support available during the diagnostic case [129]. It does not compare self-authored with generated material. Codigen's novice pre/post quiz gains likewise do not isolate the whiteboard's contribution. The connection suggests a design, while testing its learning rationale would require comparing how learners construct and use the material.

Supplied material can support learning through its use. Learners need not build the material they learn with. VocabEncounter supplies phrases in context and reports better vocabulary two days later than its word-only condition [5]. Rehearsal supplies conversational alternatives, practice, and feedback; its participants used more cooperative strategies in a later unaided conversation than participants given the same static materials without interactive practice [112]. In a study of AI assistance for incidental learning, explanations without a recommended answer improved immediate concept learning over minimal feedback [41]. Supplied material also leaves room for different uses: CoNotate participants reported both running its suggested queries and borrowing their vocabulary to write new ones [96]. The design question is which operation the supplied material helps people perform, and whether the study tests that ability.

5.6 What capability the evaluation establishes

We distinguish process, artifact, and learning outcomes, assessed in the study or reported by participants (Table 3). Each measure specifies its construct, scoring criteria, rater, comparison, timing, and available assistance. An external quality rating remains a judgment; a numeric self-rating remains self-report. Scoring a finished artifact later does not make it a learning assessment.

Product and capability measures answer different questions. In BELVEDERE, the maps showed more explicit links between evidence and hypotheses, but no difference in final prose conclusions was detected between conditions, and students could still consult their maps while writing [121]. Both outcomes describe products made with support; neither tests what students can do without the maps.

A benefit depends on the comparison and the situation. Cognitive forcing asks people to engage more deliberately with AI advice. Among 199 participants, fully correct decisions averaged 35% with simple explanations and 33% with forcing, a nonsignificant difference; on items with incorrect AI advice, forcing raised correctness from 3% to 9%, while the no-AI condition reached 18% [20]. Adaptive help raises the same issue: predicting whether a person will answer correctly differs from showing that advice based on that prediction helps them [78, 118]. Appendix A.7 gives the learning and duration profiles of the reviewed studies.

Independence concerns particular people, tasks, and resources. Useful tools need not be withdrawn. Pea distinguishes developing a person's capability from removing resources that stay useful [100]. In a study of vocational prompting, six of eight participants reached full task correctness with a prompting device, and asking a person for help lowered the score while consulting the device did not, so the score measured independence from a helper, not

Table 4. Assessed outcomes in four AI-involved comparisons.

Comparison	Assessed result	Conditions
CodeVoyager versus same-model chat [61]	Higher immediate closed-book comprehension (8.06 versus 5.44/20); no detected assisted-completion or accuracy advantage	16 developers; interface package; no delayed test
FeedbackWriter versus usual TA feedback [75]	Better revisions ($d = 0.5$); knowledge-test advantage unestablished	Randomized 354-student trial; optional posttest; test-time assistance unclear
Rehearsal added to static training [112]	Better immediate conflict-response behavior; recall and recognition advantages unestablished	Behavior and recall assess different capabilities
CodeTailor versus generated solutions [51]	Better immediate unsupported application of supported elements; total posttest advantage unestablished	Selected 14 of 18 participants; tested elements depend on practice difficulty

unaided performance [25]. A design should name which activity is meant to become independent, which resources should remain, and who prepares or maintains them.

5.7 Performance and subsequent capability

An interface can improve one assessed outcome while leaving another unresolved. Table 4 distinguishes comprehension, revision quality, enacted behavior, and application of supported skills.

These comparisons apply Norman’s distinction between performance with an artifact and the capability attributed to the person [92]. DBox’s decomposition and feedback also improved performance on an immediate similar problem solved without assistance, although effort rose [79].

Assessment conditions specify the capability demonstrated. Among the 396 AI-involved sources in the outcome appraisal, 223 reported a separate capability assessment, 31 assessed capability in a later session, 71 used new material or a changed task, and 67 reduced or removed a named training aid. These overlapping features describe the selected appraisal set (Appendix A.8).

An assessment’s task, timing, and available support each change what it shows: an immediate new problem can test application without testing retention, and a later test can still allow help. WriteAid tested pupils’ writing two to three weeks later without AI or teacher support, but on the same topic and without an untreated comparison or a pretreatment skill measure [91].

The promised use determines the informative test. Immediate performance, later retention, and application to new material need not move together. Two classroom experiments with structured LLM reflection found no significant advantage on exams two weeks later [66]. Preference is not capability either: across three AI Margin Notes studies of reader interfaces, note selection, and human and AI note-taking, comprehension two hours after reading did not differ significantly, even where preferences or task demands did [57].

An informative evaluation therefore starts from the design’s intended use. An explanation meant to prepare a new decision needs a decision task; feedback meant to change the next attempt needs that attempt observed; a practice partner meant to build interpersonal skill needs evidence of enacted behavior, while recall answers a different question. Effectiveness reviews already show that scaffolding can improve cognitive outcomes [11]; the comparisons here show which alternative and which outcome a designer can use as evidence.

6 DISCUSSION

The review shows how interfaces redistribute work: they supply material, organize alternatives, and give people ways to judge and revise them. The consequences depend on what people do through those structures. Three arguments follow for designing semantic scaffolding and evaluating its value.

6.1 Control must reach the consequences of a choice

User control is more common in GenAI papers, yet an editable interface can still leave people unable to carry out the revision they intend. A local edit may change a task assumption, train the model, or replace an output. Plume preserves selected wording during regeneration; the data-analysis systems save earlier versions but can omit manual edits from regenerated work [59, 72]. These differences change the cost of reconsidering a decision. We argue that control should be judged across a revision: can the person identify what changes, preserve what still holds, and repair the affected work?

The basis for judging that work must also be open to revision. Design rationale links questions, options, and criteria [82]. Luminate and VIVID make criteria part of what a system generates during use [28, 120]. This creates two connected decisions: whether a criterion is appropriate and whether an alternative meets it. RECAST makes the distinction consequential: lower model toxicity scores did not reliably produce better human ratings [135]. An interface that helps people satisfy its criteria should also help them question those criteria, inspect their basis, and use a different standard.

The same responsibility extends across people. Coaching interfaces carry a novice's priorities or an analysis of a tutor's activity to someone who must interpret it [17, 53]. A useful handoff keeps contributions, assumptions, and unresolved questions available for that person's judgment. Across both revisions and handoffs, people need to recover the reasons for a decision and see when a change calls it into question.

6.2 Evaluate process, artifact, and learning together

An evaluation should connect three questions: how did people work, what did they produce, and what did they become able to do? Process measures capture practice, checking, revision, time, and effort. Artifact measures assess the resulting draft, program, or decision. Learning measures assess a capability on a subsequent attempt under specified conditions. Norman's distinction between a person-artifact system and the person's own activity explains why success on one question cannot answer the others [92].

The reviewed comparisons show why all three matter. CodeVoyager improved immediate closed-book comprehension without a detected advantage in assisted completion or accuracy; FeedbackWriter improved revisions without an established advantage on its optional knowledge test [61, 75]. Rehearsal improved enacted conflict-response behavior while recall and recognition advantages remained unestablished [112]. These results identify different benefits. Choosing the learning measure requires naming the capability the design is meant to develop.

Process evidence helps explain what an intervention makes possible and what it costs. Required reflection reduced completed practice in a fixed interval; QuizBot's equal-repetition comparison allowed substantially different practice times [7, 108]. Optional support raises a different issue: the algebra-video trial tested a recommendation policy, with low uptake, so its result cannot be attributed directly to watching a video [68]. We therefore advocate evaluating the practice a design creates or displaces alongside product quality and later capability. This makes trade-offs visible without assuming that more interaction means more learning.

6.3 Support can remain while capability develops

Learning need not culminate in working without tools. Pea distinguishes scaffolding intended to fade from continuing distributed intelligence [100]; Krakauer asks what capability remains when an artifact’s immediate advantage is removed [65]. Both direct attention to what the person gains. In vocational prompting, device-supported task correctness counted differently from asking a person for help [25]. The intended independence concerned human assistance, while the device remained useful.

This permits a more deliberate division between useful automation and meaningful practice. Reiser distinguishes structuring a task from making its problems available for reasoning [105]. ExPeerience generates problem contexts and explanations while leaving SQL construction to the learner [147]. Rehearsal supplies a counterpart and feedback while the person practices responding [112]. Generating material can create an occasion for human activity as well as replace one. Designers should choose which capability deserves practice, then arrange the surrounding support to make that practice possible. The test is whether people become better at the chosen activity, with the resources their future work will actually provide.

7 RESEARCH OPPORTUNITIES

These findings suggest four directions: build support around a consequential difficulty, connect process and learning in evaluation, study how people form judgments in use, and create opportunities to learn within everyday tools.

7.1 Design for decisions that remain revisable

A promising next step is support for reconsidering work after its basis changes. AI Chains exposes intermediate dependencies, Plume preserves selected text, and CoAutoML exposes task assumptions [72, 119, 137]. Together, these cases suggest making the connection between a decision, its reasons, and its downstream effects inspectable. Green and Petre’s concerns with viscosity, hidden dependencies, and progressive evaluation give this design problem a theoretical basis [45].

Test it by changing a requirement after people have edited a result. Include one prior choice that remains valid and one that must change. Compare a history-only interface with one that identifies affected work and lets people selectively reapply valid edits. A specified correction tests the effort of editing; a changed requirement also tests whether people can discover what needs repair. Measure preservation, missed consequences, rechecking effort, and final quality. For creative tasks, first record which intentions the person wants to preserve. This would establish when revision support helps people reconsider a decision without losing work they still value.

7.2 Connect practice to artifact quality and later learning

Future evaluations should follow a design’s effect through all three outcomes in Section 6.2. Match the task, concepts, and examples while varying whether the interface supplies a solution, a partial representation, or feedback on the person’s construction. Record the work people actually perform and assess the resulting artifact. Then test the intended skill on new material immediately and after a delay, with the same declared resources available across conditions.

Practice budgets deserve explicit comparison. The reflection and QuizBot results motivate testing both equal time and equal repetitions [7, 108]. Estimate whether the design’s effect differs between those budgets. For optional support, distinguish the effect of offering it from the choices people make to use it [68]. Combine interaction traces with observations or short explanation probes to understand those choices. This would reveal whether a design’s immediate

benefit comes with more productive practice, less practice, or a different activity altogether, while later assessments establish what people learned.

7.3 Study how people form and share judgments in use

Interactive systems offer a way to study how reasoning develops around a representation. Luminate makes proposed criteria available for revision; CollabCoder brings individual interpretations into shared discussion [43, 120]. Distributed cognition provides a basis for following the work across people and representations [47]. Observe when people adopt a criterion, contest it, or carry it into a later decision, and examine how deadlines and the demands of producing an artifact shape those choices. Interface traces need observation and participants' explanations to reveal why a judgment changed.

Handoffs make this question especially concrete. Compare a coaching summary alone with the same summary carrying the learner's priorities, evidence, and open questions, allowing equal preparation time [17, 53]. Follow what the coach infers, what the learner corrects, and which issues shape the next meeting. Analyze the coach-learner group together; assess learners directly when the claim concerns their learning. The opportunity is to understand which representations help people maintain and repair a shared interpretation as work moves between them.

7.4 Create learning opportunities within everyday tools

Tools for getting work done could also offer people opportunities to develop capabilities they want to retain. An explanation before a decision, a comparison between alternatives, or feedback on a proposed revision can make a moment of ordinary work available for practice. Incidental-learning research found an immediate concept-learning benefit from explanations without a recommended answer, compared with minimal feedback [41]; ExPeerience illustrates how generated material can leave the target activity with the learner [147]. These cases motivate embedding opportunities to reason within work, instead of treating learning only as a separate training phase.

Develop this approach around a capability chosen with the user: recognizing a misleading comparison, judging a source, or repairing a program. Offer a brief prediction, explanation, or correction at a relevant decision, with control over when to participate. Compare the ordinary tool with this added opportunity over repeated use. Assess interruptions, work quality, uptake, and later performance on new cases; remove the learning prompt while retaining resources that normally belong to the task. The central question is whether meaningful practice can fit the user's work well enough to support both present goals and growing capability.

8 LIMITATIONS

Coverage. We searched 40 venues over 8 years, which favors published, English-language research [6, 71]. Technology comparisons leave out 639 papers whose AI technology could not be determined, and each analysis uses the papers coded for its fields (Table 9). The theories and empirical studies we compare against are a selection, not a census of every relevant tradition.

Reading and coding. Language models read each paper against a fixed codebook, one model per paper, and grouped the coded descriptions into categories. People have not yet checked the coding or the cross-case interpretations. Models can miss evidence shown only in figures and can share biases, and other analysts could draw the boundaries between closely related mechanisms differently.

Effects. Studies differ in tasks, participants, comparators, outcomes, and duration, and many compare whole systems. Most coded changes in how people work rest on the authors' own attribution rather than a measured contrast (75.8% of coded process shifts). The GenAI and traditional AI/ML comparisons are associations among published designs, not effects of the technology. The design space cannot rank mechanisms by effectiveness or estimate an overall learning benefit.

9 CONCLUSION

AI tools increasingly supply the structures people think with: criteria, examples, plans, and records of earlier choices. Our design space maps these semantic scaffolds across domains and connects each to the work it organizes and to what studies found. Generative-AI papers more often concern producing artifacts and include user-control mechanisms. What people can change matters, whether a specification, the AI's interpretation, the criteria, or the result, and support can reach a learner through peers, teachers, and coaches. Supplied material can support learning through the way people use it, and added interaction does not by itself make people practice the intended skill. The design space gives researchers concrete alternatives for building help that develops capability as people work, including the capability to use support that stays.

REFERENCES

- [1] Tazin Afrin, Omid Kashefi, Christopher Olshefski, Diane Litman, Rebecca Hwa, and Amanda Godley. 2021. Effective Interfaces for Student-Driven Revision Sessions for Argumentative Writing. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3411764.3445683
- [2] Shaaron Ainsworth. 2006. DeFT: A conceptual framework for considering learning with multiple representations. *Learning and Instruction* 16, 3 (June 2006), 183–198. doi:10.1016/j.learninstruc.2006.03.001
- [3] Saleema Amershi, Maya Cakmak, William Bradley Knox, and Todd Kulesza. 2014. Power to the people: The role of humans in interactive machine learning. *AI magazine* 35, 4 (2014), 105–120.
- [4] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fournery, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [5] Riku Arakawa, Hiromu Yakura, and Sosuke Kobayashi. 2022. VocabEncounter: NMT-powered Vocabulary Learning by Presenting Computer-Generated Usages of Foreign Words into Users' Daily Lives. In *CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3491102.3501839
- [6] Hilary Arksey and Lisa O'Malley. 2005. Scoping studies: towards a methodological framework. *International Journal of Social Research Methodology* 8, 1 (2005), 19–32. doi:10.1080/1364557032000119616
- [7] Michael W. Asher, Gillian Gold, and Paulo F. Carvalho. 2026. Benefit or Bottleneck? Assessing the Impact of Structured Reflection on Learning from AI-Driven Explanatory Feedback. In *Proceedings of the Thirteenth ACM Conference on Learning @ Scale*. ACM, New York, NY, USA, 1–11. doi:10.1145/3774398.3811598
- [8] Lisanne Bainbridge. 1983. Ironies of automation. *Automatica* 19, 6 (1983), 775–779. doi:10.1016/0005-1098(83)90046-8
- [9] Alex Barrett, Fengfeng Ke, Nuodi Zhang, Chih-Pu Dai, Saptarshi Bhowmik, and Xin Yuan. 2025. Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK 2025)*. ACM, New York, NY, USA, 761–770. doi:10.1145/3706468.3706570
- [10] Brian R. Belland. 2011. Distributed Cognition as a Lens to Understand the Effects of Scaffolds: The Role of Transfer of Responsibility. *Educational Psychology Review* 23, 4 (Sept. 2011), 577–600. doi:10.1007/s10648-011-9176-5
- [11] Brian R Belland, Andrew E Walker, Nam Ju Kim, and Mason Lefler. 2017. Synthesizing results from empirical research on computer-based scaffolding in STEM education: A meta-analysis. *Review of educational research* 87, 2 (2017), 309–344.
- [12] Manon Berney, Abdessalam Ouazki, Vladimir Macko, Bruno Kocher, and Adrian Holzer. 2024. Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3642296
- [13] Yali Bian and Chris North. 2021. DeepSI: Interactive Deep Learning for Semantic Interaction. In *26th International Conference on Intelligent User Interfaces (IUI '21)*. ACM, New York, NY, USA, 197–207. doi:10.1145/3397481.3450670
- [14] Michael Mose Biskjaer, Peter Dalsgaard, and Kim Halskov. 2014. A constraint-based understanding of design spaces. In *Proceedings of the 2014 conference on Designing interactive systems (DIS '14)*. ACM, New York, NY, USA, 453–462. doi:10.1145/2598510.2598533

- [15] Robert A. Bjork. 1994. Memory and Metamemory Considerations in the Training of Human Beings. In *Metacognition: Knowing about Knowing*, Janet Metcalfe and Arthur P. Shimamura (Eds.). The MIT Press, Cambridge, MA, 185–206. doi:10.7551/mitpress/4561.003.0011
- [16] Jessica Y Bo, Pan Hao, and Brian Y Lim. 2024. Incremental XAI: Memorable Understanding of AI with Incremental Explanations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3613904.3642689
- [17] Brandon M. Booth, Jennifer Jacobs, Jeffrey B. Bush, Brent Milne, Tom Fischhaber, and Sidney K. DMello. 2024. Human-tutor Coaching Technology (HTCT): Automated Discourse Analytics in a Coached Tutoring Model. In *Proceedings of the 14th Learning Analytics and Knowledge Conference (LAK '24)*. ACM, New York, NY, USA, 725–735. doi:10.1145/3636555.3636937
- [18] Marcel Borowski, Bjarke V. Fog, Carla F. Griggio, James R. Eagan, and Clemens N. Klokmoose. 2023. Between Principle and Pragmatism: Reflections on Prototyping Computational Media with Webstrates. *ACM Transactions on Computer-Human Interaction* 30, 4 (Aug. 2023), 1–53. doi:10.1145/3569895
- [19] Nadiyah Bremer and Shirley Wu. 2021. Royal Constellations. In *Data Sketches*. A K Peters/CRC Press, 116–131. doi:10.1201/9780429445019-11
- [20] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (April 2021), 1–21. doi:10.1145/3449287
- [21] Zana Bućinca, Siddharth Swaroop, Amanda E. Paluch, Finale Doshi-Velez, and Krzysztof Z. Gajos. 2025. Contrastive Explanations That Anticipate Human Misconceptions Can Improve Human Decision-Making Skills. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–25. doi:10.1145/3706598.3713229
- [22] Sabrina Caldwell, Penny Sweetser, Nicholas O'Donnell, Matthew J. Knight, Matthew Aitchison, Tom Gedeon, Daniel Johnson, Margot Brereton, Marcus Gallagher, and David Conroy. 2022. An Agile New Research Framework for Hybrid Human-AI Teaming: Trust, Transparency, and Transferability. *ACM Transactions on Interactive Intelligent Systems* 12, 3, Article 17 (jul 2022), 36 pages. doi:10.1145/3514257
- [23] Yining Cao, Peiling Jiang, and Haijun Xia. 2025. Generative and Malleable User Interfaces with Generative and Evolving Task-Driven Data Model. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3713285
- [24] Joseph Chee Chang, Nathan Hahn, and Aniket Kittur. 2020. Mesh: Scaffolding Comparison Tables for Online Decision Making. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20)*. ACM, New York, NY, USA, 391–405. doi:10.1145/3379337.3415865
- [25] Yao-Jen Chang, Wan Chih Chang, and Tsen-Yung Wang. 2009. Context-aware prompting to transition autonomously through vocational tasks for individuals with cognitive impairments. In *Proceedings of the 11th international ACM SIGACCESS conference on Computers and accessibility (ASSETS'09)*. ACM, New York, NY, USA, 19–26. doi:10.1145/1639642.1639648
- [26] Meng Chen, Akhil Iyer, and Amy Pavel. 2025. Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3663547.3746393
- [27] Michelene T. H. Chi and Ruth Wylie. 2014. The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes. *Educational Psychologist* 49, 4 (2014), 219–243. doi:10.1080/00461520.2014.965823
- [28] Seulgi Choi, Hyewon Lee, Yoonjoo Lee, and Juho Kim. 2024. VIVID: Human-AI Collaborative Authoring of Vicarious Dialogues from Lecture Videos. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642867
- [29] Yoonseo Choi, Eun Jeong Kang, Seulgi Choi, Min Kyung Lee, and Juho Kim. 2025. Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–32. doi:10.1145/3706598.3714034
- [30] Andy Clark and David Chalmers. 1998. The extended mind. *Analysis* 58, 1 (1998), 7–19. doi:10.1093/analys/58.1.7
- [31] Hai Dang, Chelse Swoopes, Daniel Buschek, and Elena L. Glassman. 2025. CorpusStudio: Surfacing Emergent Patterns In A Corpus Of Prior Work While Writing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3706598.3713974
- [32] Dorottya Demszyk and Jing Liu. 2023. M-Powering Teachers: Natural Language Processing Powered Feedback Improves 1:1 Instruction and Student Outcomes. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '23)*. ACM, New York, NY, USA, 59–69. doi:10.1145/3573051.3593379
- [33] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. doi:10.48550/ARXIV.1702.08608
- [34] Mohamed Ez-Zaouia and Yazid Boumarafi. 2025. Codigen: Enhancing Personalized Programming Education with Visual Authoring and LLMs. In *2025 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE, 272–285. doi:10.1109/vl-hcc65237.2025.00038
- [35] Ying Fang, Bo He, Zhi Liu, Sannyuya Liu, Zhonghua Yan, and Jianwen Sun. 2024. Evaluating the Design Features of an Intelligent Tutoring System for Advanced Mathematics Learning. In *Artificial Intelligence in Education (Lecture Notes in Computer Science, Vol. 14829)*. Springer, Cham, 338–350. doi:10.1007/978-3-031-64302-6_24
- [36] Li Feng, Ryan Yen, Yuzhe You, Mingming Fan, Jian Zhao, and Zhicong Lu. 2024. CoPrompt: Supporting Prompt Sharing and Referring in Collaborative Natural Language Programming. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3613904.3642212
- [37] Kasra Ferdowsi, Ruanqianqian (Lisa) Huang, Michael B. James, Nadia Polikarpova, and Sorin Lerner. 2024. Validating AI-Generated Code with Live Programming. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–8. doi:10.1145/3613904.3642495
- [38] Jonas Frich, Lindsay MacDonald Vermeulen, Christian Remy, Michael Mose Biskjaer, and Peter Dalsgaard. 2019. Mapping the Landscape of Creativity Support Tools in HCI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY,

- USA, 1–18. doi:10.1145/3290605.3300619
- [39] Y. Fu, D. Bromley, and V. Setlur. 2025. Data Weaver : Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text. *Computer Graphics Forum* 44, 3 (May 2025). doi:10.1111/cgf.70098
- [40] Vinitha Gadiraju, Annika Muehlbradt, and Shaun K. Kane. 2020. BrailleBlocks: Computational Braille Toys for Collaborative Learning. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3313831.3376295
- [41] Krzysztof Z. Gajos and Lena Mamykina. 2022. Do People Engage Cognitively with AI? Impact of AI Assistance on Incidental Learning. In *27th International Conference on Intelligent User Interfaces (IUI '22)*. ACM, New York, NY, USA, 794–806. doi:10.1145/3490099.3511138
- [42] Krzysztof Z. Gajos, Daniel S. Weld, and Jacob O. Wobbrock. 2010. Automatically generating personalized user interfaces with Supple. *Artificial Intelligence* 174, 12–13 (2010), 910–950. doi:10.1016/j.artint.2010.05.005
- [43] Jie Gao, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, and Simon Tangi Perrault. 2024. CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–29. doi:10.1145/3613904.3642002
- [44] Frederic Gmeiner, Jamie Lynn Conlin, Eric Handa Tang, Nikolas Martelaro, and Kenneth Holstein. 2024. An Evidence-based Workflow for Studying and Designing Learning Supports for Human-AI Co-creation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613905.3650763
- [45] T.R.G. Green and M. Petre. 1996. Usability Analysis of Visual Programming Environments: A 'Cognitive Dimensions' Framework. *Journal of Visual Languages & Computing* 7, 2 (June 1996), 131–174. doi:10.1006/jvlc.1996.0009
- [46] Noor Hammad, David Chuan-En Lin, Amy Smith, Max Kreminski, Erik Harpstead, and Jessica Hammer. 2026. Tracing Creativity: A Design Space For Creative Activity Traces in HCI. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–24. doi:10.1145/3772318.3791263
- [47] James Hollan, Edwin Hutchins, and David Kirsh. 2000. Distributed cognition: toward a new foundation for human-computer interaction research. *ACM Transactions on Computer-Human Interaction (TOCHI)* 7, 2 (2000), 174–196.
- [48] Jiayi Hong, Alain Trubuil, and Tobias Isenberg. 2022. LineageD: An Interactive Visual System for Plant Cell Lineage Assignments based on Correctable Machine Learning. *Computer Graphics Forum* 41, 3 (June 2022), 195–207. doi:10.1111/cgf.14533
- [49] Md Naimul Hoque, Tasfia Mashiat, Bhavya Ghai, Cecilia D. Shelton, Fanny Chevalier, Kari Kraus, and Niklas Elmqvist. 2024. The HaLLMark Effect: Supporting Provenance and Transparent Use of Large Language Models in Writing with Interactive Visualization. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3641895
- [50] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems the CHI is the limit - CHI '99 (CHI '99)*. ACM Press, New York, NY, USA, 159–166. doi:10.1145/302979.303030
- [51] Xinying Hou, Zihan Wu, Xu Wang, and Barbara J. Ericson. 2024. CodeTailor: LLM-Powered Personalized Parsons Puzzles for Engaging Support While Learning Programming. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale (L@S '24)*. ACM, New York, NY, USA, 51–62. doi:10.1145/3657604.3662032
- [52] Chao-Chun Hsu, Erin Bransom, Jenna Sparks, Bailey Kuehl, Chenhao Tan, David Wadden, Lucy Wang, and Aakanksha Naik. 2024. CHIME: LLM-Assisted Hierarchical Organization of Scientific Studies for Literature Review Support. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 118–132. doi:10.18653/v1/2024.findings-acl.8
- [53] Evey Jiaxin Huang, Matthew Easterday, and Elizabeth Gerber. 2025. AI That Helps Us Help Each Other: A Proactive System for Scaffolding Mentor-Novice Collaboration in Entrepreneurship Coaching. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (Oct. 2025), 1–30. doi:10.1145/3757549
- [54] Jeremy Zhengqi Huang, Caluã de Lacerda Patata, Liang-Yuan Wu, and Dhruv Jain. 2025. CapTune: Adapting Non-Speech Captions With Anchored Generative Models. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3663547.3746346
- [55] Edwin Hutchins. 2005. Material anchors for conceptual blends. *Journal of pragmatics* 37, 10 (2005), 1555–1577.
- [56] Peiling Jiang, Jude Rayan, Steven P. Dow, and Haijun Xia. 2023. Graphologue: Exploring Large Language Model Responses with Interactive Diagrams. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3586183.3606737
- [57] Nikhita Joshi and Daniel Vogel. 2026. Designing and Evaluating AI Margin Notes in Document Reader Software. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3772318.3790786
- [58] Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller. 2003. The Expertise Reversal Effect. *Educational Psychologist* 38, 1 (2003), 23–31. doi:10.1207/S15326985EP3801_4
- [59] Majeed Kazemitabaar, Jack Williams, Ian Drosos, Toví Grossman, Austin Zachary Henley, Carina Negreanu, and Advait Sarkar. 2024. Improving Steering and Verification in AI-Assisted Data Analysis with Interactive Task Decomposition. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3654777.3676345
- [60] Eunhye Kim, Bryan Min, Haijun Xia, and Juho Kim. 2026. Elicitive User Interfaces: Designing How Users Shape Generative Interfaces. doi:10.48550/arXiv.2609.23642 arXiv:2609.23642 Preprint, version 1, September 20, 2026.

- [61] Yeonjoon Kim, Kyochul Jang, Hyungwoo Song, Minjeong Shin, and Bongwon Suh. 2026. CodeVoyager: Integrating Interactive Visual Aids with LLMs for Code Comprehension. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26)*. ACM, New York, NY, USA, 1053–1084. doi:10.1145/3742413.3789057
- [62] David Kirsh and Paul Maglio. 1994. On Distinguishing Epistemic from Pragmatic Action. *Cognitive Science* 18, 4 (Oct. 1994), 513–549. doi:10.1207/s15516709cog1804_1
- [63] Hyung-Kwon Ko, Gwanmo Park, Hyeon Jeon, Jaemin Jo, Juho Kim, and Jinwook Seo. 2023. Large-scale Text-to-Image Generation Models for Visual Artists' Creative Works. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. ACM, New York, NY, USA, 919–933. doi:10.1145/3581641.3584078
- [64] Kenneth R. Koedinger and Vincent Alevan. 2007. Exploring the Assistance Dilemma in Experiments with Cognitive Tutors. *Educational Psychology Review* 19, 3 (2007), 239–264. doi:10.1007/s10648-007-9049-0
- [65] David C. Krakauer. 2016. Will A.I. Harm Us? Better to Ask How We'll Reckon With Our Hybrid Nature. *Nautilus*. <https://nautilus.us/will-ai-harm-us-better-to-ask-how-well-reckon-with-our-hybrid-nature-236098> Published September 5, 2016. Accessed September 26, 2026.
- [66] Harsh Kumar, Ruiwei Xiao, Benjamin Lawson, Ilya Musabirov, Jiakai Shi, Xinyuan Wang, Huayin Luo, Joseph Jay Williams, Anna N. Rafferty, John Stamper, and Michael Liut. 2024. Supporting Self-Reflection at Scale with Large Language Models: Insights from Randomized Field Experiments in Classrooms. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale (L@S '24)*. ACM, New York, NY, USA, 86–97. doi:10.1145/3657604.3662042
- [67] Tzu-Sheng Kuo, Aaron Lee Halfaker, Zirui Cheng, Jiwoo Kim, Meng-Hsin Wu, Tongshuang Wu, Kenneth Holstein, and Haiyi Zhu. 2024. Wikibench: Community-Driven Data Curation for AI Evaluation on Wikipedia. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–24. doi:10.1145/3613904.3642278
- [68] Walter L. Leite, Samrat Roy, Nilanjana Chakraborty, George Michailidis, A. Corinne Huggins-Manley, Sidney D'Mello, Mohamad Kazem Shirani Faradonbeh, Emily Jensen, Huan Kuang, and Zeyuan Jing. 2022. A novel video recommendation system for algebra: An effectiveness evaluation study. In *LAK22: 12th International Learning Analytics and Knowledge Conference (LAK22)*. ACM, New York, NY, USA, 294–303. doi:10.1145/3506860.3506906
- [69] Shahid Latif, Zheng Zhou, Yoon Kim, Fabian Beck, and Nam Wook Kim. 2022. Kori: Interactive Synthesis of Text and Charts in Data Documents. *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (Jan. 2022), 184–194. doi:10.1109/tvcg.2021.3114802
- [70] Bryan Lawson. 2006. *How Designers Think: The Design Process Demystified*. Routledge. https://books.google.com/books/about/How_Designers_Think.html?id=IPvqZJNAdG8C
- [71] Danielle Levac, Heather Colquhoun, and Kelly K. O'Brien. 2010. Scoping studies: advancing the methodology. *Implementation Science* 5, 1 (2010), 69. doi:10.1186/1748-5908-5-69
- [72] Maxim Lisnic, Vidya Setlur, and Nicole Sultanum. 2025. Plume: Scaffolding Text Composition in Dashboards. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3706598.3713580
- [73] Michael Xieyang Liu, Aniket Kittur, and Brad A. Myers. 2022. Crystalline: Lowering the Cost for Developers to Collect and Organize Information for Decision Making. In *CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3491102.3501968
- [74] Michael Xieyang Liu, Tongshuang Wu, Tianying Chen, Franklin Mingzhe Li, Aniket Kittur, and Brad A Myers. 2024. Selenite: Scaffolding Online Sensemaking with Comprehensive Overviews Elicited from Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642149
- [75] Xinyi Lu, Kexin Phyllis Ju, Mitchell Dudley, Larissa Sano, and Xu Wang. 2026. AI-Mediated Feedback Improves Student Revisions: A Randomized Trial with FeedbackWriter in a Large Undergraduate Course. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3772318.3791121
- [76] Sehi L'Yi, Astrid van den Brandt, Etowah Adams, Huyen N. Nguyen, and Nils Gehlenborg. 2024. Learnable and Expressive Visualization Authoring Through Blended Interfaces. Center for Open Science. doi:10.31219/osf.io/pjcn4
- [77] Ruofei Ma, Yu Zhao, Yuheng Shao, Yunjie Yao, and Quan Li. 2025. StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. ACM, New York, NY, USA, 1314–1334. doi:10.1145/3708359.3712088
- [78] Shuai Ma, Ying Lei, Xinru Wang, Chengbo Zheng, Chuhan Shi, Ming Yin, and Xiaojuan Ma. 2023. Who Should I Trust: AI or Myself? Leveraging Human and AI Correctness Likelihood to Promote Appropriate Trust in AI-Assisted Decision-Making. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3544548.3581058
- [79] Shuai Ma, Junling Wang, Yuanhao Zhang, Xiaojuan Ma, and April Yi Wang. 2025. DBox: Scaffolding Algorithmic Programming Learning through Learner-LLM Co-Decomposition. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3713748
- [80] Shuai Ma, Zijun Wei, Feng Tian, Xiangmin Fan, Jianming Zhang, Xiaohui Shen, Zhe Lin, Jin Huang, Radomír Měch, Dimitris Samaras, and Hongan Wang. 2019. SmartEye: Assisting Instant Photo Taking via Integrating User Preference with Deep View Proposal Network. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3290605.3300701
- [81] Wendy E. Mackay. 2000. Responding to cognitive overload: Co-adaptation between users and technology. *Intellectica* 30, 1 (2000), 177–193. doi:10.3406/intel.2000.1597

- [82] Allan MacLean, Richard Young, Victoria Bellotti, and Thomas Moran. 1991. Questions, Options, and Criteria: Elements of Design Space Analysis. *Human-Computer Interaction* 6, 3 (Sept. 1991), 201–250. doi:10.1207/s15327051hci0603&4_2
- [83] Lauren Margulieux and Richard Catrambone. 2017. Using Learners' Self-Explanations of Subgoals to Guide Initial Problem Solving in App Inventor. In *Proceedings of the 2017 ACM Conference on International Computing Education Research (ICER '17)*. ACM, New York, NY, USA, 21–29. doi:10.1145/3105726.3106168
- [84] Damien Masson, Young-Ho Kim, and Fanny Chevalier. 2025. Textoshop: Interactions Inspired by Drawing Software to Facilitate Text Editing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3706598.3713862
- [85] Justin Matejka, Tovi Grossman, and George Fitzmaurice. 2012. Citeology: Visualizing Paper Genealogy. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 181–190. doi:10.1145/2212776.2212796
- [86] Bryan Min, Allen Chen, Yining Cao, and Haijun Xia. 2025. Malleable Overview-Detail Interfaces. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–25. doi:10.1145/3706598.3714164
- [87] Bryan Min, Sangho Suh, Jim Hollan, and Haijun Xia. 2026. Making Abstraction Concrete: A Design Space and Interaction Model of Abstraction in Interactive Systems. doi:10.48550/arXiv.2605.11344 arXiv:2605.11344 Preprint, version 1, May 11, 2026.
- [88] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3287560.3287596
- [89] Yamato Miyatake. 2026. HCI Research Trails: An Interactive Citation Map for HCI Research. Interactive web visualization and open-source software. <https://hci-research-trails.vercel.app/> Source code: <https://github.com/mytkymt/paper-lineage>; accessed 2026-08-31.
- [90] Sina Mohseni, Niloofar Zarei, and Eric D. Ragan. 2021. A Multidisciplinary Survey and Framework for Design and Evaluation of Explainable AI Systems. *ACM Transactions on Interactive Intelligent Systems* 11, 3–4, Article 24 (aug 2021), 45 pages. doi:10.1145/3387166
- [91] Junho Myung, Hyunseung Lim, Hana Oh, Hyoungwook Jin, Nayeon Kang, So-Yeon Ahn, Hwajung Hong, Alice Oh, and Juho Kim. 2026. When Scaffolding Breaks: Investigating Student Interaction with LLM-Based Writing Support in Real-Time K-12 EFL Classrooms. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3772318.3791517
- [92] Donald A. Norman. 1991. Cognitive Artifacts. In *Designing Interaction: Psychology at the Human-Computer Interface*, John M. Carroll (Ed.). Cambridge University Press, 17–38. https://www.researchgate.net/publication/213799289_Cognitive_Artifacts
- [93] Donald A. Norman. 2008. Signifiers, not affordances. *Interactions* 15, 6 (2008), 18–19. doi:10.1145/1409040.1409044
- [94] Kai Nylund, Jennifer Mankoff, and Venkatesh Potluri. 2025. MatplotAlt: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks. *Computer Graphics Forum* 44, 3 (May 2025). doi:10.1111/cgf.70119
- [95] Bruno Oliveira and Carla Teixeira Lopes. 2023. From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features. In *Proceedings of the 2023 Conference on Human Information Interaction and Retrieval (CHIIR '23)*. ACM, New York, NY, USA, 338–345. doi:10.1145/3576840.3578307
- [96] Srishti Palani, Zijian Ding, Austin Nguyen, Andrew Chuang, Stephen MacNeil, and Steven P. Dow. 2021. CoNotate: Suggesting Queries Based on Notes Promotes Knowledge Discovery. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3411764.3445618
- [97] Srishti Palani, Aakanksha Naik, Doug Downey, Amy X. Zhang, Jonathan Bragg, and Joseph Chee Chang. 2023. Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3544548.3580841
- [98] Zachary A. Pados, Matthew Tang, Ioannis Anastasopoulos, Shreya K. Sheel, and Ethan Zhang. 2023. OATutor: An Open-source Adaptive Tutoring System and Curated Content Library for Learning Sciences Research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3544548.3581574
- [99] John W. Payne, James R. Bettman, and Eric J. Johnson. 1993. *The Adaptive Decision Maker*. Cambridge University Press. doi:10.1017/cbo9781139173933
- [100] Roy D. Pea. 2004. The Social and Technological Dimensions of Scaffolding and Related Theoretical Concepts for Learning, Education, and Human Activity. *Journal of the Learning Sciences* 13, 3 (July 2004), 423–451. doi:10.1207/s15327809jls1303_6
- [101] Xiaohan Peng, Janin Koch, and Wendy E. Mackay. 2025. FusAI: Composing Generative AI Visual Prompts Using Pen-based Interaction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3706598.3714027
- [102] Kevin Pu, K. J. Kevin Feng, Tovi Grossman, Tom Hope, Bhavana Dalvi Mishra, Matt Latzke, Jonathan Bragg, Joseph Chee Chang, and Pao Siangliulue. 2025. IdeaSynth: Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–31. doi:10.1145/3706598.3714057
- [103] Sadhana Puntambekar and Roland Hübcher. 2005. Tools for scaffolding students in a complex learning environment: What have we gained and what have we missed? *Educational psychologist* 40, 1 (2005), 1–12.
- [104] Chris Quintana, Brian J. Reiser, Elizabeth A. Davis, Joseph Krajcik, Eric Fretz, Ravit Golan Duncan, Eleni Kyza, Daniel Edelson, and Elliot Soloway. 2004. A Scaffolding Design Framework for Software to Support Science Inquiry. *Journal of the Learning Sciences* 13, 3 (July 2004), 337–386. doi:10.1207/s15327809jls1303_4

- [105] Brian J. Reiser. 2004. Scaffolding Complex Learning: The Mechanisms of Structuring and Problematising Student Work. *Journal of the Learning Sciences* 13, 3 (July 2004), 273–304. doi:10.1207/s15327809jls1303_2
- [106] Mohi Reza, Nathan M Laundry, Ilya Musabirov, Peter Dushniku, Zhi Yuan “Michael” Yu, Kashish Mittal, Tovii Grossman, Michael Liut, Anastasia Kuzminykh, and Joseph Jay Williams. 2024. ABScribe: Rapid Exploration & Organization of Multiple Writing Variations in Human-AI Co-Writing Tasks using Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI ’24)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3613904.3641899
- [107] Bethany Rittle-Johnson, Abbey M. Loehr, and Kelley Durkin. 2017. Promoting self-explanation to improve mathematics learning: A meta-analysis and instructional design principles. *ZDM* 49, 4 (Jan. 2017), 599–611. doi:10.1007/s11858-017-0834-z
- [108] Sherry Ruan, Liwei Jiang, Justin Xu, Bryce Joe-Kun Tham, Zhengneng Qiu, Yeshuang Zhu, Elizabeth L. Murnane, Emma Brunskill, and James A. Landay. 2019. QuizBot: A Dialogue-based Adaptive Learning System for Factual Knowledge. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI ’19)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3290605.3300587
- [109] Marcel Ruoff, Brad A. Myers, and Alexander Maedche. 2023. ONYX: Assisting Users in Teaching Natural Language Interfaces Through Multi-Modal Interactive Task Learning. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI ’23)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3544548.3580964
- [110] Mike Scaife and Yvonne Rogers. 1996. External cognition: how do graphical representations work? *International journal of human-computer studies* 45, 2 (1996), 185–213.
- [111] Tobias Schneider, Joana Hois, Alischa Rosenstein, Sabiha Ghellal, Dimitra Theofanou-Fülbier, and Ansgar R.S. Gerlicher. 2021. ExplAI Yourself! Transparency for Positive UX in Autonomous Driving. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI ’21)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3411764.3446647
- [112] Omar Shaikh, Valentino Emil Chai, Michele Gelfand, Diyi Yang, and Michael S. Bernstein. 2024. Rehearsal: Simulating Conflict to Teach Conflict Resolution. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI ’24)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3613904.3642159
- [113] Jocelyn J Shen, Jennifer King Chen, Leah Findlater, and Griffin Dietz Smith. 2025. eaSEL: Promoting Social-Emotional Learning and Parent-Child Interaction through AI-Mediated Content Consumption. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3706598.3713405
- [114] Bruce Sherin, Brian J. Reiser, and Daniel Edelson. 2004. Scaffolding Analysis: Extending the Scaffolding Metaphor to Learning Artifacts. *Journal of the Learning Sciences* 13, 3 (July 2004), 387–421. doi:10.1207/s15327809jls1303_5
- [115] Donghoon Shin, Tze-Yu Chen, Gary Hsieh, and Lucy Lu Wang. 2025. What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS ’25)*. ACM, New York, NY, USA, 1210–1227. doi:10.1145/3715336.3735686
- [116] Donghoon Shin, Lucy Lu Wang, and Gary Hsieh. 2024. From Paper to Card: Transforming Design Implications with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI ’24)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3613904.3642266
- [117] Betsy Sparrow, Jenny Liu, and Daniel M. Wegner. 2011. Google effects on memory: Cognitive consequences of having information at our fingertips. *Science* 333, 6043 (2011), 776–778. doi:10.1126/science.1207745
- [118] Moritz Spiller, Ying-Hsang Liu, Md Zakir Hossain, Tom Gedeon, Julia Geissler, and Andreas Nürnberger. 2021. Predicting Visual Search Task Success from Eye Gaze Data as a Basis for User-Adaptive Information Visualization Systems. *ACM Transactions on Interactive Intelligent Systems* 11, 2 (May 2021), 1–25. doi:10.1145/3446638
- [119] Valeria Starkova and Iris Howley. 2026. CoAutoML: User Interface Framework for Machine Learning Novices using LLM-based AutoML and Test-Driven Machine Teaching. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI ’26)*. ACM, New York, NY, USA, 755–782. doi:10.1145/3742413.3789153
- [120] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminare: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI ’24)*. ACM, New York, NY, USA, 1–26. doi:10.1145/3613904.3642400
- [121] Eva Erdosne Toth, Daniel D. Suthers, and Alan M. Lesgold. 2002. “Mapping to know”: The effects of representational guidance and reflective assessment on scientific inquiry. *Science Education* 86, 2 (Feb. 2002), 264–286. doi:10.1002/sce.10004
- [122] Andrea C. Tricco, Erin Lillie, Wasifa Zarin, Kelly K. O’Brien, Heather Colquhoun, Danielle Levac, David Moher, Micah D. J. Peters, Tanya Horsley, Laura Weeks, Susanne Hempel, Elie A. Akl, Christine Chang, Jessie McGowan, Lesley Stewart, Lisa Hartling, Adrian Aldcroft, Michael G. Wilson, Chantelle Garrity, Simon Lewin, Christina M. Godfrey, Marilyn T. Macdonald, Etienne V. Langlois, Karla Soares-Weiser, Jo Moriarty, Tammy Clifford, Özge Tunçalp, and Sharon E. Straus. 2018. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Annals of Internal Medicine* 169, 7 (2018), 467–473. doi:10.7326/M18-0850
- [123] Tushar Udeshi, Anna Khazenon, Kabir Khan, Nick Breen, R. J. Corwin, Chris DiGiano, Kodi Weatherholtz, and Marek Zaluski. 2026. Methodologies for Improving the Quality of AI Tutoring in K-12 Education. In *Artificial Intelligence in Education (Lecture Notes in Computer Science, Vol. 16582)*. Springer, Cham, 201–214. doi:10.1007/978-3-032-29755-6_14
- [124] Janneke van de Pol, Monique Volman, and Jos Beishuizen. 2010. Scaffolding in Teacher–Student Interaction: A Decade of Research. *Educational Psychology Review* 22, 3 (April 2010), 271–296. doi:10.1007/s10648-010-9127-6

- [125] David Wadden, Kejian Shi, Jacob Morrison, Alan Li, Aakanksha Naik, Shruti Singh, Nitzan Barzilay, Kyle Lo, Tom Hope, Luca Soldaini, Shannon Zejiang Shen, Doug Downey, Hannaneh Hajishirzi, and Arman Cohan. 2025. SciRIF: A Resource to Enhance Language Model Instruction-Following over Scientific Literature. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 6072–6109. doi:10.18653/v1/2025.emnlp-main.310
- [126] Thiemo Wambsganss, Tobias Kueng, Matthias Soellner, and Jan Marco Leimeister. 2021. ArgueTutor: An Adaptive Dialog-Based Learning System for Argumentation Skills. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3411764.3445781
- [127] Chenglong Wang, Yu Feng, Rastislav Bodik, Isil Dillig, Alvin Cheung, and Amy J Ko. 2021. Falx: Synthesis-Powered Visualization Authoring. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3411764.3445249
- [128] Leijie Wang, Kathryn Yurechko, Pranati Dani, Quan Ze Chen, and Amy X. Zhang. 2025. End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3706598.3713691
- [129] Minhong Wang, Bian Wu, Paul A. Kirschner, and J. Michael Spector. 2018. Using cognitive mapping to foster deeper learning with complex problems in a computer-based environment. *Computers in Human Behavior* 87 (Oct. 2018), 450–458. doi:10.1016/j.chb.2018.01.024
- [130] Zifeng Wang, Lang Cao, Qiao Jin, Joey Chan, Nicholas Wan, Behdad Afzali, Hyun-Jin Cho, Chang-In Choi, Mehdi Emamverdi, Manjot K. Gill, Sun-Hyung Kim, Yijia Li, Yi Liu, Yiming Luo, Hanley Ong, Justin F. Rousseau, Irfan Sheikh, Jenny J. Wei, Ziyang Xu, Christopher M. Zallek, Kyungsang Kim, Yifan Peng, Zhiyong Lu, and Jimeng Sun. 2025. A foundation model for human-AI collaboration in medical literature mining. *Nature Communications* 16, 1 (2025), 8361. doi:10.1038/s41467-025-62058-5
- [131] Florian Weber, Thiemo Wambsganss, Seyed Parsa Neshaei, and Matthias Soellner. 2024. LegalWriter: An Intelligent Writing Support System for Structured and Persuasive Legal Case Writing for Novice Law Students. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–23. doi:10.1145/3613904.3642743
- [132] Jacob O. Wobbrock, Shaun K. Kane, Krzysztof Z. Gajos, Susumu Harada, and Jon Froehlich. 2011. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing* 3, 3, Article 9 (2011), 27 pages. doi:10.1145/1952383.1952384
- [133] Claes Wohlin. 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/2601248.2601268
- [134] David Wood, Jerome S. Bruner, and Gail Ross. 1976. The Role of Tutoring in Problem Solving. *Journal of Child Psychology and Psychiatry* 17, 2 (April 1976), 89–100. doi:10.1111/j.1469-7610.1976.tb00381.x
- [135] Austin P. Wright, Omar Shaikh, Haekyu Park, Will Epperson, Muhammed Ahmed, Stephane Pinel, Duen Horng (Polo) Chau, and Diyi Yang. 2021. RECAST: Enabling User Recourse and Interpretability of Toxicity Detection Models with Interactive Visualization. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (April 2021), 1–26. doi:10.1145/3449280
- [136] Tongshuang Wu, Ellen Jiang, Aaron Donsbach, Jeff Gray, Alejandra Molina, Michael Terry, and Carrie J Cai. 2022. PromptChainer: Chaining Large Language Model Prompts through Visual Programming. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (*CHI EA '22*). Association for Computing Machinery, New York, NY, USA, Article 359, 10 pages. doi:10.1145/3491101.3519729
- [137] Tongshuang Wu, Michael Terry, and Carrie Jun Cai. 2022. AI Chains: Transparent and Controllable Human-AI Interaction by Chaining Large Language Model Prompts. In *CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–22. doi:10.1145/3491102.3517582
- [138] Meng Xia, Reshika Palaniyappan Velumani, Yong Wang, Huamin Qu, and Xiaojuan Ma. 2021. QLens: Visual Analytics of Multi-step Problem-solving Behaviors for Improving Question Design. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (Feb. 2021), 870–880. doi:10.1109/tvcg.2020.3030337
- [139] Iman Yeckehzaare, Tirdad Barghi, and Paul Resnick. 2020. QMaps: Engaging Students in Voluntary Question Generation and Linking. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3313831.3376882
- [140] Weitao You, Yinyu Lu, Zirui Ma, Nan Li, Mingxu Zhou, Xue Zhao, Pei Chen, and Lingyun Sun. 2025. DesignManager: An Agent-Powered Copilot for Designers to Integrate AI Design Tools into Creative Workflows. *ACM Transactions on Graphics* 44, 4 (July 2025), 1–26. doi:10.1145/3730919
- [141] J.D. Zamfirescu-Pereira, Eunice Jun, Michael Terry, Qian Yang, and Bjoern Hartmann. 2025. Beyond Code Generation: LLM-supported Exploration of the Program Design Space. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3706598.3714154
- [142] Siyu Zha, Yujia Liu, Chengbo Zheng, Jiaqi Xu, Fuze Yu, Jiangtao Gong, and Yingqing Xu. 2025. Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–22. doi:10.1145/3706598.3713952
- [143] Ashley Ge Zhang, Yan Chen, and Steve Oney. 2023. VizProg: Identifying Misunderstandings By Visualizing Students' Coding Progress. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3544548.3581516
- [144] Jiajie Zhang and Donald A. Norman. 1994. Representations in Distributed Cognitive Tasks. *Cognitive Science* 18, 1 (Jan. 1994), 87–122. doi:10.1207/s15516709cog1801_3

- [145] Xinlei Zhang, Takashi Miyaki, and Jun Rekimoto. 2020. WithYou: Automated Adaptive Speech Tutoring With Context-Dependent Speech Recognition. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA, 1–12. doi:10.1145/3313831.3376322
- [146] Yu Zhang, Jingwei Sun, Li Feng, Cen Yao, Mingming Fan, Liuxin Zhang, Qianying Wang, Xin Geng, and Yong Rui. 2024. See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–24. doi:10.1145/3613904.3642545
- [147] Yuzhe Zhou, Prithvi Manjunatha Babu, Udayan Pandey, Alejandra J. Magana, and Tianyi Li. 2026. ExPeerience: Towards AI-Assisted Learnersourcing to Bridge Conceptual Understanding and Problem Solving in Database Programming Education. In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26)*. ACM, New York, NY, USA, 1178–1201. doi:10.1145/3742413.3789139
- [148] Zhongyi Zhou, Jing Jin, Vrushank Phadnis, Xiuxiu Yuan, Jun Jiang, Xun Qian, Kristen Wright, Mark Sherwood, Jason Mayes, Jingtao Zhou, Yiyi Huang, Zheng Xu, Yinda Zhang, Johnny Lee, Alex Olwal, David Kim, Ram Iyengar, Na Li, and Ruofei Du. 2025. InstructPipe: Generating Visual Blocks Pipelines with Human Instructions and LLMs. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–22. doi:10.1145/3706598.3713905
- [149] Jonathan Zong, Isabella Pedraza Pineros, Mengzhu (Katie) Chen, Daniel Hajas, and Arvind Satyanarayan. 2025. Semantic Scaffolding: Augmenting Textual Structures with Domain-Specific Groupings for Accessible Data Exploration. arXiv:2506.15883. <https://arxiv.org/abs/2506.15883>
- [150] Alexandra Zytek, Dongyu Liu, Rhema Vaithianathan, and Kalyan Veeramachaneni. 2022. Sibyl: Understanding and Addressing the Usability Challenges of Machine Learning In High-Stakes Decision Making. *IEEE Transactions on Visualization and Computer Graphics* 28, 1 (Jan. 2022), 1161–1171. doi:10.1109/tvcg.2021.3114864

A EVIDENCE EXTRACTION AND CODING

This appendix holds what a reader needs to check or reuse the coding: the fields of each evidence card, how the closed and open fields were coded, the agreement checks, the data behind each analysis, how sources for the outcome comparisons were chosen, and how cited sources were matched.

A.1 Card Fields

Each Stream 1 paper has a mechanism card (Table 5), and 2,058 papers also have a theory-lineage card (Table 6); each Stream 2 source has a knowledge-claim card (Table 7). The sectioned layout follows Model Cards [88], and AI-assisted design-card generation is a precedent for turning source findings into cards [116].

Table 5. Stream 1 mechanism card: work and human activity first, then scaffold design and evaluation.

Card section	Fields recorded
Workflow changes (RQ1)	Process before and after support; mechanism credited; attribution basis; source of the baseline.
Human activity (RQ1)	Activity externalized to the tool or representation; activity retained by people; work still required.
Scaffold design (RQ2)	Author term; facet and value, with their merged codes; what makes each mechanism separate; supporting passage and page.
Learning and support (RQ2)	Learning measurement, direction, and evidence; reliance; changes in support and their grounds; observation horizon.
Measures and comparisons (RQ2)	Recorded outcome layer (process, artifact, learning); author claim, claim scope, contribution type, study design, participants, comparison, statistics, and claim-evidence alignment.
Conditions and limitations	Author-stated limitations and boundary conditions.
Recorded codes	Work domain, intent, expertise, capability gap, support properties, AI involvement, and technology era.
Source	Paper ID, DOI, venue, year, page count, the PDF read, the full-text decision, a map of relevant pages, and codebook notes.

Table 6. Stream 1 theory-lineage card: how the paper uses the sources it cites (RQ2).

Card section	Fields recorded
0. Identity and position	Link to the mechanism card; overall position toward prior work; position summary; human-facing rationale; lineage depth.
1. Documented sources (RQ2)	One entry per cited work: citation identity, relation to the focal paper, use, enactment, and supporting passage and page.
2. What was read	Introduction and related-work pages; design or evaluation pages checked for application of a cited idea.
3. Synthesis note	A short account of what the paper’s use of prior work shows.

A.2 Closed Fields

Closed fields take values from a fixed list. Intent records whether the support targets immediate performance or sensemaking, learning or skill development, or both. Measurement records what the evaluation assessed, so an intention to support learning stays apart from a measure of it; the direction of a result and its observation horizon are recorded

Table 7. Stream 2 card: one to eight field-level claims, linked to the research questions they inform.

Card section	Fields recorded
0. Identity and position	Card and work IDs; publication details; discovery routes; contribution types; position summary; overall uncertainty.
1. Page-grounded claims	One entry per claim: author wording and reviewer interpretation; question links; applicability; affected human work; outcome evidence; evidence basis; counterevidence and limits; supporting excerpts; uncertainty.
2. Evidence store	Verbatim excerpts with pages and locators; links recording which claim each excerpt supports, its role, and the reviewer paraphrase.

with the measure. Support properties are coded separately and can co-occur: structuring, problematizing, contingency, adaptation, fading, responsibility transfer, and distributed support. Contingency responds to the person's activity or need. Fading reduces support as competence grows; a fixed reduction on a schedule is not by itself fading. Responsibility transfer concerns who performs or regulates the activity, not whether the interface remains available. Technology is classified once per paper, from the AI in its system, an experimental condition, training, or evaluation, so a paper classified as generative AI can still contain mechanisms that involve no generation.

A.3 Open Fields

The 21 open fields hold 142,877 observations in the papers' own terms. Equivalent descriptions within a field were merged into 1,290 codes, and the codes were grouped into 223 categories and 92 families across 18 fields, with subcategories only where they add a distinction. Each grouping keeps links to its observations, and fields are never pooled. Codes that record missing information, such as an unspecified baseline or an unreported measure, stay apart from substantive categories. When one free-text value lists several limitations, its single code summarizes them rather than listing each. Appendix B defines the families and categories of the six fields the framework uses.

A.4 Agreement Checks

To test the open-code grouping, a separate model pass coded extracted observations with the existing codes hidden, and we compared the two groupings field by field (Table 8). An observation entered the comparison only when its identifier, source card, and field value matched exactly. Two codings need not use the same words, so we compared partitions: whether each pair of observations shares a code in each coding. The adjusted Rand index (ARI) corrects pair agreement for chance, and positive-pair F1 summarizes overlap among the pairs either coding groups together; neither treats one coding as correct.

Raw agreement is high because both codings keep most pairs apart: for facet, 19,083 of 19,503 pairs are apart in both and 32 are together in both. The blind pass used fewer codes than the existing coding in five fields and more in learning evidence. It reported medium or high uncertainty on most assignments in five fields (for example, 659 of 695 values) and on 7 of 24 in learning evidence. The learning-evidence row covers 24 of 48 existing observations; the other 24 share a repeated not-measured value and fell outside the comparison's selection rule.

A separate check asked only whether AI was involved: three fresh model sessions labeled 532 title-and-abstract records added in September and gave the same label to 529 (99.4%; $\kappa = .988$).

Table 8. Blind open-code comparison on exactly matched observations. Codes gives existing / blind code counts; Rand is raw pair agreement, and pair F1 compares the pairs grouped together. Unmatched counts existing observations outside the comparison. Papers overlap across fields, so the 1,571 matched observations are not 1,571 distinct papers.

Field	Obs. (papers)	Unmatched	Codes	Rand	ARI	Pair F1
Externalized to tool	110 (110)	18	64 / 46	96.2%	.167	.186
Facet	198 (48)	15	132 / 76	98.0%	.132	.142
Process before	125 (48)	15	79 / 34	95.8%	.445	.462
Process after	419 (160)	0	262 / 60	97.0%	.079	.088
Value	695 (144)	0	517 / 95	98.1%	.082	.087
Learning evidence	24 (24)	24	5 / 16	50.4%	.108	.217

Table 9. Data behind each analysis. Counts refer to the stated unit.

Data	Count and unit	Analysis
Stream 1 corpus	4,511 papers on 26 September 2026; 474 non-AI comparison papers outside the systematic counts	Corpus counts (Section 3)
Main figures	4,086 papers (4,115 records) with usable coding	Annual, technology, and mechanism figures
Closed fields	2,891 papers coded by 28 August 2026; 301 measure learning	Intent, learning-measurement, and process counts
Open fields	142,877 observations in 21 fields from the 3,295 papers coded by 28 August 2026, of which 2,847 remain in the corpus	Design space and framework values; counts of papers carrying a value
Complete mappings	2,315 papers with complete facet-value codes; 2,311 also with remaining-work codes	Mechanism-by-work matrix (Appendix C)
Theory lineage	2,058 papers, including non-AI comparison papers; 51,815 cited-source records	Roles of prior work (Section 5.5)
Stream 2	512 sources; 2,867 claims	Prior findings and theory
Citation network	4,751 papers from both streams	Citation atlas and co-citation map (Section 5.5)

A.5 Validation Status of the Readings

For part of the evidence, the records of the original reading runs have not been fully validated, and we mark that evidence for provisional use. On 26 September 2026, such evidence appeared in 319 of the 1,076 learning papers and 1,252 of the 4,086 papers behind the main figures. The counts mark incomplete validation records, not errors, and the marked papers stay in the analyses.

A.6 Data Used for Each Analysis

Each analysis uses the papers coded for the fields it needs (Table 9).

All 2,847 open-coded papers have complete domain and retained-work codes; 532 lack complete facet-value codes. The closed-field population includes 44 papers outside the open-field profile and two with unresolved support-property coding.

A.7 Learning Measurement

Among the 2,891 closed-field papers, 301 (10.4%) measured learning. Two-thirds of those (199, 66.1%) measured it within a single session; 35 used repeated sessions, 38 followed people over weeks, and 25 measured learning in deployed use.

A.8 Selection of Sources for Outcome Comparison

The study comparisons in Section 5 draw on sources chosen for their outcome evidence. A learning-scope assessment of 4,714 records proposed 1,226 within scope. We appraised the 422 of these whose cards recorded that learning was measured (twelve more had been appraised earlier and were analyzed separately), plus 14 performance cases found among 53 descriptions that needed interpretation: 436 records, or 434 distinct PDFs. The 190 sources flagged as having a documented capability comparison entered the detailed synthesis.

A.9 Source Matching and Coverage

A.9.1 Source-use coding. Each theory-lineage record links one cited work to its role in one citing paper’s design, argument, or measure (Section 5.5), so a work cited by two papers yields two records.

A.9.2 Citation metadata. The bibliographic map joins the two streams by DOI or Semantic Scholar paper ID and counts a paper found in both once. It uses Semantic Scholar data retrieved on 9 September 2026 for 4,331 Stream 1 papers and 448 Stream 2 works (28 in both). Reference lists for 3,233 papers name 106,523 cited sources; two with conflicting identities are withheld. Of the remaining 106,521, 5,748 are cited by at least five review papers and 157 by at least 40 (Figure 15). A reference list shows that a paper cites a work; its lineage card shows how the paper uses it.

The co-citation map counts each citing paper once per source and keeps sources with at least five citers and pairs with at least three shared citers. It weights a pair by association strength: the pair’s shared citers times the number of papers, divided by the product of the two sources’ citer counts. Louvain clustering (seed 42) forms the neighbourhoods, and a maximum spanning forest supplies the drawn links; stem, branches, and leaves mark how widely a source is shared. Less-cited sources sit near their most shared neighbourhood without joining it. The interactive atlas supports zooming into a neighbourhood and inspecting each source.²

A.9.3 Matching and unavailable sources. Cited works without a DOI are matched by title and year. Title variants can split one work into several identities, which overcounts distinct works and undercounts shared references; descriptive labels and conflicting identities stay out of the shared-reference map. Papers without a reference list, and full texts we could not obtain or read, are outside the source-use analysis.

B THE FRAMEWORK’S CODEBOOK

The framework combines mechanism design with five dimensions of human work and evidence. Tables 10 and 11 define those five dimensions; Appendix C.1 gives the mechanism categories and examples. Counts use the 28 August 2026 coding populations in Appendix A.6; a paper may carry several values. Each group names the paper denominator for its percentages. Daggers identify absent or unspecified values, as defined in each row.

B.1 What follows for the person, the work and the evidence

²Interactive citation atlas: the interactive citation atlas .

Table 10. The three dimensions about the person: what the person still does, keeps, and hands to the tool, with the share and number of papers carrying each value. A dagger marks the absence of a value.

Dimension and value	Definition	Share	Papers
What the person still does (n=2,842 papers)			
The task the tool did not do	Work of the task itself that the system does not do: performing the substantive task, making sense of what is shown, and checking and repairing what comes back.	51.1%	1,451
New work the tool created	Obligations that exist only because the system exists: feeding and instructing it, and operating and coping with it.	29.9%	851
The whole task kept by the person	The person feeds the system and judges what comes back, across three or more stages, with no stage removed.	10.6%	300
Final authority kept by design	The final decision, the reliance decision, values-laden judgement and accountability stay with the person as a matter of design or law, not as a step in a workflow.	8.3%	237
Duty asserted, nothing specified†	The value says responsibility remains with the person without naming any obligation.	0.1%	3
What the person keeps (n=2,847 papers)			
Judgement and decision	Interpreting and inferring, deciding and authorising, setting direction, bringing knowledge.	48.0%	1,367
Doing and revising the work	Doing the work, repairing and revising, steering the loop, improving the system.	35.7%	1,016
Checking the output	Checking output for errors, calibrating reliance, judging fit to goal, verifying against outside evidence.	9.1%	260
Responsibility, social and self-directed work	Coordinating and deliberating with others, consent and disclosure, ethical judgement, communicating outward, holding accountability, care work; and learning, reflection and sustained attention to the system.	6.9%	197
Not stated†	The paper does not say what the person retains.	0.2%	7
What moves into the tool (n=2,326 papers)			
Computation and judgement	The tool predicts, simulates, fits models, calculates, optimises, estimates uncertainty, computes geometry; and attributes behaviour, classifies, detects errors, critiques, scores, infers a person's state, ranks and verifies.	30.5%	709
Representation and translation	The tool gives external form, converts between formats and languages, lays material out, summarises, verbalises, encodes data visually, coordinates views, compiles specifications; and finds groupings, patterns, similarities, aggregates and comparisons.	23.6%	549
Decision, guidance and action	The tool recommends, selects the next step or decides outright; supplies templates, procedures, decompositions, reference material, instruction and prompts; propagates judgments, does clerical work, operates environments, enforces constraints, fabricates; stands in for people, coordinates them, relays communication.	15.6%	364
Perception, search and memory	The tool senses, extracts, recognises, segments, transcribes and flags; finds and narrows material; records, stores, recalls and reminds; watches and tracks.	15.0%	349
Content generation	The tool generates new artifacts (alternatives, images, prose, code, 3D forms, audio) and responses (explanations, questions, answers, utterances, conversational replies, revisions of the person's material).	14.9%	346
Offloading that failed or was withheld	The offloading failed or was incomplete, or the tool deliberately stopped at a proposal and left the judgment with the person.	0.3%	8
Nothing offloaded, or not stated†	The value states that nothing moved to the tool, or names no operation at all.	0.0%	1

Table 11. The two dimensions about the work and the evidence: what happens to the work and how learning is evidenced, with the share and number of papers carrying each value. A dagger marks the absence of a value.

Dimension and value	Definition	Share	Papers
What happens to the work (n=2,643 papers)			
The person's work changes form	The person still works, differently: reading new representations, navigating and searching, steering and specifying at a higher level, reviewing and selecting instead of producing, checking and repairing, following a guided procedure, judging and reflecting on new grounds, coordinating through shared artifacts.	92.2%	2,437
A step becomes automatic	A step the person used to do is now done, or mostly done, by the system: automatic extraction or generation, exception-only attention, the same task made cheaper, automatic checking of the person's own work, effort freed for other work.	26.4%	697
Study condition or proposal, not practice	The value describes a study arm, protocol or manipulation, or a proposed or recommended change that was not observed in use.	16.1%	425
Work newly possible	The person can now do something they could not: work in a new setting or moment, at a larger scale or scope, or a task that was out of reach.	4.6%	121
New work created by the system	Labour that did not exist before the system: setting up, configuring and maintaining it, added steps or obligations, burden or unproductive activity it caused.	4.0%	105
Work unchanged	The practice did not change: the person keeps the core task and the system only informs it, under-uses or works around it, or nothing changed at all.	2.0%	54
How learning is evidenced (n=2,800 papers)			
A proxy measured instead	Capability stood in for by a proxy: subjective judgements about the system, usability instruments, qualitative accounts, attitudes or engagement; in-session task performance, behaviour or rated work products; or the quality of the system's own output, including the machine's learning.	37.6%	1,054
Learning measured	An objective capability measure was taken: immediately with support present, later or with support withdrawn, or with a null, mixed or adverse result, or compromised by confounding or deliberate suppression.	9.4%	263
Learning self-reported only	A learning or skill gain is asserted on self-report alone, or gestured at in remarks while the paper says learning itself was not evaluated.	4.6%	130
Learning not measured†	Nothing about the person's learning, transfer, retention or capability was measured: the value names no measure and no substitute, the authors say so themselves or defer it, there were no participants at all, or learning is the stated purpose yet never evaluated.	48.3%	1,353

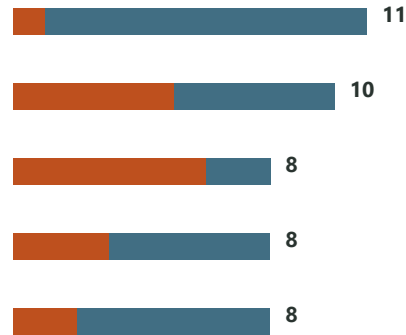
Where prior work is used

1,614 distinct papers · 1,669 PDF-matched lineage cards

Most shared identifier-linked sources

Distinct papers with recorded use

Manifold: A model-agnostic framework for interpretation and diagnosis of machine learning models
 Rulematrix: visualizing and understanding classifiers with rules
 The eyes have it: A task by data type taxonomy for information visualizations
 Kori: Interactive Synthesis of Text and Charts in Data Documents
 NL4DV: A toolkit for generating analytic specifications for data visualization from natural language queries



Follow a source into later designs

Highest-ranked source with an unambiguous recorded year; every use is shown.

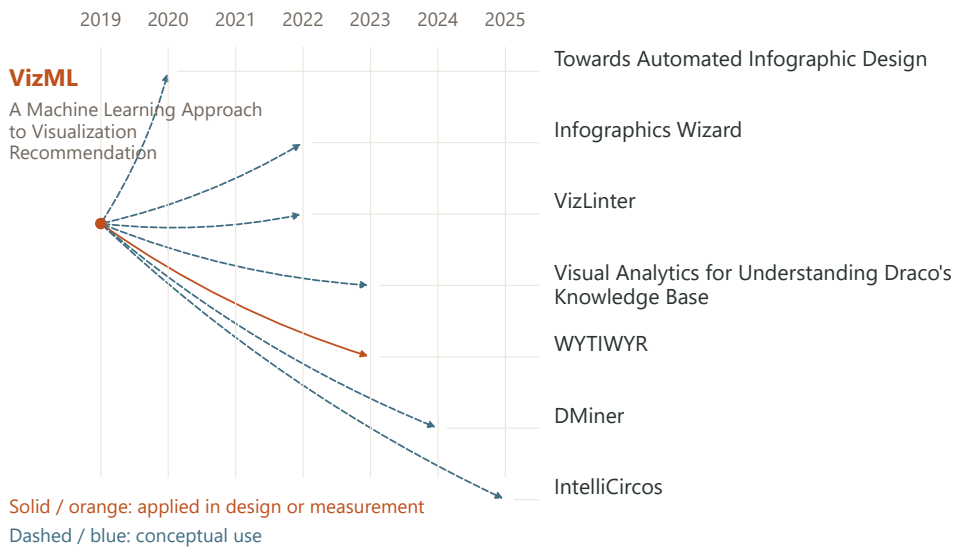
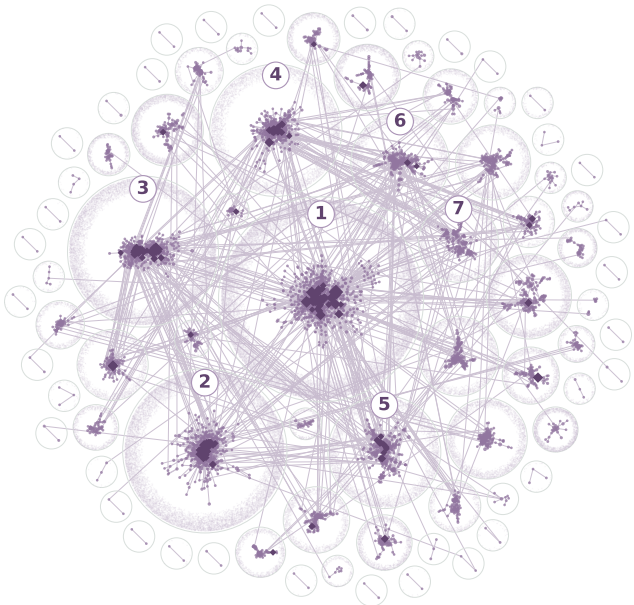


Fig. 14. **How papers use prior work.** Across 1,669 lineage cards for papers in the analysis, the upper panel ranks sources with a DOI or other identifier by the number of papers that use them, split into applied and conceptual-only use. Below, the top-ranked source with a known publication year connects to later designs. Solid arrows show application in design or measurement; dashed arrows show conceptual use. The companion website links all 41,266 source–paper pairs to passages and cards.

Shared sources across the two review streams

106,521 cited sources · 5,748 cited by five or more review papers



Seven largest groups

- 1 Generative AI and human-AI interaction
789 shared sources
 - 2 Explainability, trust and AI decisions
567 shared sources
 - 3 Deep learning and model inspection
492 shared sources
 - 4 Visualization authoring and exploration
460 shared sources
 - 5 Responsible AI and research practice
322 shared sources
 - 6 Programming and data work with AI
247 shared sources
 - 7 Uncertainty and visual communication
165 shared sources
- Leaves · 1-4 · 100,773

◆ Stem · 40+ citers · 157 ● Branches · 5-39 · 5,591

Reference coverage: 3,233 of 4,751 distinct review papers · 9 September 2026

Groups follow co-citation; topic labels summarize prominent member titles.
7,369 sources without group connections remain available in the online source list.

Fig. 15. **Shared-reference neighborhoods.** Reference metadata covers 3,233 of 4,751 papers, from bibliographic data of 9 September 2026, before the corpus was limited to AI papers. Numbers label the seven largest groups of sources cited together.

C DETAILED MECHANISM AND EVIDENCE PROFILES

Coding and populations. This appendix uses the coding of the mechanism cards completed by 28 August 2026, restricted to the systematic corpus (Section 3): 2,847 open-coded papers and 2,891 closed-field papers. Appendix A.6 lists the population behind each profile.

The complete facet catalogue supports the main mechanism analysis (Section 5); complementary evidence profiles support the workflow analysis (Section 4).

C.1 Category Maps and Mechanism Examples

The eighteen category maps cover all 110 mechanism facets in five families. Branches group design operations that systems may combine. Schematics explain the choices; source figures show cited implementations. Variations are recorded implementations, not settings available in every system. Map order and size carry no frequency or effectiveness scale.

Search the full definitions. Links beneath each map open its source mechanisms.

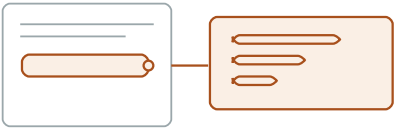
C.1.1 Explanation and visibility. What can a person inspect and understand?

FC-C01 / Explanation and visibility

Explaining system behaviour

Which aspect of the system is made inspectable?

Makes a system's outputs, behaviour, uncertainty or current state inspectable, so people can understand what it is doing and judge its results.



DESIGN CHOICES

RETAINED DISTINCTIONS

A particular result



FC001
output rationale exposed

FC002
feature attribution for an output

Process and sequence



FC006
live process-state visibility

FC008
stepwise reasoning or execution trace

Model and internals



FC003
global model-behaviour summary

FC007
internal or latent state surfaced

Reliability and limits



FC004
uncertainty and confidence exposed

FC005
capability and knowledge-boundary disclosure

Recorded form · FC001: Rationale for a specific machine output (VA001).

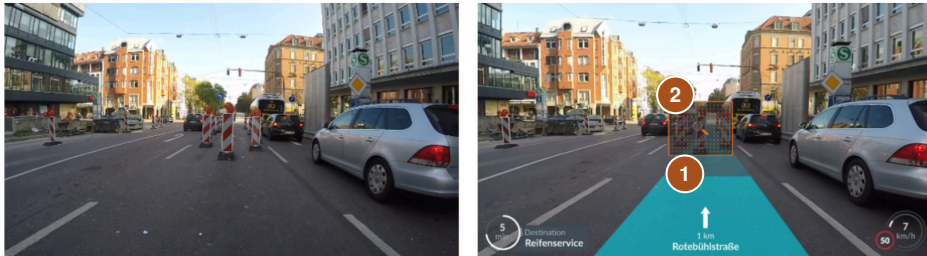
Recorded form · FC002: Feature or factor contribution to an outcome (VA002).

Fig. 16. Explaining system behaviour. Which aspect of the system is made inspectable? Separate the explanation of one result, the running process, the model across cases, and the limits of what it can establish. Category definitions; Browse category mechanisms. Recorded forms: Rationale for a specific machine output; Feature or factor contribution to an outcome.

FC-C01 / SOURCE ILLUSTRATION

Explaining system behaviour

PAPER ILLUSTRATION · FC006



1. Intended route overlaid on the road 2. Detected obstacle marked in the scene

Live overlays expose the vehicle's planned route and detected objects in the road scene.

ExplAIn Yourself! Transparency for Positive UX in Autonomous Driving - Figure 2, PDF page 4

Schneider et al.; CC BY; cropped; numbered annotations added by this review.

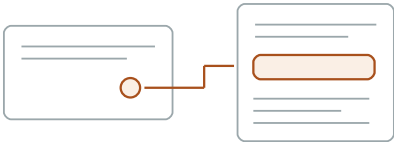
Fig. 17. Explaining system behaviour: source illustration. Figure 2, PDF page 4 in [111]. Live overlays expose the vehicle's planned route and detected objects in the road scene. Inspect the illustrated mechanism.

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FC-C02 / Explanation and visibility
Sources and evidence

How can a result be traced or checked?

Connects a result to its sources, supporting evidence or underlying cases, so people can trace where it came from and check its basis.



DESIGN CHOICES

RETAINED DISTINCTIONS

Origin and derivation

FC009

output-to-source provenance link

FC012

analytic-history provenance capture

Underlying cases

FC010

aggregate-to-instance drill-down

Evidence for a claim

FC011

evidence presented for verification

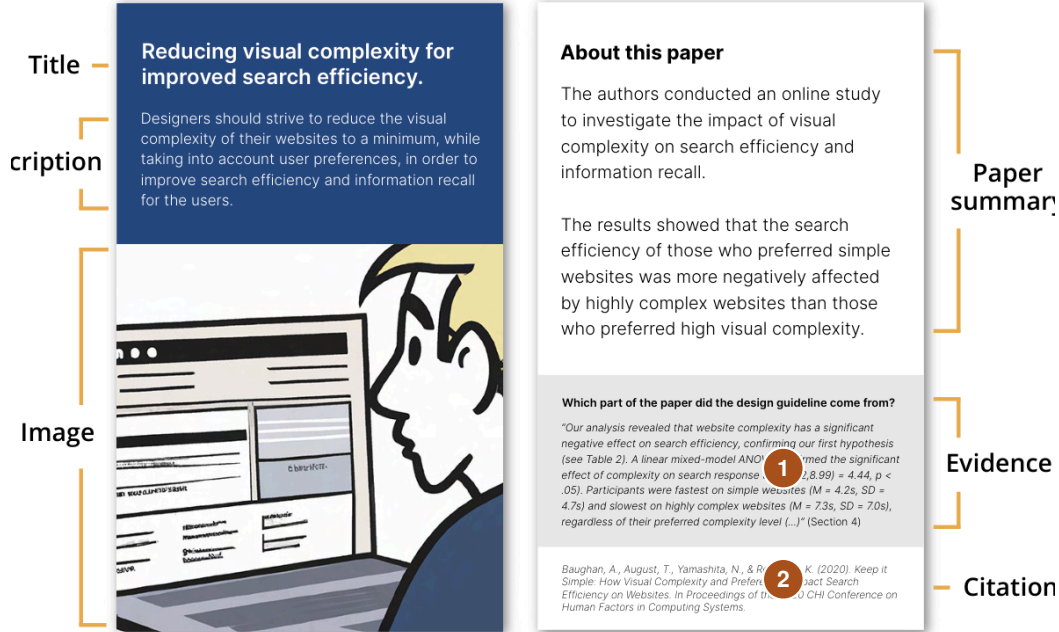
Recorded form · FC010: Drill-down from summary to underlying instances (VA012).

Fig. 18. Sources and evidence. How can a result be traced or checked? Tracing origin and derivation, opening underlying cases, and assessing evidence answer different checking questions. Category definitions; Browse category mechanisms. Recorded forms: Drill-down from summary to underlying instances.

FC-C02 / SOURCE ILLUSTRATION

Sources and evidence

PAPER ILLUSTRATION · FC009



1. Supporting passage 2. Citation back to the paper

The design card places a supporting source passage and its citation beside the implication.

From Paper to Card: Transforming Design Implications with Generative AI · Figure 1, PDF page 2

Shin et al.; CC BY; cropped; numbered annotations added by this review.

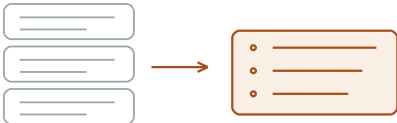
Fig. 19. Sources and evidence: source illustration. Figure 1, PDF page 2 in [116]. The design card places a supporting source passage and its citation beside the implication. Inspect the illustrated mechanism.

FC-C03 / Explanation and visibility

Summarising and labelling content

How is content condensed, interpreted or recast?

Condenses or interprets content through summaries, labels, profiles or a change of representation, so people can grasp what it means.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Condense content	FC107 summarisation and condensation	
Characterise an item	FC105 machine-generated semantic labelling	FC109 entity or concept profile
Change its form	FC108 semantic transformation into another form	

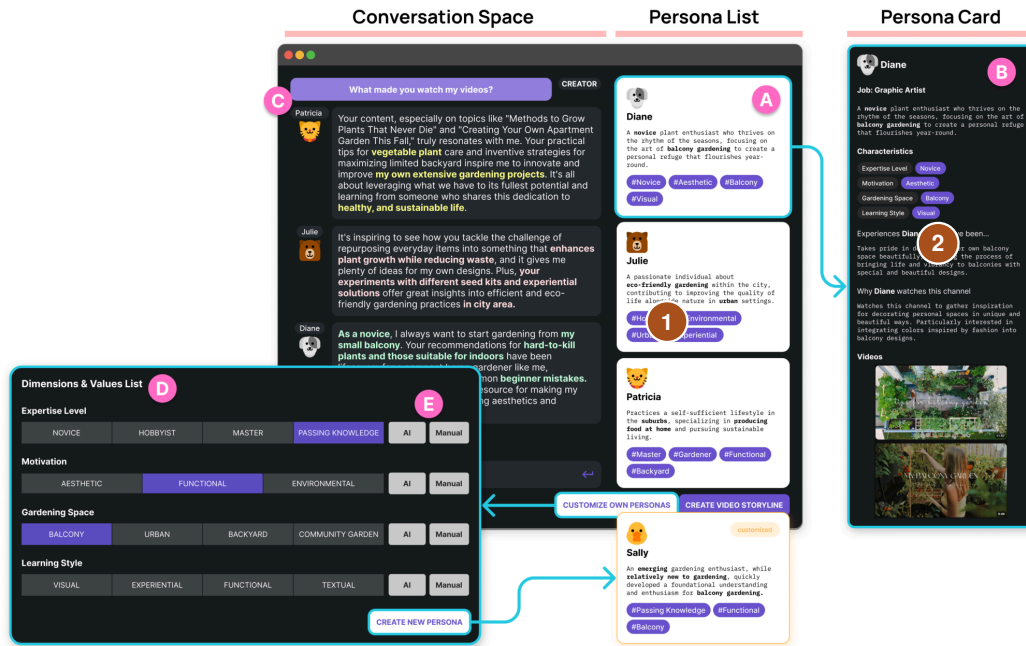
Recorded form · FC107: Condensed summary of a single item (VA097).

Fig. 20. Summarising and labelling content. How is content condensed, interpreted or recast? Distinguish reducing a body of content, characterising an item, and changing representational kind. Category definitions; Browse category mechanisms. Recorded forms: Condensed summary of a single item.

FC-C03 / SOURCE ILLUSTRATION

Summarising and labelling content

PAPER ILLUSTRATION · FC109



1. Compact audience profiles 2. Expanded persona details

Audience segments become profile cards, with detailed experiences and motivations available on selection.

Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas · Figure 2, PDF page 6

Choi et al.; CC BY; cropped; numbered annotations added by this review.

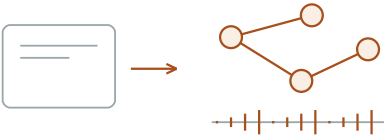
Fig. 21. Summarising and labelling content: source illustration. Figure 2, PDF page 6 in [29]. Audience segments become profile cards, with detailed experiences and motivations available on selection. Inspect the illustrated mechanism.

FC-C04 / Explanation and visibility

Visual and sensory representation

Through which form can a person perceive meaning?

Expresses meaning through visual form, language, sound, touch or physical action, including translations and adaptations that make it accessible.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Perceptual carrier	FC085 visual encoding and graphical mapping	FC083 non-visual sensory encoding	FC086 embodied or physical enactment
Translation of meaning	FC084 cross-modal translation		FC089 verbal description of perceptual content
Adaptation and reframing	FC088 accessible or simplified presentation		FC087 metaphorical or symbolic re-representation

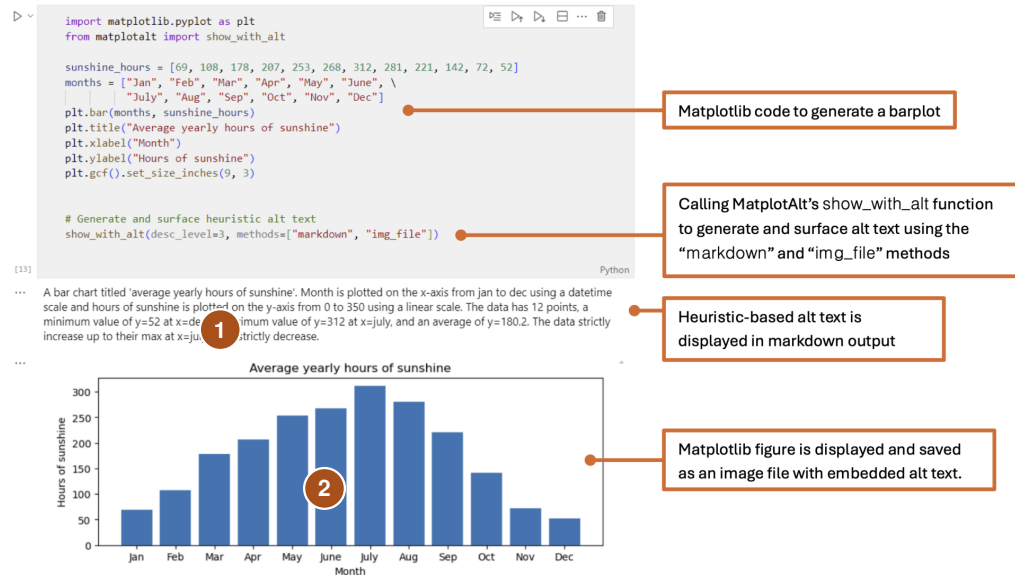
Recorded form · FC083: Spoken or audio rendering of content (VA071).

Fig. 22. Visual and sensory representation. Through which form can a person perceive meaning? Separate the perceptual carrier, translation between carriers, and adaptations that change how the meaning can be grasped. Category definitions; Browse category mechanisms. Recorded forms: Spoken or audio rendering of content.

FC-C04 / SOURCE ILLUSTRATION

Visual and sensory representation

PAPER ILLUSTRATION · FC088



1. Description surfaced as text 2. The same chart remains visible

A notebook presents a chart and its generated description together; the library also supports alternative output placements.

MatplotlibAlt: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks · Figure 1, PDF page 1

Nylund et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 23. Visual and sensory representation: source illustration. Figure 1, PDF page 1 in [94]. A notebook presents a chart and its generated description together; the library also supports alternative output placements. Inspect the illustrated mechanism.

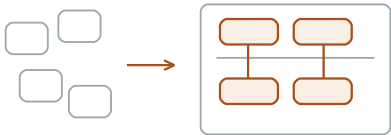
C.1.2 Organisation and comparison. How can a person organise and compare material?

FC-C05 / Organisation and comparison

Organising and structuring

Which relationships give material its structure?

Arranges information into relationships, groups, levels or sequences, so people can understand its structure and move between the whole and its parts.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Links and containment	FC015 relational or network map	FC013 hierarchical or nested organisation	
Groups and categories	FC016 similarity grouping and clustering	FC017 categorisation scheme or controlled vocabulary	FC106 human semantic coding and categorisation
Whole, parts and detail	FC014 overview or aggregate gist	FC018 decomposition into parts	FC019 multiscale detail and semantic zoom
Position and alignment	FC020 temporal arrangement of material	FC021 spatial arrangement of material	FC022 cross-representation correspondence
Quantitative structure	FC023 distributional or statistical structure display		

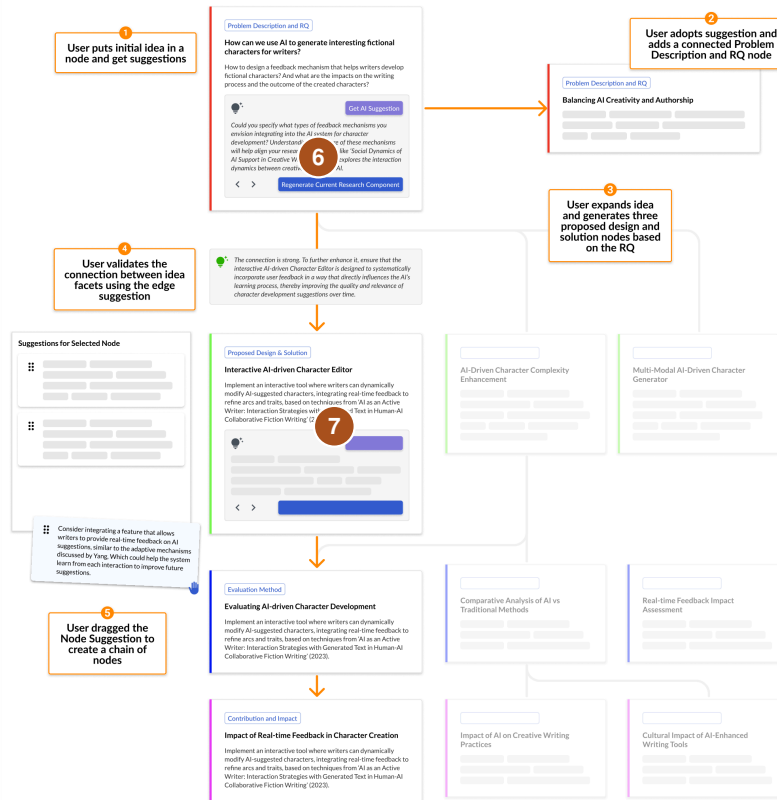
Recorded form - FC015: Relational graph or dependency network (VA019).

Fig. 24. Organising and structuring. Which relationships give material its structure? Relations, category membership, granularity, position and statistical structure organise different aspects of the same material and may be combined. Category definitions; Browse category mechanisms. Recorded forms: Relational graph or dependency network.

FC-C05 / SOURCE ILLUSTRATION

Organising and structuring

PAPER ILLUSTRATION · FC018



6. Research question node 7. Linked solution node

An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes.

Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback · Figure 4, PDF page 8

Pu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 25. Organising and structuring: source illustration. Figure 4, PDF page 8 in [102]. An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes. Inspect the illustrated mechanism.

FC-C06 / Organisation and comparison

Comparing alternatives

What is placed in a common comparison frame?

Places alternatives, cases or states in a common frame so people can examine their similarities, differences and trade-offs.

	A	B

DESIGN CHOICES

RETAINED DISTINCTIONS

Separate candidates

FC024
side-by-side alternative comparison

Changed states or versions

FC025
contrastive or counterfactual comparison

FC028
change-over-time comparison

Standards and populations

FC026
benchmark or peer-referenced comparison

FC027
subgroup or population comparison

Recorded form · FC024: Side-by-side comparison of alternatives (VA023).

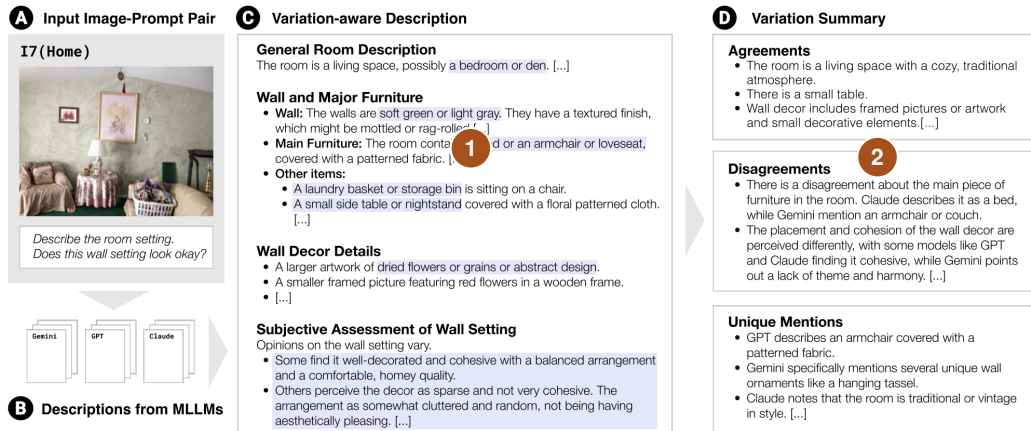
Recorded form · FC025: Before-and-after or version comparison (VA025).

Fig. 26. Comparing alternatives. What is placed in a common comparison frame? The comparand distinguishes candidate selection, variation of the same object, and comparison against groups or reference points. Category definitions; Browse category mechanisms. Recorded forms: Side-by-side comparison of alternatives; Before-and-after or version comparison.

FC-C06 / SOURCE ILLUSTRATION

Comparing alternatives

PAPER ILLUSTRATION · FC024



1. Detailed merged description 2. Agreements and disagreements

The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary.

Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions · Figure 2, PDF page 5

Chen et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 27. Comparing alternatives: source illustration. Figure 2, PDF page 5 in [26]. The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary. Inspect the illustrated mechanism.

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FC-C07 / Organisation and comparison
Search, filtering and ranking

How is relevant material found and brought into view?

Helps people find relevant material and control what is in view through search, navigation, filters, rankings and scope choices.

The diagram illustrates a search interface. It features a search bar at the top with a magnifying glass icon. Below the search bar are three filter buttons. The top filter button is highlighted with an orange border, while the other two have a light blue border.

DESIGN CHOICES

RETAINED DISTINCTIONS

Define the request

FC030
query construction and search specification

FC033
analytic scope setting

Select and order

FC029
relevance filtering of a corpus

FC032
ranked or prioritised surfacing

Navigate and look up

FC034
navigation and wayfinding through structure

FC031
on-demand lookup of related information

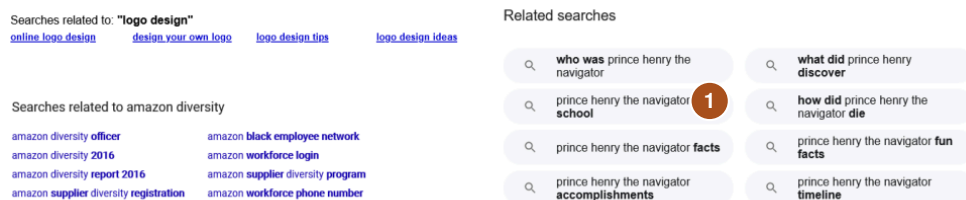
Recorded form · FC030: Filter, query or search over a set (VA028).

Fig. 28. Search, filtering and ranking. How is relevant material found and brought into view? Defining the request, selecting or ordering results, and moving through material are distinct retrieval operations. Category definitions; Browse category mechanisms. Recorded forms: Filter, query or search over a set.

FC-C07 / SOURCE ILLUSTRATION

Search, filtering and ranking

PAPER ILLUSTRATION · FC030



1. Follow-on query formulations

Related-search suggestions provide alternative formulations for continuing a search.

From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features · Figure 16, PDF page 5

Oliveira et al.; CC BY; cropped; numbered annotations added by this review.

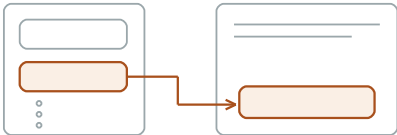
Fig. 29. Search, filtering and ranking: source illustration. Figure 16, PDF page 5 in [95]. Related-search suggestions provide alternative formulations for continuing a search. Inspect the illustrated mechanism.

FC-C08 / Organisation and comparison

Notes, history and memory

What is kept available for continuing the work?

Keeps information and work history available through notes, annotations, memory aids and replay, so people can revisit and continue their work.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Express and capture	FC077 externalising internal structure into an artifact	FC080 capture of activity as it happens
Mark and arrange	FC078 annotation onto material	FC082 reorganisable working workspace
Recall and revisit	FC079 external memory and recall cue	FC081 replay and history navigation

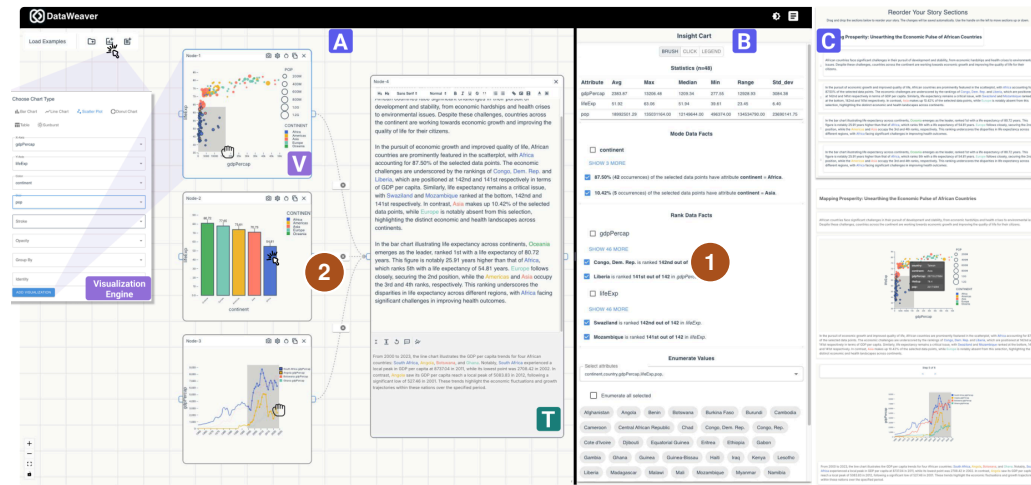
Recorded form · FC078: Annotation and comment capture on content (VA048).

Fig. 30. Notes, history and memory. What is kept available for continuing the work? Separate making or capturing a record, arranging material at hand, and returning to an earlier state. Category definitions; Browse category mechanisms. Recorded forms: Annotation and comment capture on content.

FC-C08 / SOURCE ILLUSTRATION

Notes, history and memory

PAPER ILLUSTRATION · FC082



1. Facts held in the Insight Cart 2. Connected story components

An Insight Cart holds candidate facts for later selection into a connected data story.

DATAWEAVER: Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text · Figure 2, PDF page 5

Fu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 31. Notes, history and memory: source illustration. Figure 2, PDF page 5 in [39]. An Insight Cart holds candidate facts for later selection into a connected data story. Inspect the illustrated mechanism.

C.1.3 User control and expression. What can a person express, change or try?

FC-C09 / User control and expression

Goals and constraints

How does a person specify what the system should do?

Lets people state what they want through goals, rules, criteria, examples or structured inputs that the system can act on.

DESIGN CHOICES

RETAINED DISTINCTIONS

Words and demonstrations

FC053
natural-language or conversational specification

FC054
example or demonstration specification

Structured and executable input

FC057
structured input form or schema

FC058
authoring a formal or executable representation

Conditions and criteria

FC055
explicit constraint or rule specification

FC056
criteria and rubric specification

Recorded form · FC055: Rule, criterion or rubric authoring (VA042).

Fig. 32. Goals and constraints. How does a person specify what the system should do? Distinguish conveying intent, providing a structured or executable specification, and declaring constraints or evaluation criteria. Category definitions; Browse category mechanisms. Recorded forms: Rule, criterion or rubric authoring.

FC-C09 / SOURCE ILLUSTRATION

Goals and constraints

PAPER ILLUSTRATION · FC054

☐ Show only Comments labeled as "Remove" 30 / 30 labeled examples; 14 example(s) labeled as "Remove".

Those people who pull out guns just because they get upset are called gang banger thug losers. 99% of their guns are illegally possessed. Legal gun owners aren't taking out their guns during an argument at parties.

Remove **1** Keep

So I should get rid of my gun because some idiot killed.

Remove Keep

Idiot sandwich 🥪

Remove Keep

Load More Examples Create **2** your Filter

1. Keep or remove this example 2. Train the filter from labels

Situated keep/remove judgments supply examples of the content a person wants the classifier to retain.

End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting · Figure 2, PDF page 6

Wang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 33. Goals and constraints: source illustration. Figure 2, PDF page 6 in [128]. Situated keep/remove judgments supply examples of the content a person wants the classifier to retain. Inspect the illustrated mechanism.

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FC-C10 / User control and expression

Editing and steering

Where can a person intervene in the work?

Lets people change a representation, revise an output or redirect a process through edits, settings, parameters and reversible actions.



DESIGN CHOICES

RETAINED DISTINCTIONS

Object and output



FC047
direct manipulation of a semantic object

FC048
editable machine output

Settings and quantities



FC050
configuration of system behaviour

FC049
continuous parameter or dial control

Direction and commitment



FC051
steering an ongoing process

FC052
reversibility and safe experimentation

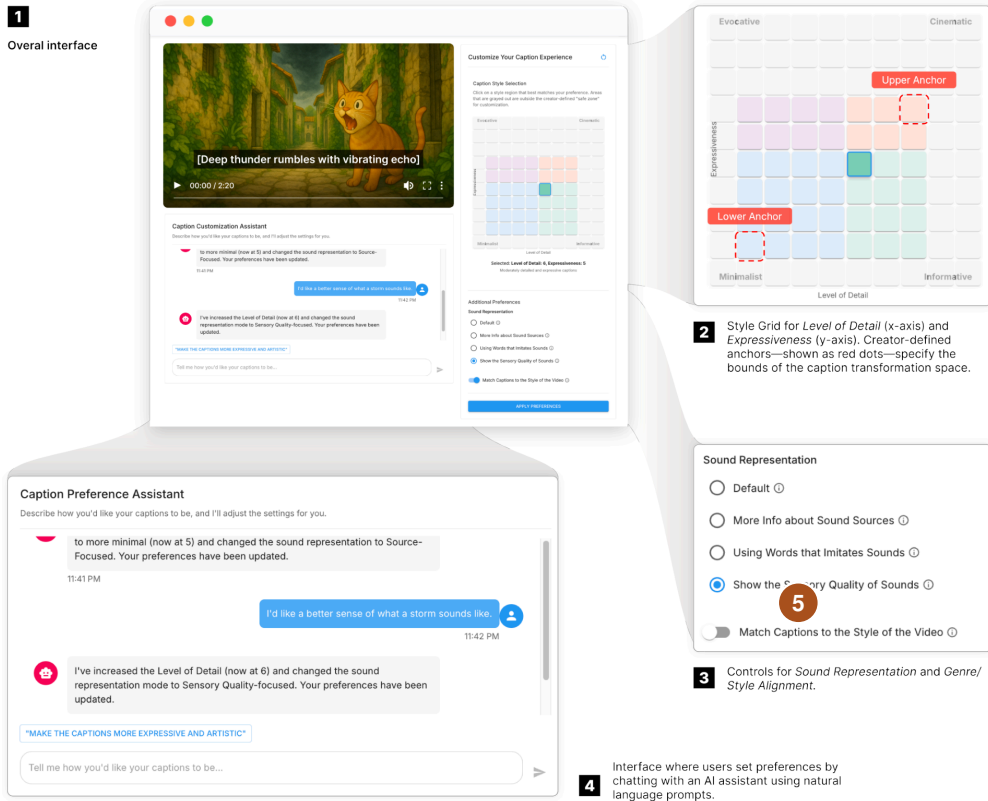
Recorded form · FC049: Parameter, threshold or weight controls (VA039).

Fig. 34. Editing and steering. Where can a person intervene in the work? Editing an object, configuring behaviour, steering the current process and controlling commitment operate at different points. Category definitions; Browse category mechanisms. Recorded forms: Parameter, threshold or weight controls.

FC-C10 / SOURCE ILLUSTRATION

Editing and steering

PAPER ILLUSTRATION · FC050



5. Choose how the sound is represented

Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material.

CapTune: Adapting Non-Speech Captions With Anchored Generative Models · Figure 4, PDF page 8

Huang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 35. Editing and steering: source illustration. Figure 4, PDF page 8 in [54]. Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material. Inspect the illustrated mechanism.

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FC-C11 / User control and expression
Generating and choosing alternatives

How are possibilities offered and shaped?

Provides candidate outputs, continuations or a space of possibilities that people can explore, constrain and select from.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Present a choice space	FC035 candidate or option set to choose from	FC038 option-space landscape or design space
Generate or continue	FC036 generated alternatives to explore	FC110 completion and continuation of content
Bound generation	FC037 user-constrained generation	

Recorded form · FC036: Gallery of generated alternatives to choose among (VA033).
Recorded form · FC110: Completion or continuation of what the person began (VA036).

Fig. 36. Generating and choosing alternatives. How are possibilities offered and shaped? Separate presenting a choice space, producing new content, and bounding the generation process. A continuation is not necessarily a set of alternatives. Category definitions; Browse category mechanisms. Recorded forms: Gallery of generated alternatives to choose among; Completion or continuation of what the person began.

FC-C11 / SOURCE ILLUSTRATION

Generating and choosing alternatives

PAPER ILLUSTRATION · FC110

Double-click the snippet to edit it directly. **THIS IS SUGGESTED TEXT**

Adjust role

Explain interaction × ⓘ

Adjust formatting

heading drophead body **note**

Get recommendations

Highlight interactive elements and guide the audience's interaction flow using moderately complex sentences. Use imperative sentences to instruct users on actions they can perform. Depending on the complexity, aim for 10-25 words, a readability level appropriate for someone with a 6th to 10th grade education, and a word variety of 50-70%.

Word count **5** ⓘ

Word variety **40%** ⓘ

Readability score **69** ⓘ

Your text is likely to be understood by someone who has at least a 8th-grade education.

Generate text

↻ Re-generate ✂ Shorten **1** 📄 Simplify

Add context to incorporate into generation

What should we consider when generating this text?

🗑 Delete snippet

1. Regenerate, shorten or simplify

A text snippet exposes explicit actions for generating a revised version while the author continues composing.

Plume: Scaffolding Text Composition in Dashboards · Figure 5, PDF page 11

Lisnic et al.; CC BY; cropped; numbered annotations added by this review.

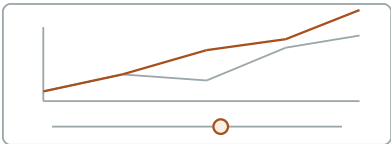
Fig. 37. Generating and choosing alternatives: source illustration. Figure 5, PDF page 11 in [72]. A text snippet exposes explicit actions for generating a revised version while the author continues composing. Inspect the illustrated mechanism.

FC-C12 / User control and expression

What-if exploration and rehearsal

What can be tried before acting in the real setting?

Lets people try hypothetical changes, probe a system's behaviour or practise in a simulated setting and inspect what happens.



DESIGN CHOICES

RETAINED DISTINCTIONS

Probe the system

FC098
system-behaviour probing

Construct a scenario

FC097
scenario construction and what-if exploration

Rehearse an activity

FC096
simulated environment for rehearsal

Recorded form - FC096: Practice, rehearsal or simulation environment (VA058).

Fig. 38. What-if exploration and rehearsal. What can be tried before acting in the real setting? Separate learning how a system responds, constructing a hypothetical world, and rehearsing a person's activity. Category definitions; Browse category mechanisms. Recorded forms: Practice, rehearsal or simulation environment.

FC-C12 / SOURCE ILLUSTRATION

What-if exploration and rehearsal

PAPER ILLUSTRATION · FC096



1. Dialogue history 2. Teacher question entry

Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history.

Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents · Figure 1, PDF page 3

Barrett et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 39. What-if exploration and rehearsal: source illustration. Figure 1, PDF page 3 in [9]. Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history. Inspect the illustrated mechanism.

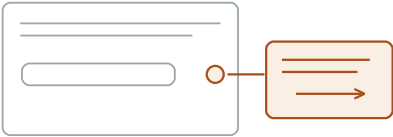
C.1.4 Guidance and feedback. How is the next action or reflection supported?

FC-C13 / Guidance and feedback

Recommendations and guidance

What helps a person decide where to look or what to do?

Directs attention or suggests an action, resource or procedure, including corrective advice, consequence previews and warnings.



DESIGN CHOICES

RETAINED DISTINCTIONS

Notice and hold back

FC043
attention direction and salience
cueing

FC044
risk alert or warning

Preview a
consequence

FC046
prospective consequence preview

Choose an action or
resource

FC040
next-step recommendation

FC042
item or content recommendation

Perform or repair

FC045
procedural how-to instruction

FC041
corrective or remediation guidance

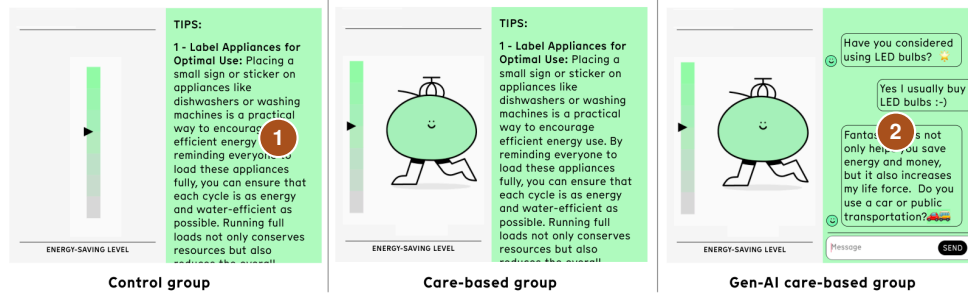
Recorded form · FC046: Preview of the consequence before committing (VA037).

Fig. 40. Recommendations and guidance. What helps a person decide where to look or what to do? Attention and warnings, previews, recommendations, and procedural or corrective guidance contribute different information to an action. Category definitions; Browse category mechanisms. Recorded forms: Preview of the consequence before committing.

FC-C13 / SOURCE ILLUSTRATION

Recommendations and guidance

PAPER ILLUSTRATION · FC040



1. Fixed tips 2. Conversational condition

The study contrasts fixed energy-saving tips with a conversational condition that elicits household context.

Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment · Figure 6, PDF page 6

Berney et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 41. Recommendations and guidance: source illustration. Figure 6, PDF page 6 in [12]. The study contrasts fixed energy-saving tips with a conversational condition that elicits household context. Inspect the illustrated mechanism.

FC-C14 / Guidance and feedback

Feedback and checking

What basis is provided for judging the work?

Shows how work is progressing or how a result meets a goal, criterion or reference. It includes feedback on attempts, problem diagnosis, quality assessments and checks.



DESIGN CHOICES

RETAINED DISTINCTIONS

Attempt and progress

FC065
formative outcome feedback

FC066
progress and status tracking

Problem, quality and status

FC063
Identifying problems in the work

FC064
evaluative judgment or quality rating

FC067
correctness or status marking

Check against a reference

FC068
verification and checking against a reference

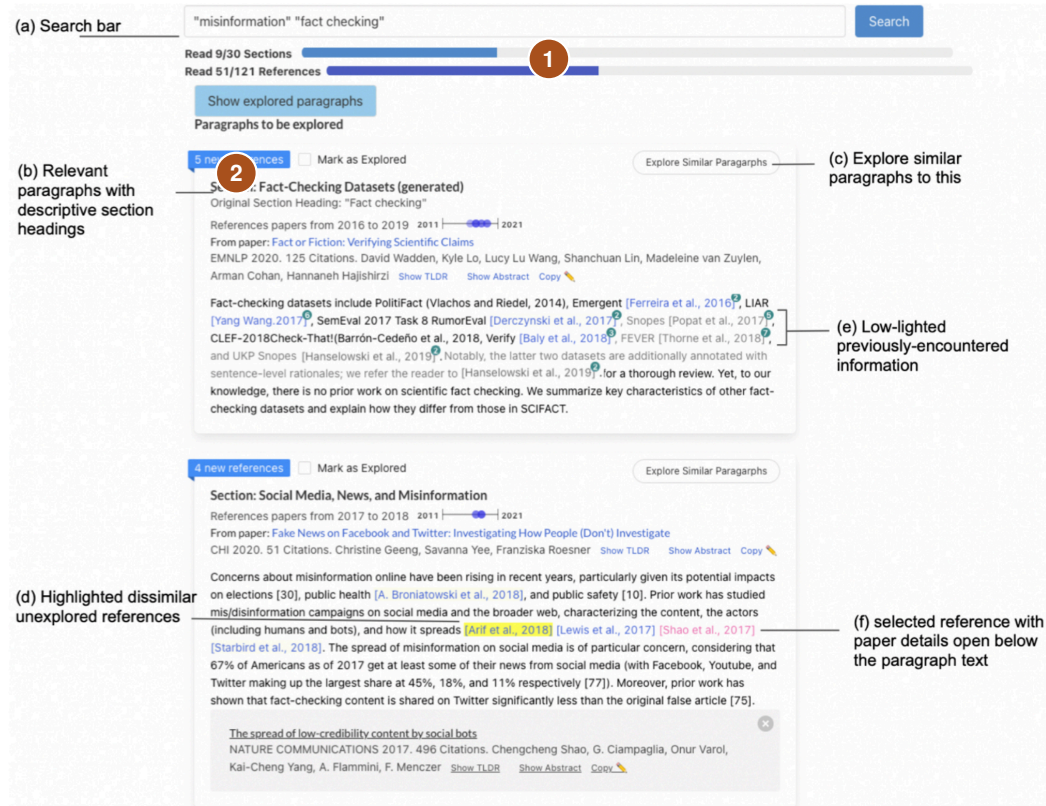
Recorded form · FC063: Error and failure diagnosis (VA007).

Fig. 42. Feedback and checking. What basis is provided for judging the work? Separate feedback about an attempt or progress, appraisals and status marks, and an active check against a reference. Category definitions; Browse category mechanisms. Recorded forms: Error and failure diagnosis.

FC-C14 / SOURCE ILLUSTRATION

Feedback and checking

PAPER ILLUSTRATION · FC066



1. Paragraph and reference coverage 2. Unread-reference count

Persistent coverage bars and local unread-reference counts make progress through the literature visible.

Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections · Figure 1, PDF page 2

Palani et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 43. Feedback and checking: source illustration. Figure 1, PDF page 2 in [97]. Persistent coverage bars and local unread-reference counts make progress through the literature visible. Inspect the illustrated mechanism.

FC-C15 / Guidance and feedback

Instruction and paced help

What help is available, and when is it available?

Provides explanations, examples and orientation, and controls when or how much help is available through staged disclosure, learning paths or changing support.



DESIGN CHOICES

RETAINED DISTINCTIONS

Explain, exemplify and orient

FC101
conceptual explanation of domain meaning

FC102
worked example or exemplar

FC103
orientation and onboarding

Change amount or release

FC099
graduated or faded support

FC100
progressive disclosure of detail

FC039
deliberate restriction or withdrawal of support

Sequence learning sessions

FC104
curriculum and learning-path structuring

Recorded form · FC099: Graduated help that escalates or fades by response (VA093).

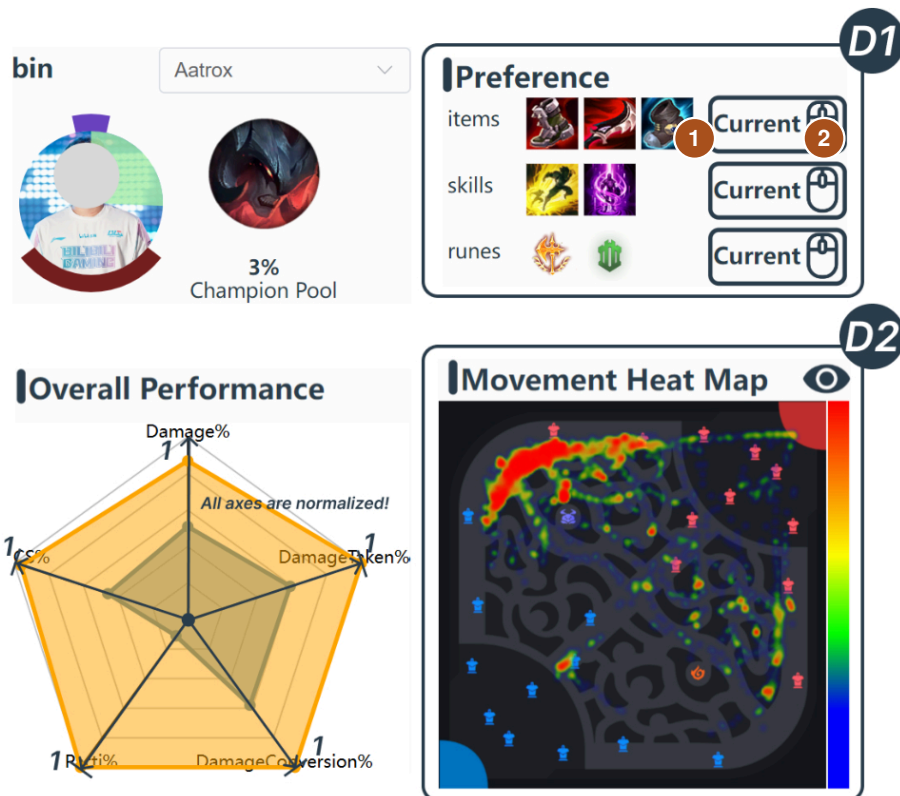
Recorded form · FC102: Worked or contrastive examples as instruction (VA057).

Fig. 44. Instruction and paced help. What help is available, and when is it available? Instructional content, the amount or timing of support, and progression across sessions are distinct design choices. Fading is not automatically evidence of learning. Category definitions; Browse category mechanisms. Recorded forms: Graduated help that escalates or fades by response; Worked or contrastive examples as instruction.

FC-C15 / SOURCE ILLUSTRATION

Instruction and paced help

PAPER ILLUSTRATION · FC102



1. Historical equipment choices 2. Switch to current choices

Historical equipment, skill and rune choices can be inspected beside controls revealing the current match.

StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports · Figure 8, PDF page 12

Ma et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 45. Instruction and paced help: source illustration. Figure 8, PDF page 12 in [77]. Historical equipment, skill and rune choices can be inspected beside controls revealing the current match. Inspect the illustrated mechanism.

FC-C16 / Guidance and feedback

Structuring the workflow

How are stages, responsibilities and continuity arranged?

Organises work into stages, loops, roles and decision points, with ways to coordinate actions, hand work over, recover from problems and resume later.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Sequence and recurrence	—	FC069 staged or phased workflow	FC070 iterative revision loop
Responsibility and routing	—	FC071 human decision checkpoint or gate	FC072 division of labour across agents
			FC074 orchestration and routing of actions
Continuity and recovery	—	FC073 handoff of work state	FC075 interaction repair and recovery
			FC076 resumable and persistent work state

Recorded form · FC075: System move that clarifies, confirms or repairs (VA038).

Fig. 46. Structuring the workflow. How are stages, responsibilities and continuity arranged? Separate progression through work, allocation of actions and authority, and continuity across interruptions or handoffs. Category definitions; Browse category mechanisms. Recorded forms: System move that clarifies, confirms or repairs.

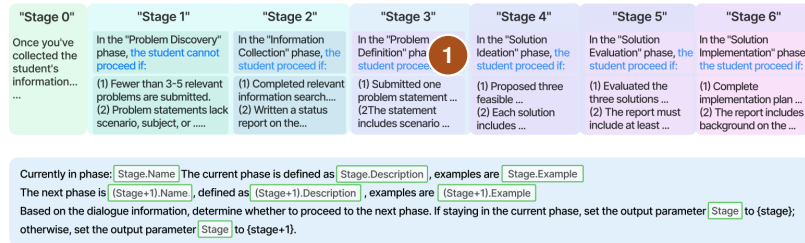
FC-C16 / SOURCE ILLUSTRATION

Structuring the workflow

PAPER ILLUSTRATION · FC074

Controller Agent Prompts

[Stage Decision Prompt]



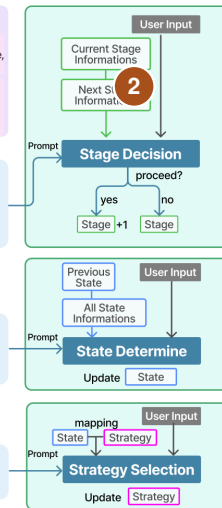
[State Determine Prompt]

If not moving to the next phase, the student's status may be: `State.Name` `State.Description`
 The student's previous status list is `[ State ]`. Remove all '0' and '1', and based on the new dialogue record, update and output the student's status list state.
 If moving to the next phase, the student's status will be "start," and update and output the student's status list as `state=[0]`.

[Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: `State.Name` `State.StrategyMapping` `Strategy.name` `Strategy.Description`
 Based on the dialogue information, output the most urgent problem ID as `urge_state` and the corresponding best strategy ID as `strategy`

Controller Agent Pipeline



1. Stage-specific completion criteria 2. Decision to advance or remain

Stage criteria and a controller decision determine whether the creative problem-solving workflow advances.

Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process · Figure 2, PDF page 9

Zha et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 47. Structuring the workflow: source illustration. Figure 2, PDF page 9 in [142]. Stage criteria and a controller decision determine whether the creative problem-solving workflow advances. Inspect the illustrated mechanism.

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FC-C17 / Guidance and feedback
Questions and reflection prompts

Which part of a person's thinking is elicited?

Asks people to express their goals, questions, explanations or reflections, making their own reasoning available to the work.

?

DESIGN CHOICES	RETAINED DISTINCTIONS
Needs and intentions	FC059 intent and need elicitation
Account and reasoning	FC062 narrative and reasoning elicitation
Questions and hypotheses	FC061 inquiry and hypothesis prompting
Reflection on thinking	FC060 reflective and metacognitive prompting

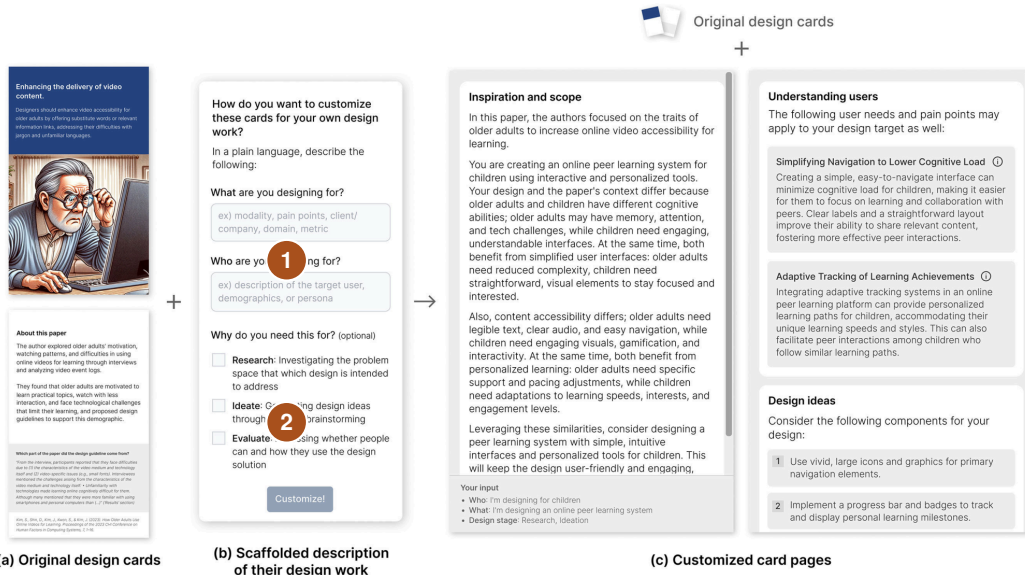
Recorded form · FC060: Reflection or self-assessment prompt (VA056).

Fig. 48. Questions and reflection prompts. Which part of a person’s thinking is elicited? The object of the prompt distinguishes needs, an account of reasoning, inquiry about the material, and reflection on one’s own thinking. Category definitions; Browse category mechanisms. Recorded forms: Reflection or self-assessment prompt.

FC-C17 / SOURCE ILLUSTRATION

Questions and reflection prompts

PAPER ILLUSTRATION · FC059



1. Who and what is being designed 2. Select the design stage

A guided form elicits design context before the system customizes research cards.

What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts · Figure 1, PDF page 7

Shin et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 49. Questions and reflection prompts: source illustration. Figure 1, PDF page 7 in [115]. A guided form elicits design context before the system customizes research cards. Inspect the illustrated mechanism.

C.1.5 Collaboration and perspective. How can people share work and perspectives?

FC-C18 / Collaboration and perspective

Shared views and perspectives

How do contributions and viewpoints become shared?

Makes people's contributions and viewpoints available to one another through shared representations, structured discussion, role framing and control over what is disclosed.



DESIGN CHOICES

RETAINED DISTINCTIONS

Shared work and awareness

FC090
shared co-constructed representation

FC091
visibility of others' state and judgment

Viewpoints and exchange

FC092
perspective-taking and viewpoint framing

FC094
role or persona framing of the interaction

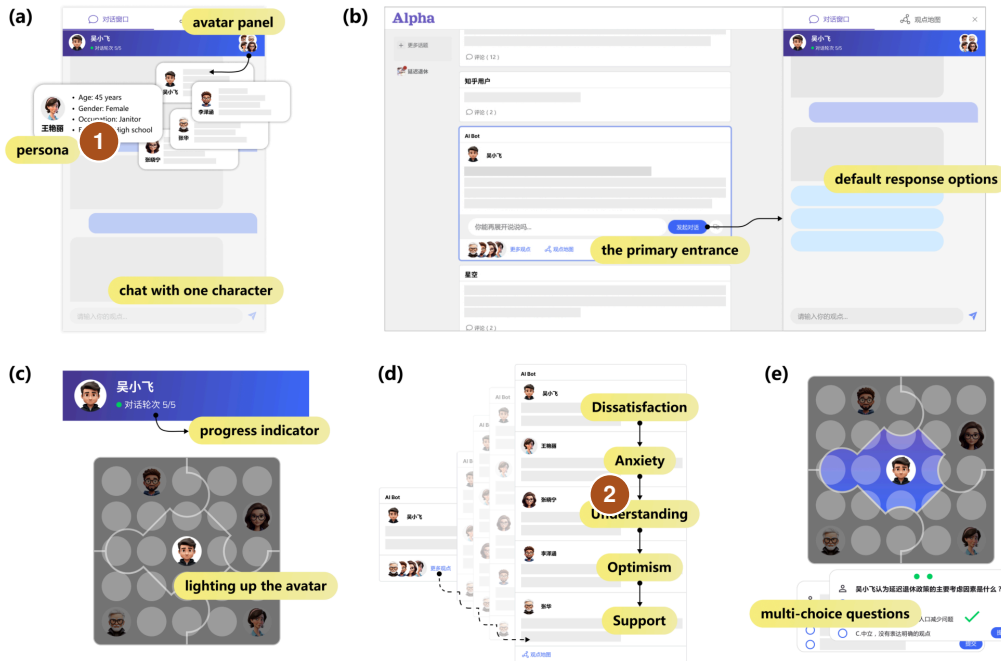
FC093
structured deliberation and perspective exchange

Control of disclosure

FC095
disclosure and privacy control

Recorded form - FC090: Shared workspace for joint work (VA065).

Fig. 50. Shared views and perspectives. How do contributions and viewpoints become shared? Separate shared artifacts and awareness, the framing and exchange of perspectives, and control of disclosure. Category definitions; Browse category mechanisms. Recorded forms: Shared workspace for joint work.

FC-C18 / SOURCE ILLUSTRATION**Shared views and perspectives****PAPER ILLUSTRATION · FC092**

1. Distinct stakeholder profiles 2. Progressive viewpoint sequence

Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure.

See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles · Figure 3, PDF page 7

Zhang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 51. Shared views and perspectives: source illustration. Figure 3, PDF page 7 in [146]. Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure. Inspect the illustrated mechanism.

C.2 Complementary Evidence Profiles

The learning-by-family matrix uses the open-code classification and its 2,761-paper matched population. It is distinct from the closed-field learning count of 301 among 2,891 included papers.

Claim support and developmental risk indicators

Each row is a mutually exclusive coding over the same 2,891 included cards.

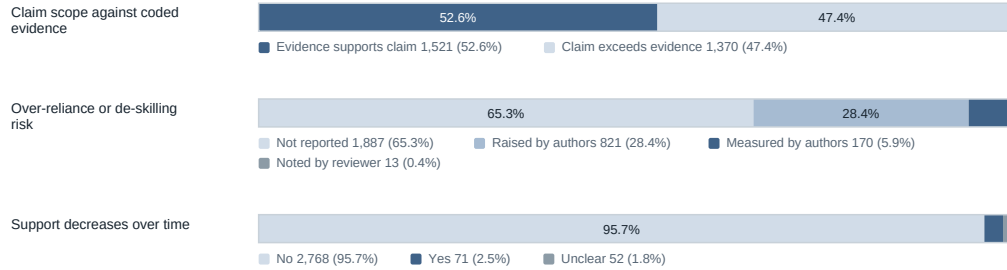


Fig. 52. **Claim scope exceeds coded evidence in nearly half of included cards, over-reliance is raised far more often than it is measured, and decreasing support is rare.** Each row is a mutually exclusive coding over the same 2,891 included cards, and every segment prints its own count and share. A claim-evidence gap records the distance between a card's stated claim and its linked evidence; it is not an assessment of the underlying paper. Examples, one or two cards per value: claim within evidence: InstructPipe [148].

Support profiles within learning-evidence groups

share of the row's papers carrying each family; 2,761 papers coded in both fields

	what the interface makes reachable				
	Organisation and comparison	Guidance and feedback	Explanation and visibility	User control and expression	Collaboration and perspective
learning evidence					
Learning not measured n = 1,334 papers	73.8% 985	66.9% 893	64.2% 856	52.5% 701	11.4% 152
A proxy measured instead n = 1,039 papers	71.3% 741	74.1% 770	61.5% 639	50.0% 520	14.6% 152
Learning measured n = 258 papers	60.9% 157	82.9% 214	57.4% 148	41.5% 107	14.3% 37
Learning self-reported only n = 130 papers	71.5% 93	84.6% 110	63.8% 83	54.6% 71	20.0% 26
all papers in the matrix n = 2,761 papers	71.6%	72.0%	62.5%	50.7%	13.3%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 53. **Guidance and feedback appear in 83% of capability-measured papers.** Rows: the learning-evidence families, including the papers that measured no capability, with the number of papers in each. Columns: the facet families. Each cell is the share of the row's papers carrying the column family, with the count; colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

C.3 Context and Evidence Profiles

The 2,891 closed-field papers record 2,071 empirical, 691 artifact, 123 methodological, four dataset, and two theoretical contributions. Contribution labels do not identify research methods or target users; study designs and participants are reported in Section 4.2.

In the design space, the domain mix shows no simple annual shift: technical, analytic, and creative professions account for 54.8% in 2019 and 55.5% in 2025, ranging from 53.4% to 61.6% across these years. Annual totals also reflect reading coverage and the incomplete 2026 publication window, so they do not estimate field-wide growth.

What is made reachable, against what the person still does

share of the row's papers in each column; 2,311 papers coded in both fields

what the interface makes reachable	what the person still does			
	The task the tool did not do	New work the tool created	The whole task kept by the person	Final authority kept by design
Organisation and comparison n = 1,639 papers	52.2% 856	29.1% 477	11.4% 187	7.1% 117
Guidance and feedback n = 1,679 papers	49.9% 838	30.1% 506	11.1% 187	8.8% 147
Explanation and visibility n = 1,452 papers	53.2% 772	28.0% 407	10.1% 146	8.7% 126
User control and expression n = 1,170 papers	42.0% 491	35.8% 419	14.2% 166	7.9% 93
Collaboration and perspective n = 311 papers	56.3% 175	19.0% 59	12.9% 40	11.9% 37
all papers in the matrix n = 2,311 papers	51.4%	30.2%	10.1%	8.2%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 54. **The profiles share an ordering, with different shares of new work.** The matrix uses the 2,311 systematic papers with complete facet-value and remaining-work assignments. Rows: the facet families, with the number of papers carrying each. Columns: what the person still does. Each cell is the share of the row's papers whose remaining work falls in the column family, with the count. Papers can belong to several mechanism-family rows; each has one remaining-work summary category in this analysis. Colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

Study design is recorded for 2,071 papers: 691 within-subject designs (33.4%), 328 single-condition usability studies (15.8%), and 119 field deployments (5.7%). The 2,072 entries include one paper with two designs; percentages use papers. Usability studies reveal practical difficulties; deployments can expose work missed by short comparisons.

The participant field also covers 2,071 papers: mixed groups account for 540 (26.1%), target practitioners 499 (24.1%), and crowdworkers 251 (12.1%); 382 (18.4%) are unspecified. Participants need not represent a system's intended users, and missing descriptions do not establish broad applicability.

C.3.1 Most process shifts rest on an author's attribution. Of 8,045 process shifts across 2,891 papers, 6,102 (75.8%) are author-attributed, 1,746 (21.7%) use measured contrasts, and 197 are reviewer inferences. Baselines also differ: 1,940 use a control condition, 869 state none, and the remainder refer to prior practice or an author's assertion. Describing a changed workflow does not establish its effect size.

C.4 Mechanism Families and Remaining Work

The detailed profile complements the control/work comparison in Section 4.2.

D ANNUAL TRENDS AND VENUE RATES

These tests and counts support the comparisons in Section 4 and Section 5. Trends use AI-involved papers with usable codes for each field. Technology follows the AI a paper builds or uses, independently of publication year: generative AI includes hybrids and earlier neural generators; traditional AI/ML includes predictive, discriminative, and symbolic systems. The technology projection covers 3,447 classified papers; 639 remain unassigned. Each annual percentage uses papers with assigned technology and usable coding for that dimension in the same year. Count summaries combine

2019–2026; the annual lines separate the years. Alias disagreements are withheld for the affected field. Adjusted contrasts account for publication year and venue as described below.

D.1 Research-context trends

We model binary context membership with year and venue fixed effects, using venue-clustered standard errors with a finite-sample correction and Student’s t reference distribution. The linear probability coefficient estimates average within-venue change in percentage points per year. Holm adjustment controls the family-wise error rate across the 16 directional tests. Intervals are pointwise 95% confidence intervals. Separate categorical-year tests check temporal variation and departures from a straight trend; these are available with the analysis data. Results describe the coded review frame.

Table 12. Venue-adjusted annual changes in research contexts, in percentage points per year. Primary estimates use 2019–2025; the last column includes available 2026 papers. Sensitivity intervals and adjusted tests are provided online.

Context	Annual change [95% CI]	Adjusted p	With 2026
AI and machine learning practice	-0.59 [-1.18, +0.00]	0.725	-0.67
Data analysis and visualization	-0.34 [-0.89, +0.22]	1.000	-0.44
Software engineering and programming	-0.34 [-0.72, +0.04]	0.894	-0.27
Scientific research practice	+0.35 [-0.03, +0.73]	0.894	+0.40
Information seeking and knowledge work	+0.42 [+0.14, +0.69]	0.064	+0.36
Writing, reading and language	+0.22 [-0.10, +0.55]	1.000	+0.31
Design practice	-0.17 [-0.60, +0.26]	1.000	-0.06
Creative and media production	+0.16 [-0.34, +0.65]	1.000	+0.27
Education and learning	+0.11 [-0.33, +0.54]	1.000	+0.11
Health and clinical care	+0.29 [-0.24, +0.83]	1.000	+0.27
Accessibility and disability	+0.28 [+0.06, +0.50]	0.210	+0.04
Everyday life and personal decisions	-0.28 [-0.61, +0.04]	0.894	-0.10
Online platforms and civic information	-0.10 [-0.49, +0.28]	1.000	-0.07
Physical and embodied activity	-0.17 [-0.60, +0.25]	1.000	-0.09
Business, industry and work	+0.28 [-0.03, +0.60]	0.894	+0.14
Public sector, law and societal systems	-0.04 [-0.20, +0.12]	1.000	-0.12

D.2 Technology comparisons

We compare binary code membership using a linear probability model with technology, categorical publication year, and venue fixed effects. Primary estimates use 2019–2025, and sensitivity models restrict the analysis to 2023–2025 or add partial 2026. Venue-clustered standard errors use a finite-sample correction and a Student t reference with one fewer degree of freedom than the number of venues, and model intervals are pointwise 95% confidence intervals. Holm correction covers all 51 context, activity, family, and category contrasts jointly, with each sensitivity window and venue omission corrected separately. We omit six major venues one at a time: CHI, CSCW, EMNLP, IUI, TVCG, and VIS, each of which supplies at least 5% of the coded papers for workflow, families, or categories in 2019–2025. The choice rests on coding denominators, not outcomes. Weighting checks reweight shared year–venue cells with at least one or at least

Table 13. User control after omitting each major venue and on shared year–venue cells. Gaps are GenAI minus traditional AI/ML in percentage points. Venue-omission gaps are adjusted for year and venue, with pointwise 95% confidence intervals and Holm correction over all 51 outcomes in each fit. Shared-cell gaps are point estimates. Equal-venue gaps also have 95% venue-bootstrap intervals. The one-paper threshold keeps 82.7% of the coded papers, the five-paper threshold 41.0%.

Venue omission	Gap [95% CI]	Holm p
Primary	+13.7 [+8.6, +18.7]	< .001
Omit CHI	+12.6 [+6.5, +18.7]	0.009
Omit CSCW	+13.4 [+8.1, +18.7]	< .001
Omit EMNLP	+14.2 [+9.1, +19.2]	< .001
Omit IUI	+13.7 [+8.2, +19.3]	< .001
Omit TVCG	+11.9 [+7.1, +16.8]	< .001
Omit VIS	+14.3 [+9.4, +19.3]	< .001
Shared-cell weight	At least one per group	At least five per group
Pooled papers	+10.8	+17.3
Within-cell overlap	+12.5	+17.1
Equal cells	+8.2	+16.6
Equal venues	+7.8	+15.1
95% interval	[-2.9, +18.5]	[+9.3, +21.1]
Equal years	+10.0	+18.8

five coded papers per technology by pooled paper count, within-cell overlap, equal cells, equal venues, or equal years. Intervals for equal-venue gaps come from 10,000 resamples of venues.

User control. User control and expression occurs in 633/1,071 GenAI papers (59.1%) and 623/1,373 traditional AI/ML papers (45.4%) in 2019–2025. Omitting each major venue gives gaps from +11.9 to +14.3 points, each with Holm-adjusted support (Table 13). On the 2,022 papers in shared year–venue cells, the gap is +12.5 with overlap weights and +7.8 when venues receive equal weight. Every weighting and cell threshold leaves it positive.

Facets and work codes. The facet and work-code gaps in the main text use the same model on each paper’s latest coding, with a normal reference for intervals. Of the 2,988 technology-classified papers from 2019–2025, 2,532 have facet codes (1,143 GenAI and 1,389 traditional AI/ML), 2,119 have codes for delegated work, and 2,526 have codes for the work people keep. Holm correction runs across the 101 facets coded on at least 40 papers and, separately, across the 24 work codes on at least 30 papers. A gap gets a direction in the text only if it has Holm-adjusted support, keeps its sign under all five weightings at both thresholds, and has an equal-venue interval that excludes zero at one threshold or both (Table 13 and Table 14). Fitted separately for the two models that coded most papers, the control gap is +13.9 (95% CI 8.8 to 19.0; 2,160 papers) and +11.1 (–1.1 to 23.3; 203 papers).

D.3 Temporal change in mechanism families

We distinguish average annual change in family prevalence from annual change in the GenAI–traditional AI/ML difference. Both come from a linear probability model with technology, publication year, their interaction and venue fixed effects. Average change holds the technology mixture fixed at 43.75% GenAI (the original primary cohort’s reference share), separating it from shifts in the composition of the corpus. Positive gap change means movement toward a more positive GenAI difference; it need not enlarge the absolute gap. Venue-clustered intervals use finite-sample correction and a Student t reference. Holm correction covers all ten tests within each window; the 95% confidence intervals are

Table 14. The other gaps given a direction in the text. Gaps are GenAI minus traditional AI/ML in percentage points. Primary gaps are adjusted for year and venue, with 95% confidence intervals. Equal-venue gaps use shared year-venue cells with at least one or at least five papers per technology, with 95% venue-bootstrap intervals. The two thresholds keep 82.7–85.4% and 40.3–46.6% of each coded population. Every gap keeps its sign under all five weightings at both thresholds.

Gap	Primary [95% CI]	Equal venues, papers per group	
		At least one	At least five
Tool generates new artifacts	+16.6 [+12.7, +20.5]	+13.8 [+6.6, +22.3]	+17.7 [+9.3, +29.6]
Tool generates responses and revisions	+9.4 [+5.7, +13.2]	+10.2 [+4.7, +17.1]	+7.6 [+0.8, +14.1]
Tool calculates and models	-11.5 [-15.5, -7.5]	-16.2 [-25.8, -7.2]	-16.2 [-25.6, -7.9]
Tool evaluates and classifies	-9.7 [-14.6, -4.7]	-2.6 [-12.7, +7.9]	-13.2 [-21.5, -2.7]
Tool compares and finds patterns	-6.7 [-9.7, -3.7]	-7.0 [-14.3, -1.1]	-1.2 [-9.5, +8.4]
Generated alternatives to explore	+4.8 [+3.0, +6.6]	+4.3 [+0.9, +8.5]	+1.5 [-1.7, +4.5]
Completion and continuation of content	+3.0 [+1.4, +4.6]	+2.1 [-0.6, +5.3]	+2.1 [+0.6, +3.9]
Person steers the loop	+3.4 [+1.2, +5.6]	+6.7 [+2.4, +11.7]	+0.3 [-5.0, +4.9]
Person improves the system	-5.9 [-7.7, -4.0]	-5.3 [-9.3, -1.8]	-6.4 [-10.9, -2.7]
Organization and comparison (family)	-8.0 [-12.4, -3.7]	-11.5 [-19.9, -3.2]	-7.9 [-17.2, +1.8]
Ranked or prioritized surfacing	-6.5 [-8.3, -4.7]	-3.4 [-7.6, +0.3]	-9.7 [-18.0, -3.7]

Table 15. Mechanism-family trends, 2019–2025: 2,444 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	+1.16 [−0.70, +3.03]	1.000	+1.41 [−0.97, +3.79]	1.000
Organisation and comparison	+0.06 [−1.73, +1.84]	1.000	+0.43 [−2.08, +2.95]	1.000
User control and expression	−1.02 [−2.44, +0.41]	0.938	+0.53 [−1.17, +2.23]	1.000
Guidance and feedback	+1.58 [+0.70, +2.47]	0.008	+1.87 [−0.44, +4.19]	0.771
Collaboration and perspective	+1.10 [+0.52, +1.68]	0.005	+2.24 [+0.03, +4.46]	0.379

Table 16. Mechanism-family trends, 2023–2025: 1,390 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	−0.21 [−5.64, +5.22]	1.000	+1.18 [−6.50, +8.87]	1.000
Organisation and comparison	−2.88 [−6.68, +0.93]	0.941	−2.82 [−9.62, +3.99]	1.000
User control and expression	−3.32 [−6.48, −0.16]	0.322	−0.60 [−6.51, +5.31]	1.000
Guidance and feedback	+1.87 [−2.73, +6.47]	1.000	+3.60 [−3.74, +10.94]	1.000
Collaboration and perspective	+2.67 [+1.20, +4.14]	0.007	+6.64 [+1.90, +11.38]	0.066

pointwise. These are linear summaries of coded design prevalence, not tests of learning or causal technology effects. Partial 2026 is excluded.

D.4 Annual distributions

Research contexts: volume and annual share

3,266 coded papers with assigned AI technology · categories overlap

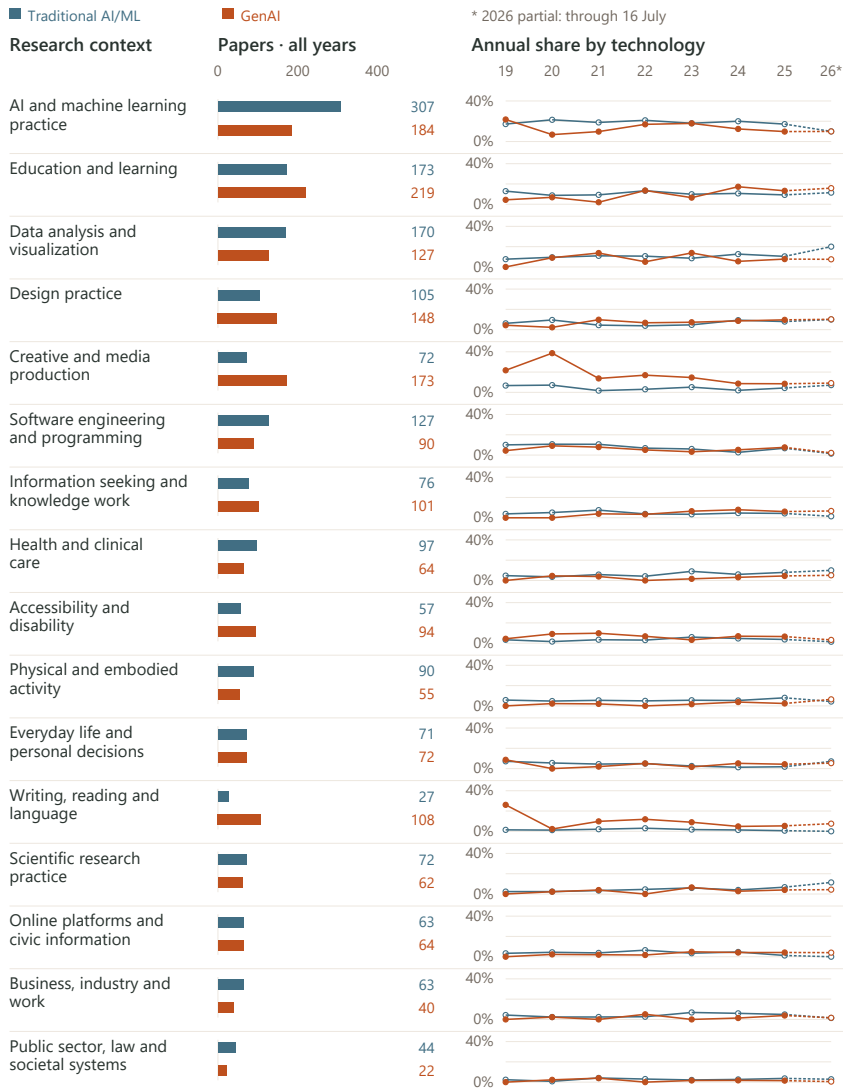


Fig. 55. Research contexts by AI technology. Bars show paper counts across all years; lines show each context's share of its technology group per year, with the coded papers behind those shares below. Contexts overlap. Shares are not adjusted; Table 12 gives the venue-adjusted tests. Only papers with usable context coding and a known technology are included.

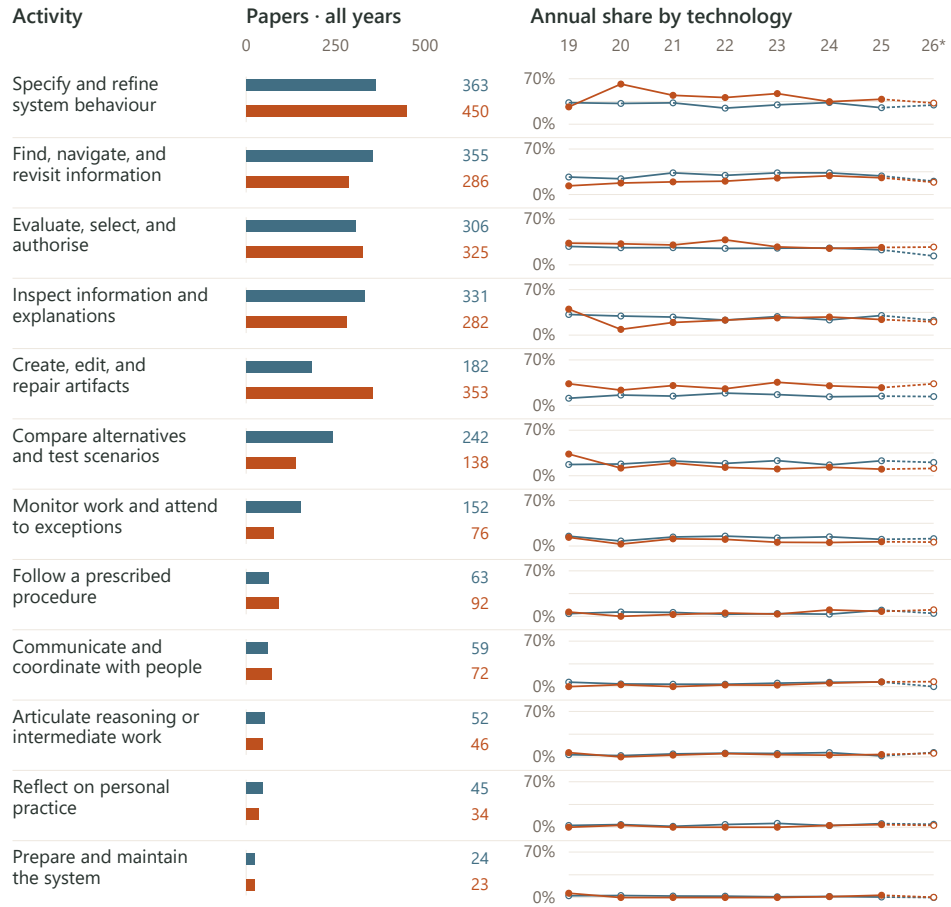
Workflow activities: volume and annual share

2,390 coded papers with assigned AI technology - categories overlap

■ Traditional AI/ML

■ GenAI

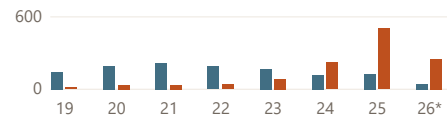
* 2026 partial: through 16 July



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 56. Workflow activities by AI technology. Bars show counts across all years; annual lines show shares within each technology group. Denominators count papers with usable activity codes and a known technology. Activities overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue.

Mechanism families: volume and annual share

2,845 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

■ GenAI

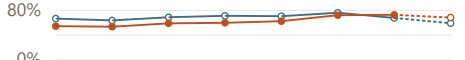
* 2026 partial: through 16 July

Mechanism family

Papers · all years

Annual share by technology

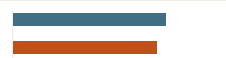
Guidance and feedback



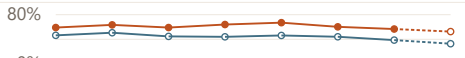
Organisation and comparison



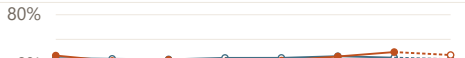
Explanation and visibility



User control and expression



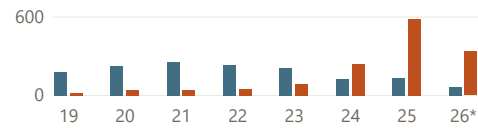
Collaboration and perspective



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 57. Mechanism families by AI technology. Bars encode counts across all years; lines compare annual shares within each technology group. The lower bars show annual coding denominators. Families overlap. Only papers with usable family codes and a known technology are included. Shares are not adjusted; Table 15 gives the venue-adjusted trends.

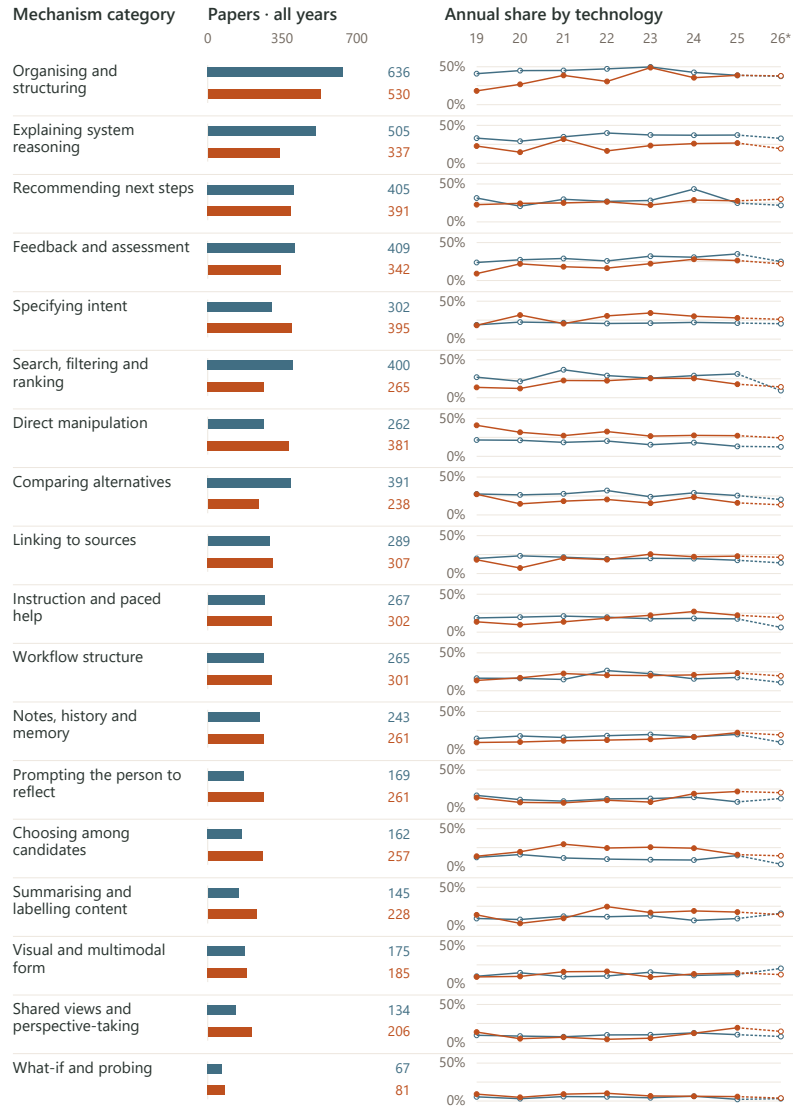
Mechanism categories: volume and annual share

2,839 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

■ GenAI

* 2026 partial: through 16 July



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

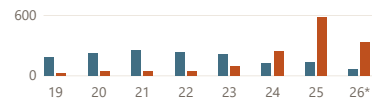


Fig. 58. Mechanism categories by AI technology. Bars encode counts across all years; lines show annual shares within each technology group. Categories overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue. Only papers with usable category codes and a known technology are included.

D.5 Learning overlays

These counts cover the 3,378 papers with usable category coding, of the 4,086 papers in the analysis at 26 September 2026, including papers without an assigned AI technology. Learning scope means the paper is in the learning review; measured means the paper measured learning explicitly. Learning scope and Measured are counted separately within each category; missing or conflicting labels are omitted from the corresponding column.

Table 17. Papers per mechanism category: all papers, papers in learning scope, and papers that measured learning. Counts are distinct papers; categories overlap.

Category	Papers	Learning scope	Measured
Explaining system reasoning	971	213	97
Linking to sources	675	119	33
Summarising and labelling content	414	107	37
Visual and multimodal form	464	155	54
Organising and structuring	1,372	293	93
Comparing alternatives	725	166	60
Search, filtering and ranking	769	177	70
Notes, history and memory	603	170	65
Specifying intent	815	161	55
Direct manipulation	736	138	55
Choosing among candidates	484	104	36
What-if and probing	169	79	32
Recommending next steps	986	330	133
Feedback and assessment	898	318	121
Instruction and paced help	681	347	154
Workflow structure	655	173	69
Prompting the person to reflect	496	242	90
Shared views and perspective-taking	405	163	57

D.6 Venue candidates and current inclusions

Table 18. Venue candidates and AI-involved inclusions in the 16 July 2026 frame, assessed through 26 September 2026. The 37,978 venue-year records represent 37,660 candidate identities. Inclusion (%) is Included divided by Initial candidates. Papers without AI, and papers whose identity or scope is unresolved, are not in Included. Venue memberships can overlap; group and overall totals count each identity once.

Venue	Initial candidates	Included	Inclusion%
HCI & interaction	17,436	2,546	14.6%
56.0% of included corpus			
CHI	7,639	1,147	15.0%
UIST	860	156	18.1%

Venue	Initial candidates	Included	Inclusion%
IUI	687	317	46.1%
DIS	1,471	166	11.3%
CSCW	2,412	212	8.8%
C&C	301	44	14.6%
TOCHI	478	37	7.7%
TiiS	230	95	41.3%
EICS	174	23	13.2%
HAI	214	26	12.1%
ECSCW	205	4	2.0%
GROUP	142	6	4.2%
ASSETS	387	59	15.2%
HRI	630	94	14.9%
IVA	222	27	12.2%
CHIIR	312	40	12.8%
HCOMP	112	28	25.0%
IJHCS	960	65	6.8%
Design & creativity	2,502	102	4.1%
2.2% of included corpus			
ICCC	422	28	6.6%
Design Science	232	15	6.5%
Design Studies	234	1	0.4%
IDETC	1,614	58	3.6%
Visualization	6,336	895	14.1%
19.7% of included corpus			
VIS	1,772	349	19.7%
EuroVis	537	84	15.6%
TVCG	4,073	445	10.9%
VL/HCC	255	65	25.5%
Graphics	2,385	115	4.8%
2.5% of included corpus			
SIGGRAPH	1,673	87	5.2%
SIGGRAPH Asia	712	28	3.9%
Learning & education	2,834	226	8.0%
5.0% of included corpus			
L@S	181	34	18.8%
LAK	655	85	13.0%
AIED	1,168	41	3.5%
IJAIED	347	34	9.8%

Venue	Initial candidates	Included	Inclusion%
EDM	483	32	6.6%
Software engineering 7.5% of included corpus	3,944	343	8.7%
CHASE	249	18	7.2%
ICSE	1,038	125	12.0%
FSE	1,297	88	6.8%
ASE	1,360	112	8.2%
AI & language 7.0% of included corpus	2,223	317	14.3%
ACL	731	133	18.2%
EMNLP	794	146	18.4%
RecSys	698	38	5.4%
Overall (distinct identities)	37,660	4,544	12.1%

E VOCABULARY GROWTH DURING READING

These curves show how each vocabulary grew as cards were read, so a reader can judge how complete the codes are. The vocabulary built from the first 3,295 mechanism cards gained 12 codes across the next 1,754 papers, and each new code was assigned to an existing category and family. All open fields were checked on 821 of those papers; the other batches checked mechanism facets, values, and domains (Figure 59).

E.1 Reading the Curves

We count distinct values after each batch of ten cards, in reading order, and scale each curve to its own final vocabulary. Codes, categories, and families are drawn where those levels exist; categories and families were assigned after coding, so they can reach their final size early in a curve. Reference lists in the lineage cards are open-ended, while their relation options are fixed by the card format. A flat segment means that no new label appeared in those cards, not that unread papers would add none.

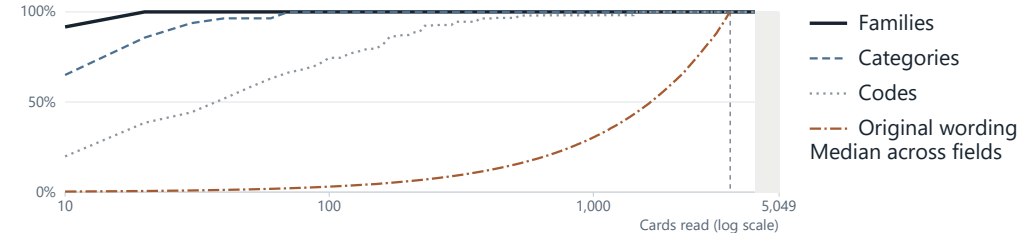
E.2 Stream 1 mechanism cards: Interface and Outcome Fields

These grids show the variation the overview hides. Every panel states its field and final count, and lines are drawn only for the levels present in that field.

Vocabulary growth across three card types

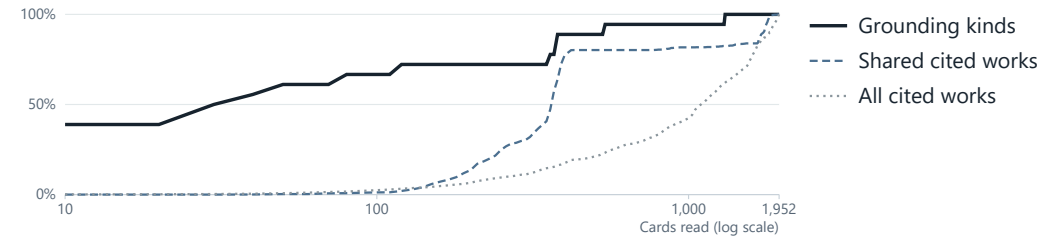
Share of the recorded vocabulary encountered; paper counts use a log scale

Stream 1 · Mechanism cards · 5,049



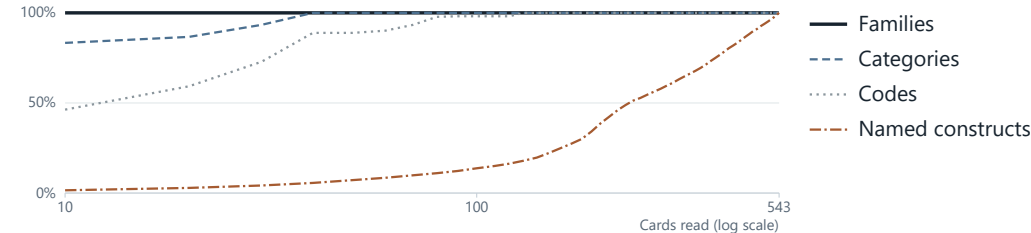
After 3,295: 12 new codes in 1,754 cards; field coverage varies in the shaded region.

Stream 1 · Theory-lineage cards · 1,952



Cited works remain an open vocabulary; grounding kinds are a fixed schema.

Stream 2 · Knowledge-claim cards · 543



Frozen coding covers 542 cards; named constructs cover all 543. Curves stop at their coverage.

Fig. 59. Vocabulary growth across the three card types. Panels count cards in reading order on logarithmic axes: 5,049 mechanism cards, 1,952 theory-lineage cards, and 543 Stream 2 cards. The Stream 1 panels include cards of non-AI comparison papers. Shading marks uneven field coverage, and median curves stop before it.

Codes, categories and families reached, by cards read, every field



Fig. 60. Stream 1, mechanism cards: vocabulary growth in each open field. Each panel separates codes, categories, and families where present, in reading order.

E.3 Stream 1 theory-lineage cards: Cited Sources and Their Roles

The curve covers the 1,952 newly written lineage cards; reused cards appear in the gallery but not in this sequence.

Stream 1 theory-lineage cards: references and relations

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Relations, positions and grounding kinds use fixed options: their flat tails are not evidence of open-code saturation.

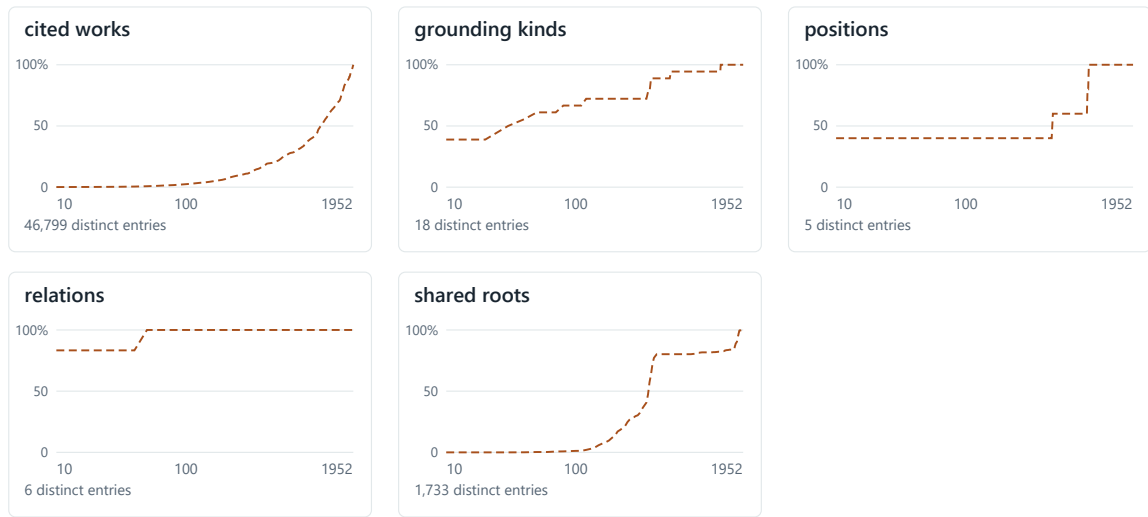


Fig. 61. Stream 1, theory-lineage cards: reference lists keep growing after relation options are exhausted. Fixed options and open-ended lists are separate panels because they close in different ways.

E.4 Stream 2: Claims and Conditions

Stream 2: each coded field

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Dotted: raw values. Dashed: canonical codes. Solid: categories. Thick: families, where present.

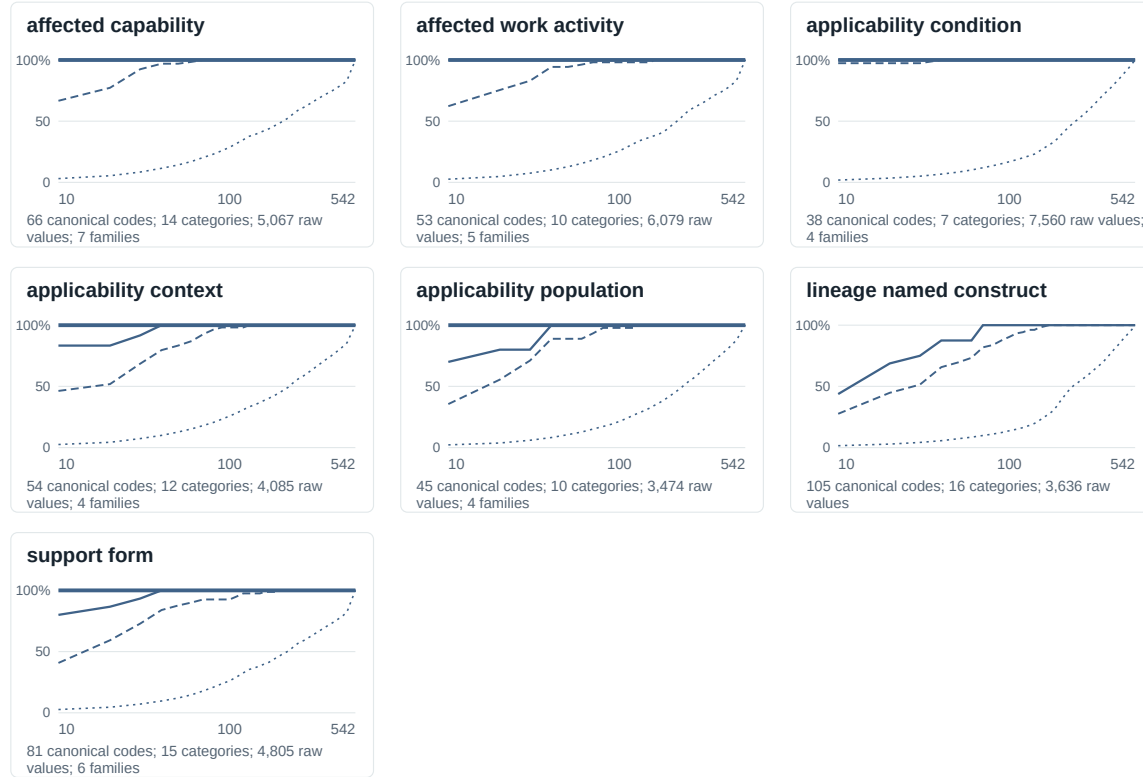


Fig. 62. **Stream 2: vocabulary growth differs by coded field.** Codes, categories, and families are drawn only where that level exists.

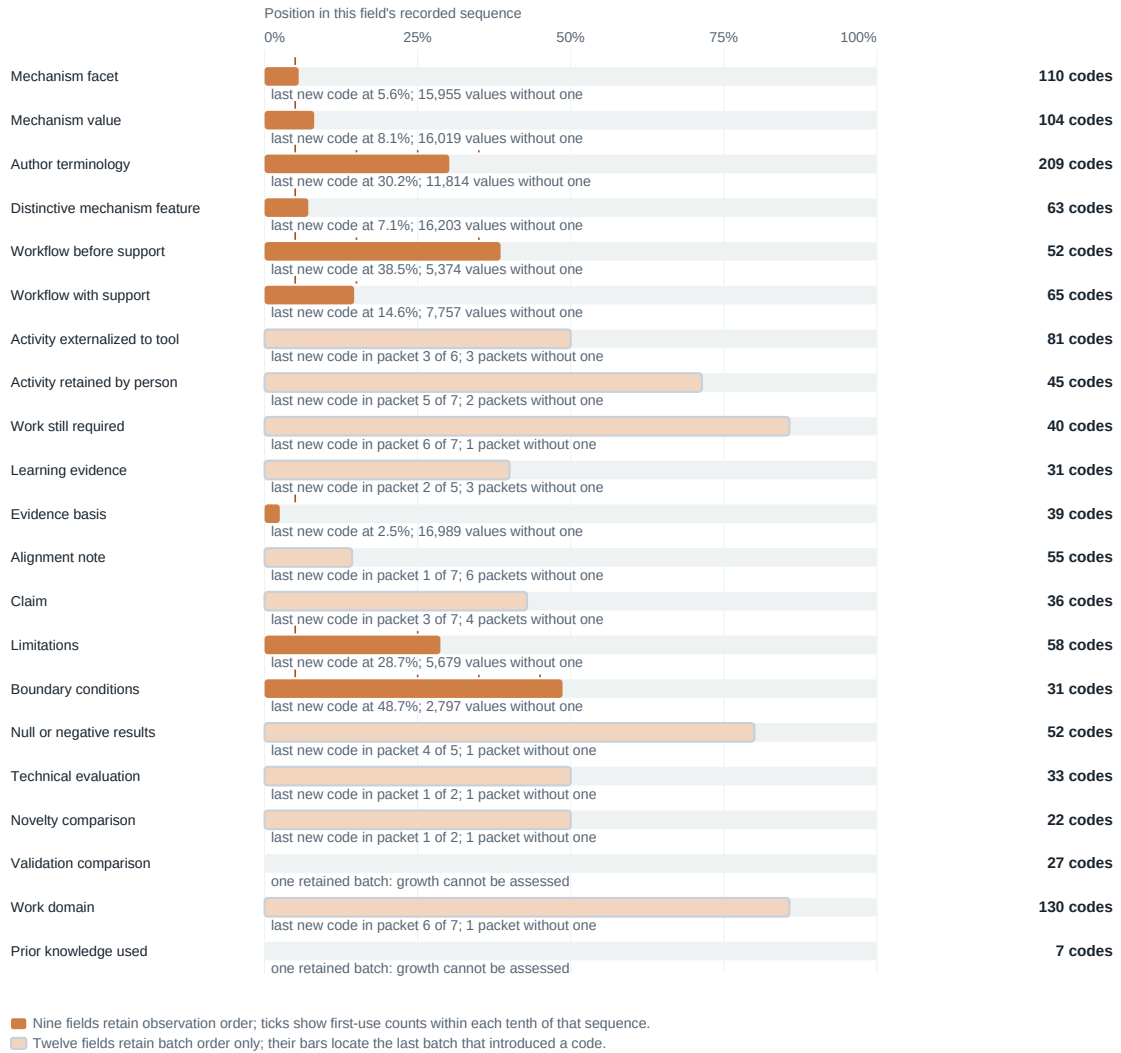
Stream 2's seven open fields cover 542 cards and 35,699 observations. Of the 449 codes, 448 are used, and no new code entered in the last 16 coding batches. A flat curve can still hide a missing code: no population code names a studied unit that is or includes a machine, and 18 assignments flag this while using codes for people or artifacts.

E.5 New Codes Within the First Coding

Within the first coding of each open field, the last new code appeared between 2.5% and 48.7% of the way through the fields whose coding order was kept (Figure 63). Three fields still have low-confidence assignments for more than a quarter of their observations, so a vocabulary can stop growing and remain too coarse.

Where did the recorded vocabulary stop expanding?

Colour ends at the last new code. The pale remainder reuses codes already encountered.



This checks label reuse within the recorded sequence, not coverage of every meaningful distinction. Stable labels can still contain low-confidence or overly broad assignments.

Fig. 63. **New-code creation within each field of the first coding.** Bars locate the last new code in the recorded coding sequence; ticks show new codes by tenth of the sequence. Where only batch order survives, the figure reports batch resolution.